ImPRESS — Biomarker Model Fit Assessment

ImPRESS Statistical Team

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Contents

1 Introduction

This document provides a pre-unblinding assessment of model fit for the **33 ADLB analytes** from Bio-Plex 19-plex and 14-plex serum panels in **54 subjects** (cohort 2).

The primary analysis (`make_rdlb.R`) fits two competing LMMs per analyte:

Model	Response	Interpretation
Linear	`chg = aval - base`	Mean change from Screening baseline (pg/mL)
Log-ratio	`log(aval/base)  log_chg`	Mean log-fold change from Screening baseline

For both models the fixed effects are: baseline value, stepped-wedge dose (`trtcd`), and visit day (`avisitn`), with a random intercept per subject. AIC selects the preferred model per analyte.

This report assesses:

1. Data completeness and out-of-range proportions
2. Concentration distributions (raw and log scale)
3. Change-score distributions
4. Residual diagnostics (both models) per analyte
5. AIC comparison summary and recommendations

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2 Data overview

Table 2: Table 1: Data completeness by analyte (analysis cohort)

PARAMCD	Parameter	Panel	N obs	N subjects	% missing AVAL	N >ULOQ	N <LLOQ
FRACTKN	Fractalkine	14-PLEX	191	54	0.0	0	0
GZMB	Granzyme B	14-PLEX	191	54	0.0	0	0
IFNG	IFN-g	14-PLEX	191	54	0.0	0	0
IL1B	IL-1b	14-PLEX	191	54	0.0	0	0
IL6	IL-6	14-PLEX	191	54	0.0	0	0
IL8	IL-8	14-PLEX	191	54	0.0	0	0
MCP1	MCP-1	14-PLEX	191	54	0.0	0	0
MMP10	MMP-10	14-PLEX	191	54	0.0	0	0
PDGFDD	PDGF-DD	14-PLEX	191	54	0.0	0	0
PECAM1	PECAM-1	14-PLEX	191	54	0.5	0	0
TGFA	TGF-alpha	14-PLEX	191	54	0.0	0	0
THBS2	Thrombospondin-2	14-PLEX	191	54	0.0	0	0
VEGFA	VEGF-A	14-PLEX	191	54	0.0	0	0
VEGFC	VEGF-C	14-PLEX	191	54	0.0	0	0
ANGPT1	Angiopoietin-1	19-PLEX	191	54	0.0	0	0
ANGPT2	Angiopoietin-2	19-PLEX	191	54	0.0	0	0
ANGPTL4	ANGPTL4	19-PLEX	191	54	0.0	0	0
CA9	CA9	19-PLEX	191	54	0.0	0	0
CD163	CD163	19-PLEX	191	54	0.0	0	0
COLIVA1	Collagen IV alpha 1	19-PLEX	191	54	0.5	0	0
EGF	EGF	19-PLEX	191	54	0.0	0	0
ERBB2	ErbB2	19-PLEX	191	54	0.0	0	0
FGFB	FGF Basic	19-PLEX	191	54	0.0	0	0
IL12P70	IL-12 p70	19-PLEX	191	54	0.0	0	0
IL2	IL-2	19-PLEX	191	54	0.0	0	0

PARAMCD	Parameter	Panel	N obs	N subjects	% missing AVAL	N >ULOQ	N <LLOQ
IP10	IP-10 (CXCL10)	19-PLEX	191	54	0.5	0	0
ITAC	I-TAC (CXCL11)	19-PLEX	191	54	0.0	0	0
MIG	MIG (CXCL9)	19-PLEX	191	54	0.0	0	0
PIGF	PIGF	19-PLEX	191	54	0.0	0	0
TENASC	Tenascin C	19-PLEX	191	54	0.0	0	0
TNFRSF18	TNFRSF18 (GITR)	19-PLEX	191	54	0.0	0	0
TNFSF6	TNFSF6 (FasL)	19-PLEX	191	54	0.0	0	0
VEGFR2	VEGFR2	19-PLEX	191	54	0.0	0	0

Note: >ULOQ values are set to NA in `aval`. If >ULOQ exceeds ~20% of observations for an analyte, consider a Tobit (censored) mixed model for that analyte.

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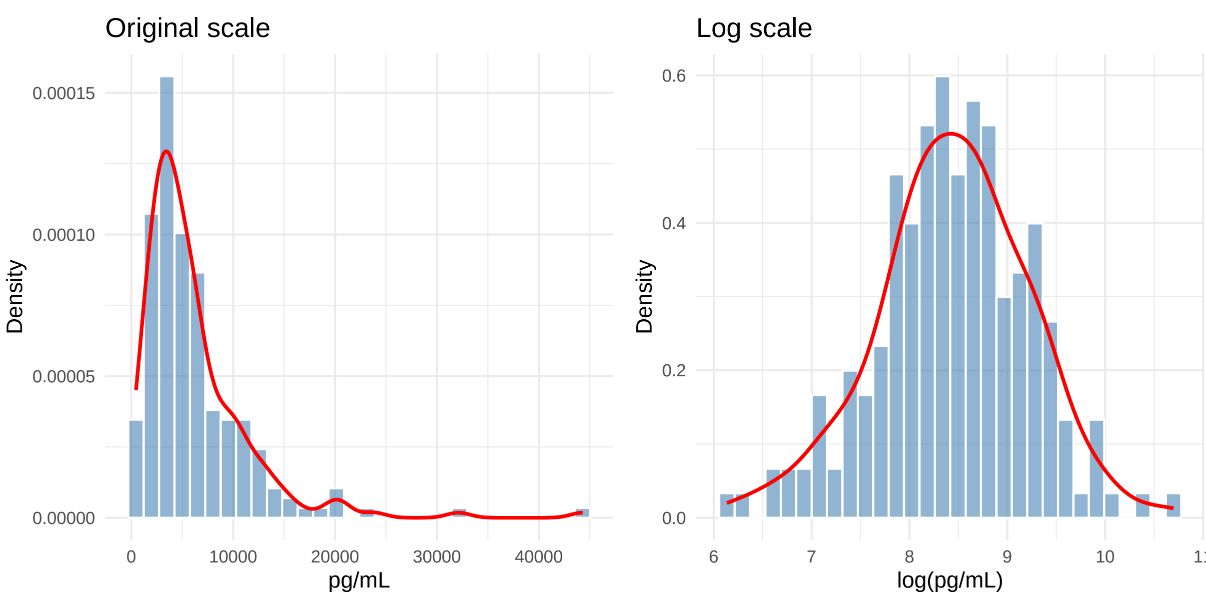
3 Concentration distributions

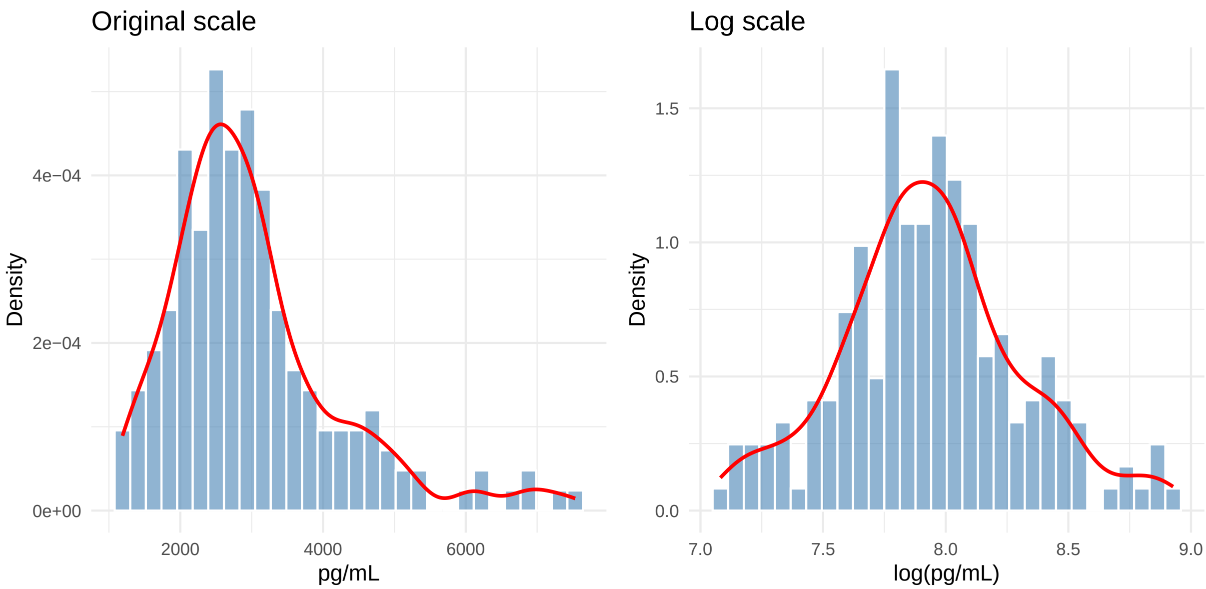
3.1 Angiopoietin-1 (`ANGPT1`)

Skewness: original scale = 3.02 | log scale = -0.19



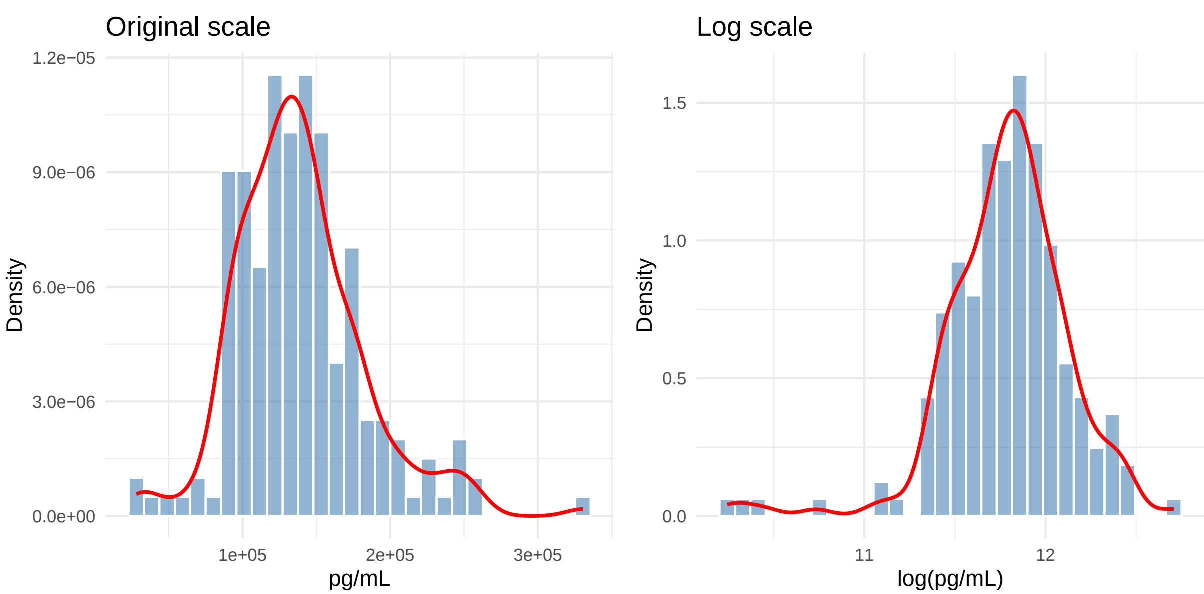
3.2 Angiopoietin-2 (`ANGPT2`)

Skewness: original scale = 1.39 | log scale = 0.21



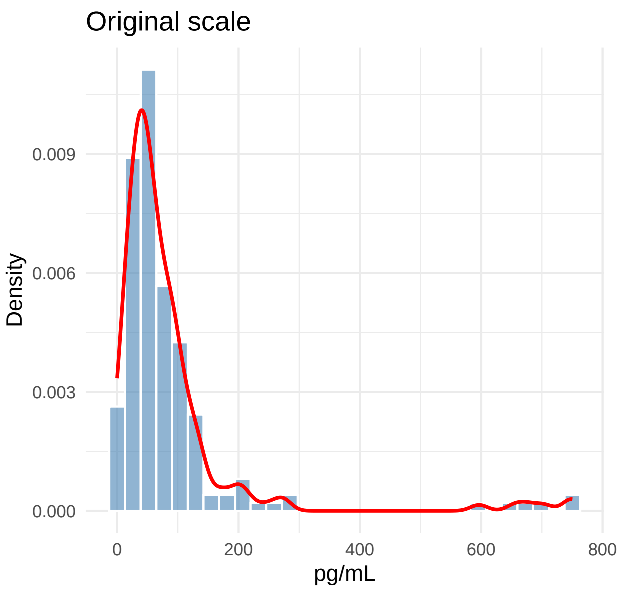
3.3 ANGPTL4 (`ANGPTL4`)

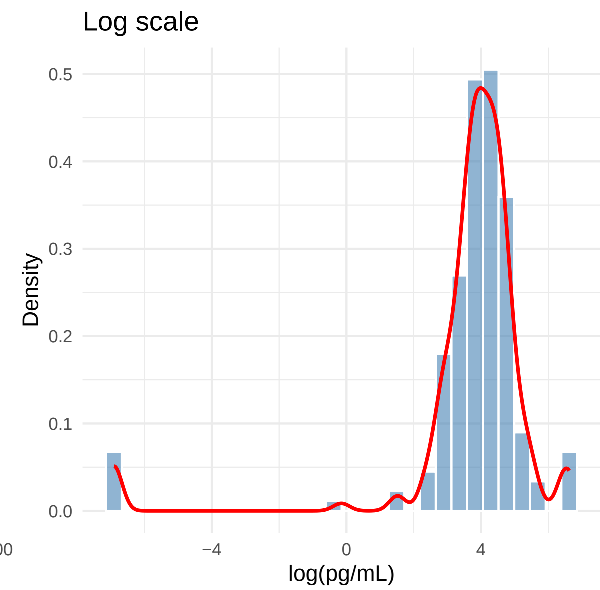
Skewness: original scale = 0.81 | log scale = -1.1



3.4 CA9 (`CA9`)

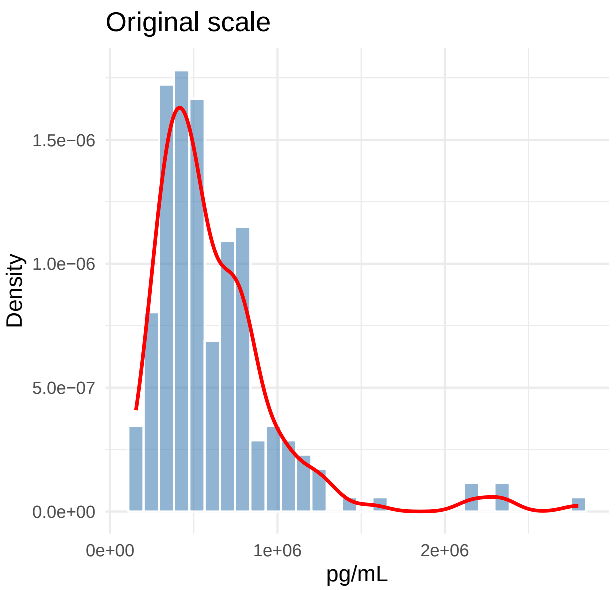
Skewness: original scale = 4.08 | log scale = -3.89

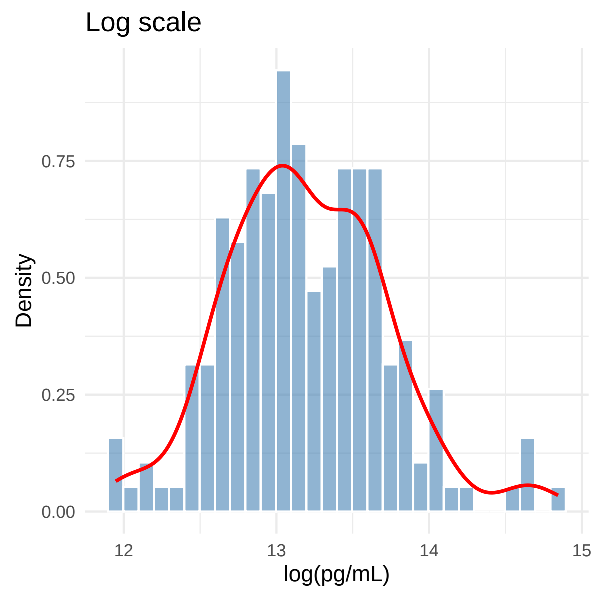




3.5 CD163 (`CD163`)

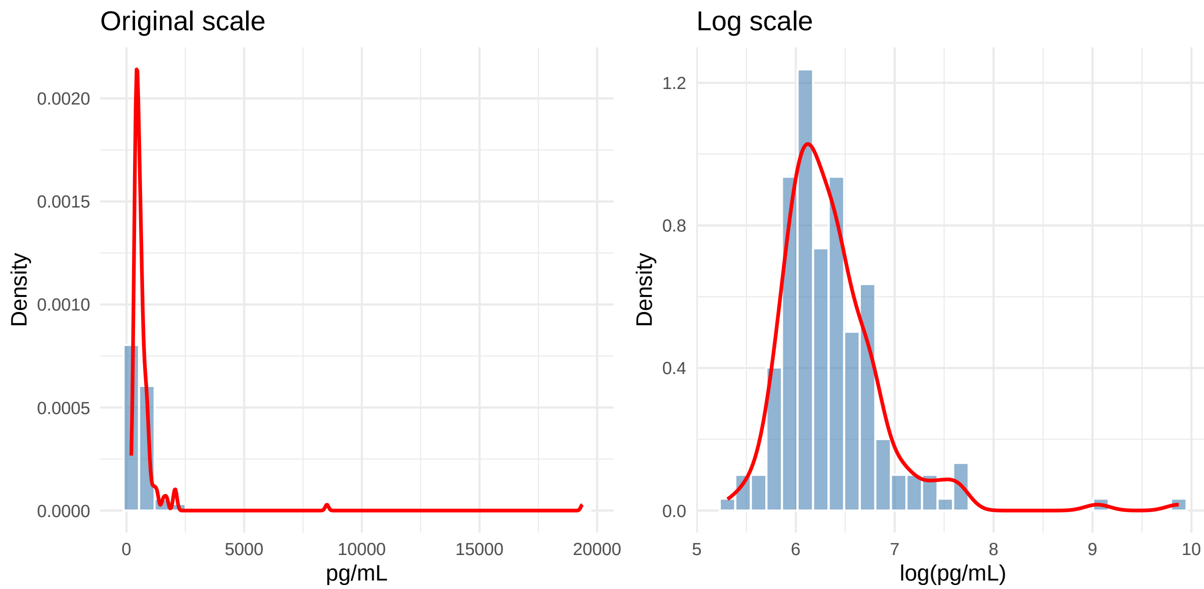
Skewness: original scale = 2.55 | log scale = 0.32





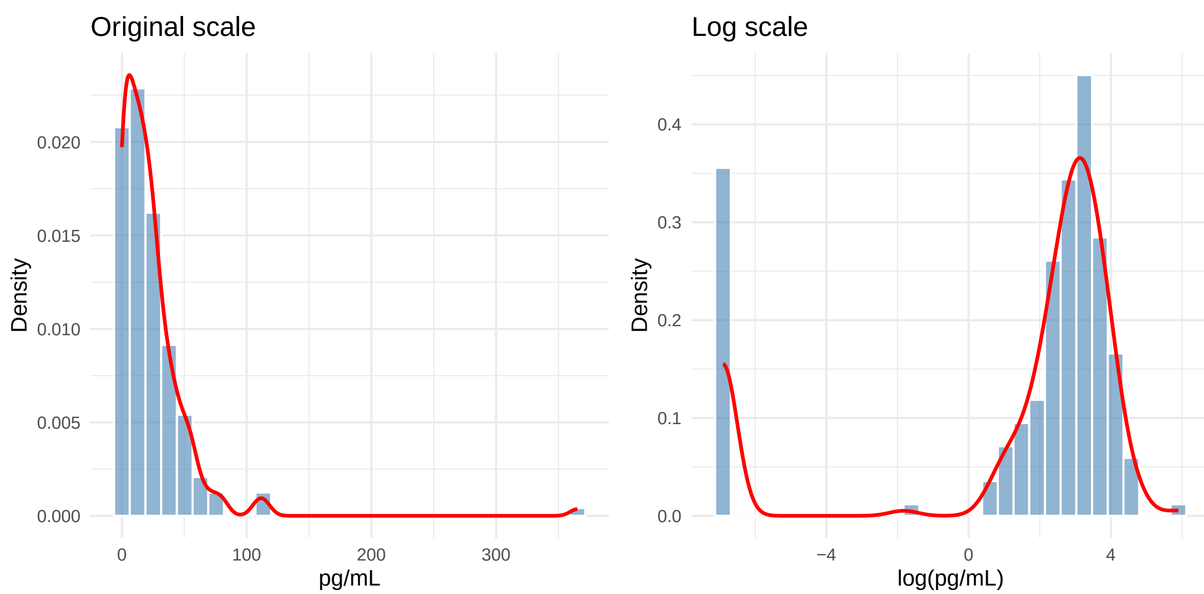
3.6 Collagen IV alpha 1 (`COLIVA1`)

Skewness: original scale = 10.56 | log scale = 2.44



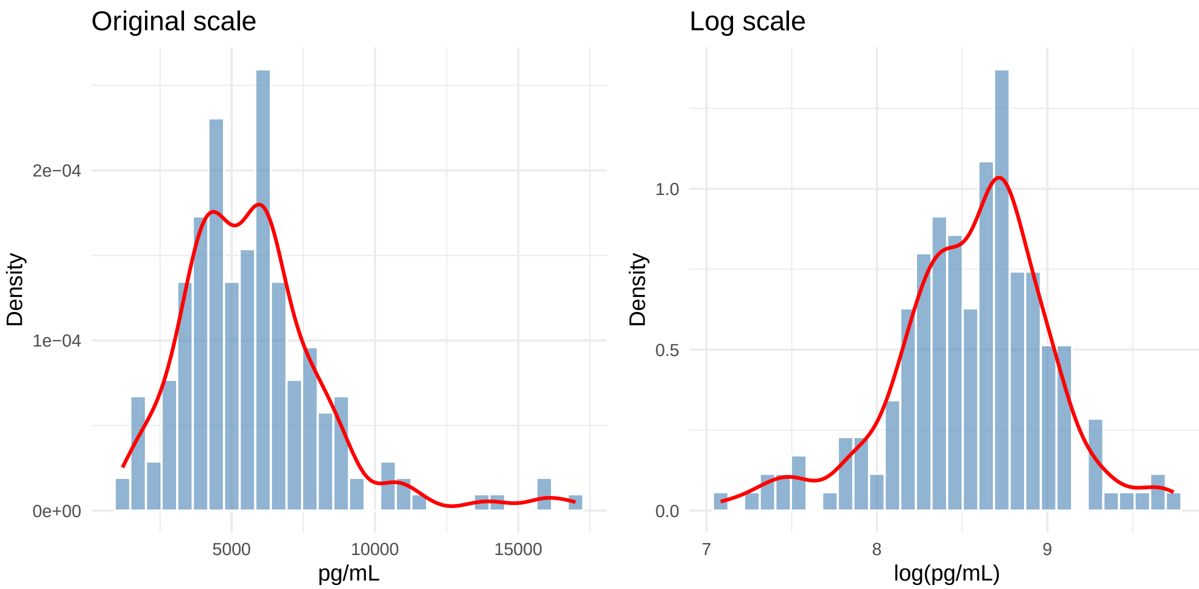
3.7 EGF (`EGF`)

Skewness: original scale = 6.35 | log scale = -1.65



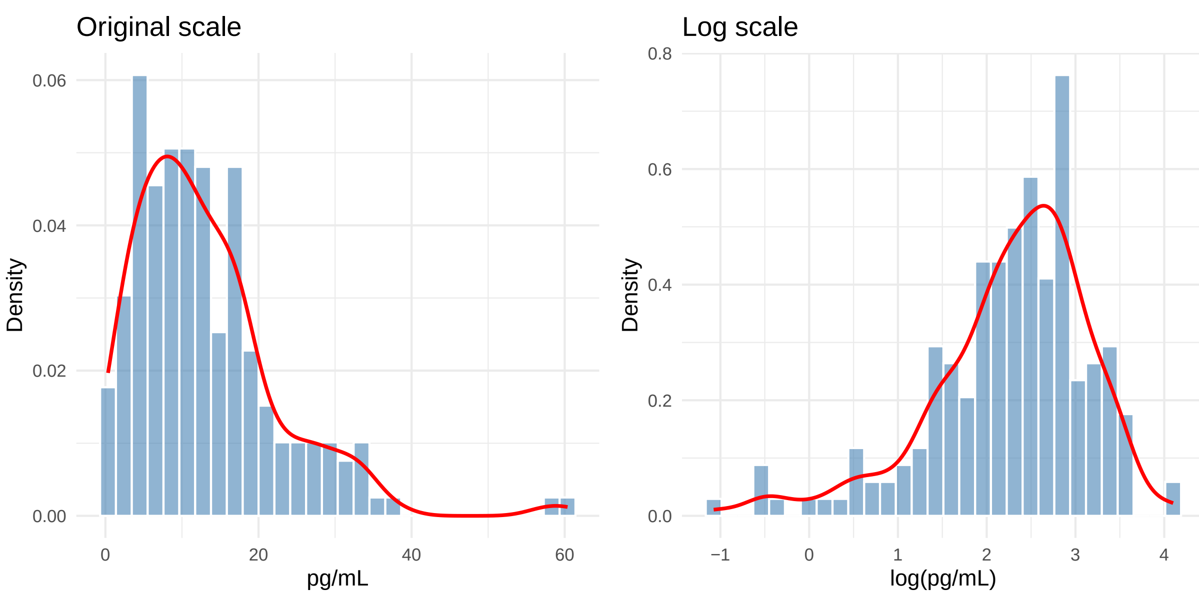
3.8 ErbB2 (`ERBB2`)

Skewness: original scale = 1.43 | log scale = -0.44



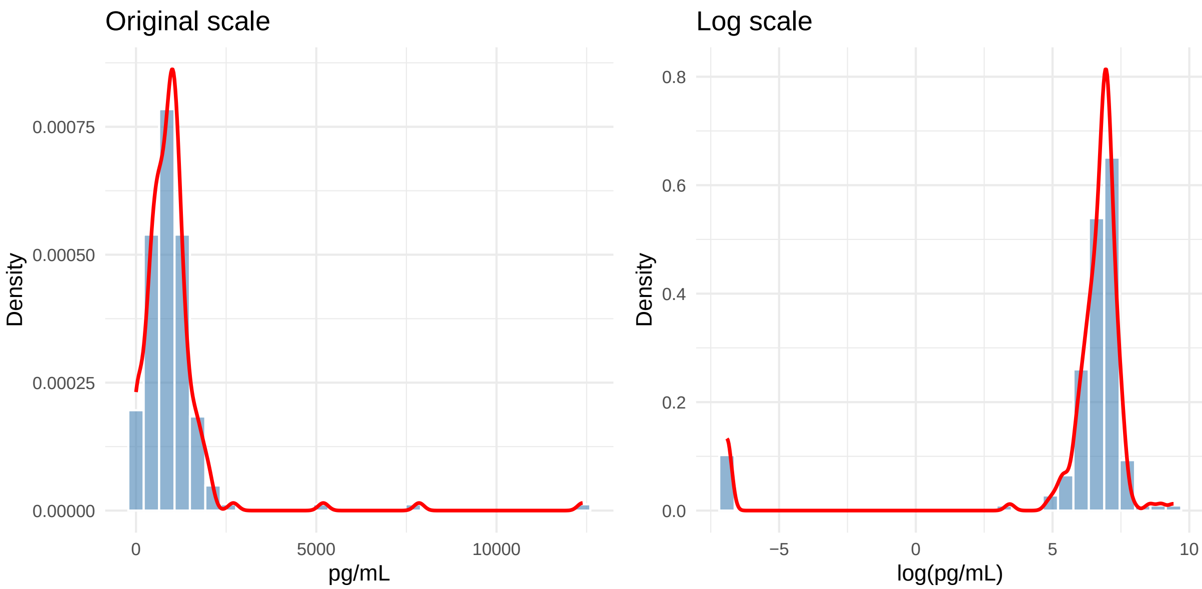
3.9 FGF Basic (FGFB)

Skewness: original scale = 1.59 | log scale = -1.04



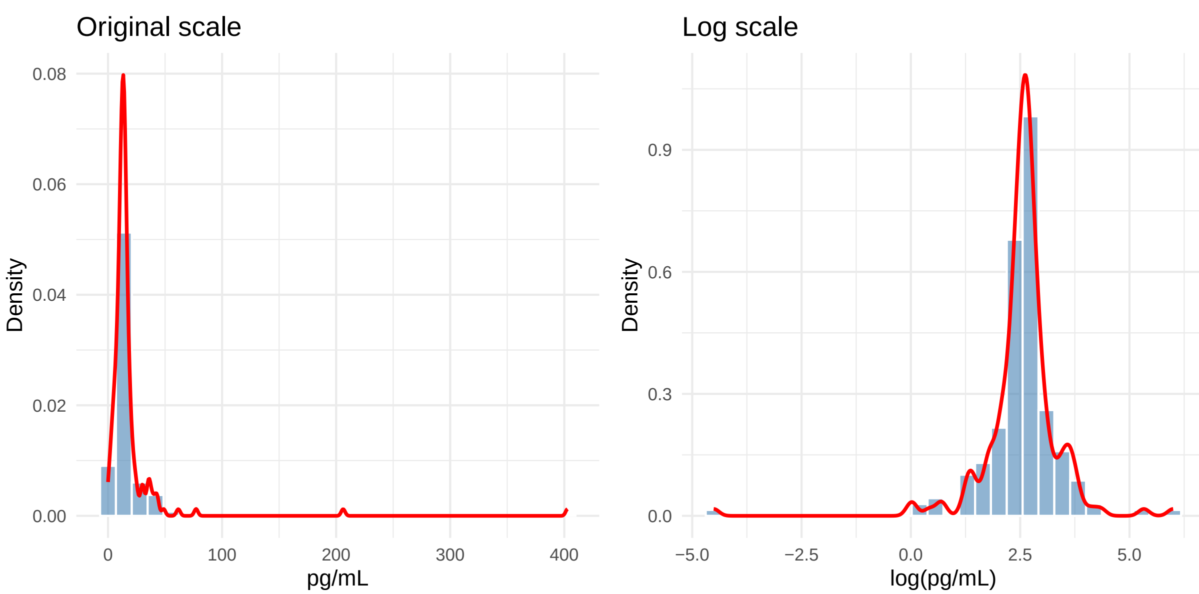
3.10 Fractalkine (FRACTKN)

Skewness: original scale = 6.82 | log scale = -3.53



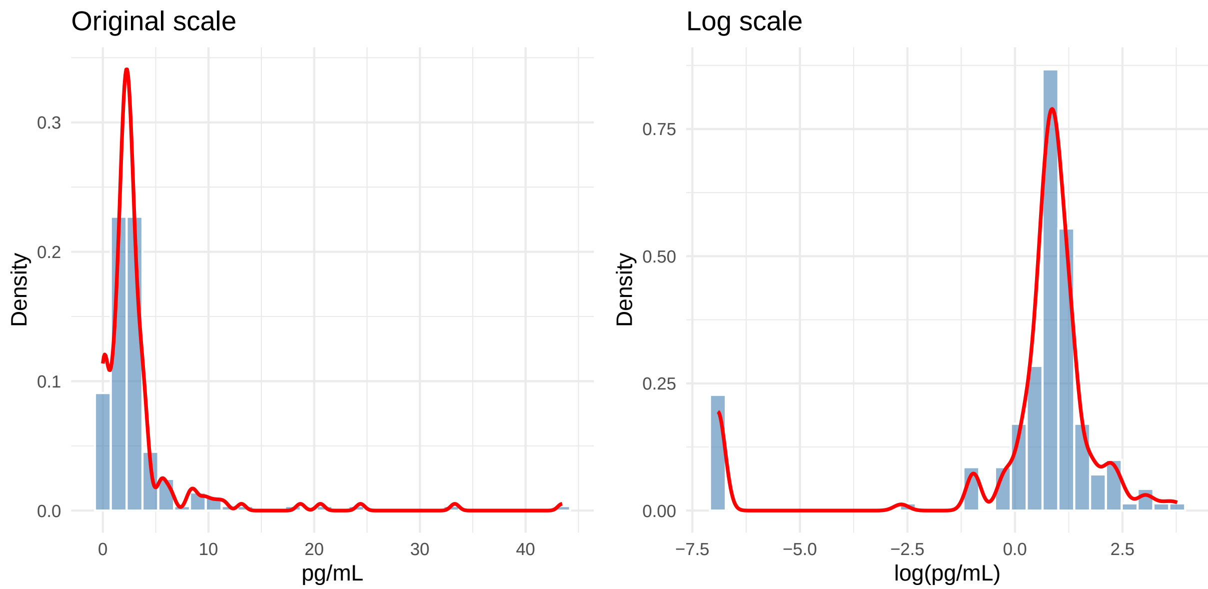
3.11 Granzyme B (`GZMB`)

Skewness: original scale = 9.45 | log scale = -2.56



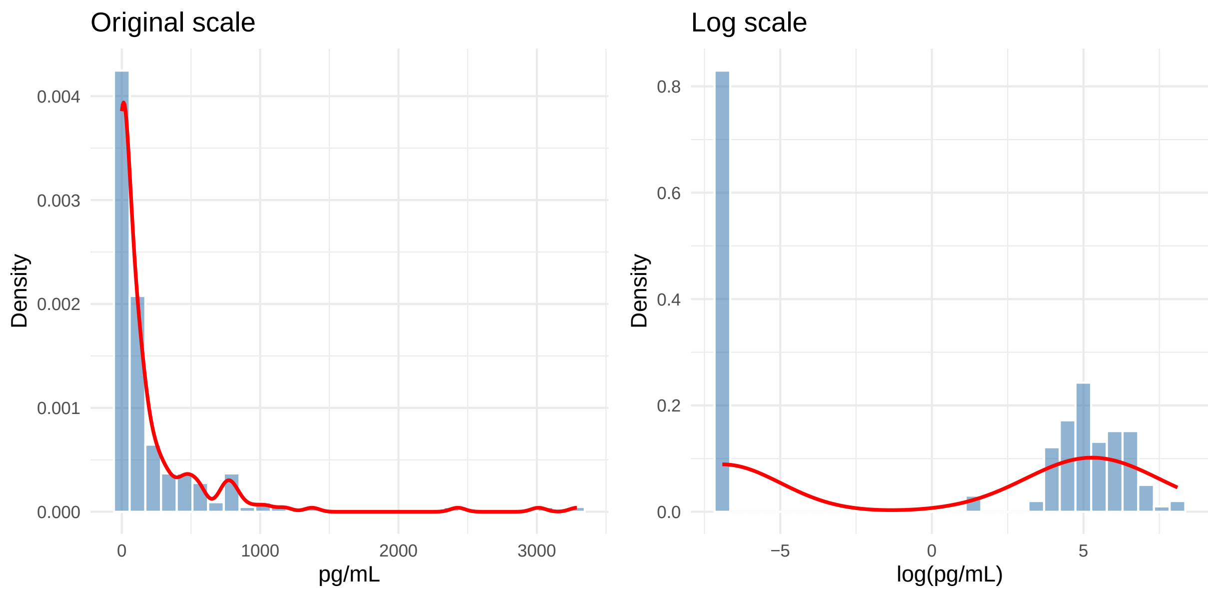
3.12 IFN-g (`IFNG`)

Skewness: original scale = 5.16 | log scale = -2.41



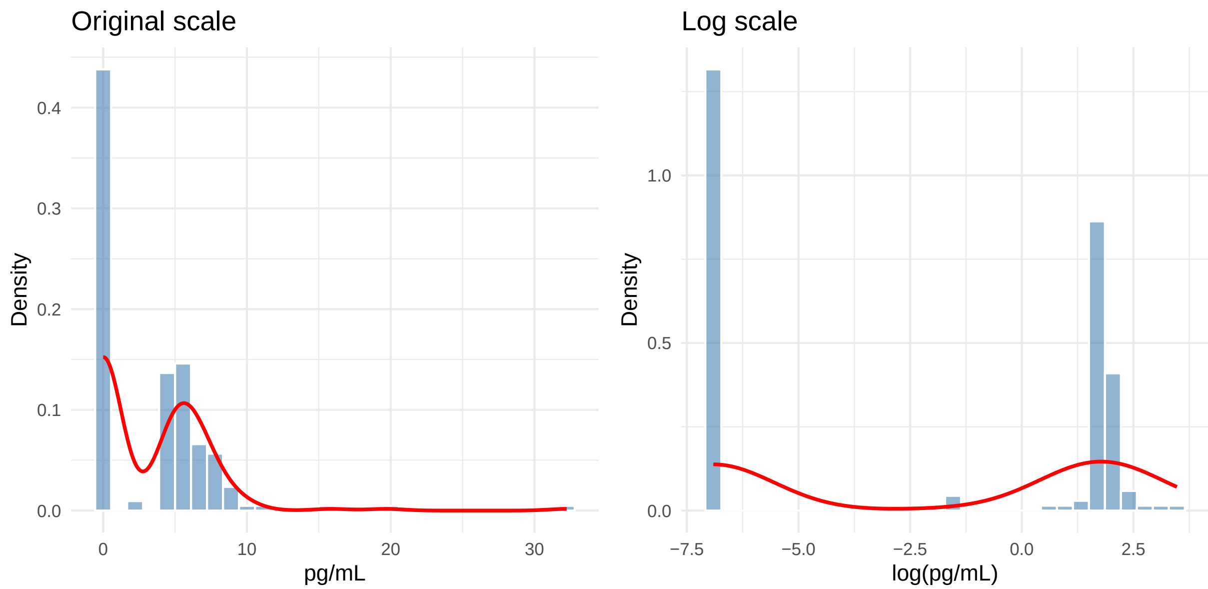
3.13 IL-12 p70 (`IL12P70`)

Skewness: original scale = 4.4 | log scale = -0.22



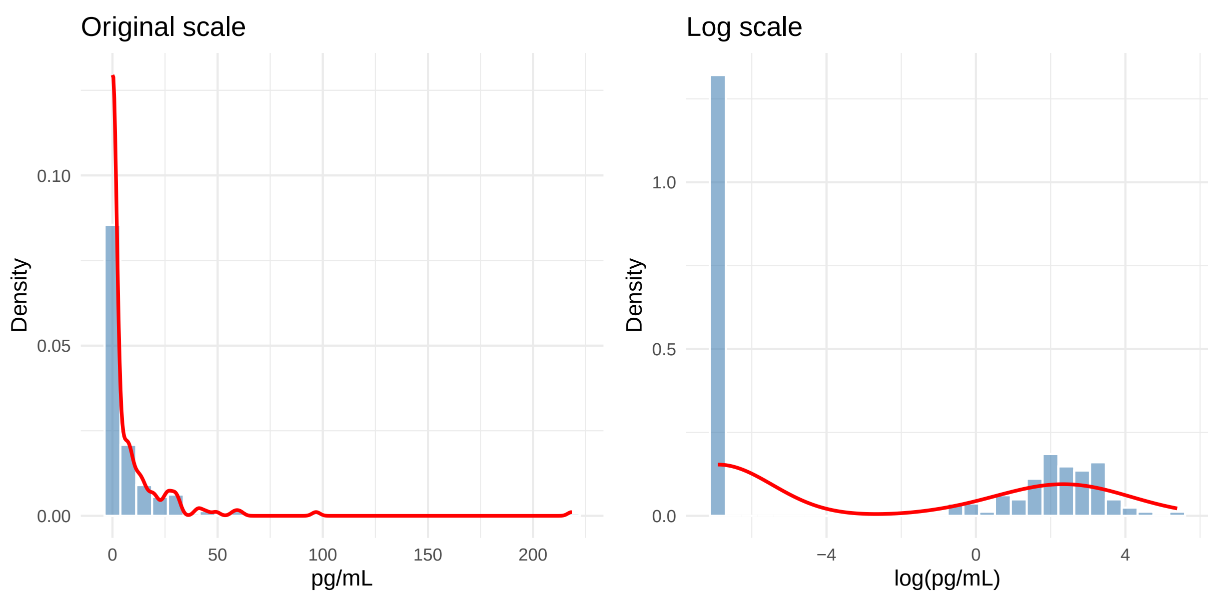
3.14 IL-1b (`IL1B`)

Skewness: original scale = 2.38 | log scale = -0.08



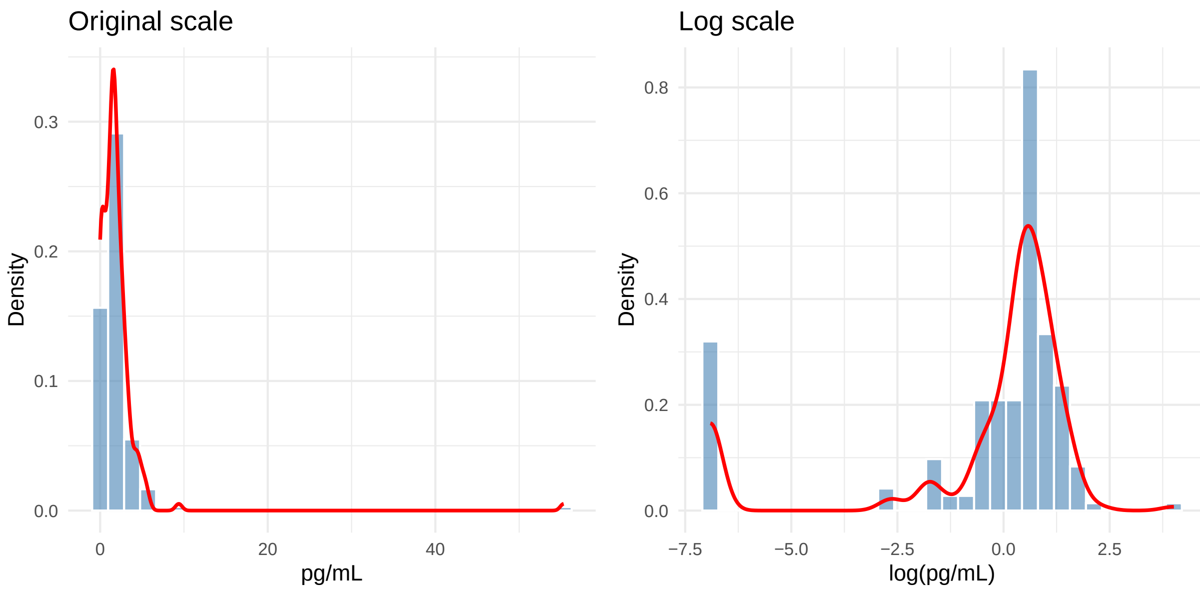
3.15 IL-2 (`IL2`)

Skewness: original scale = 6.86 | log scale = 0.32



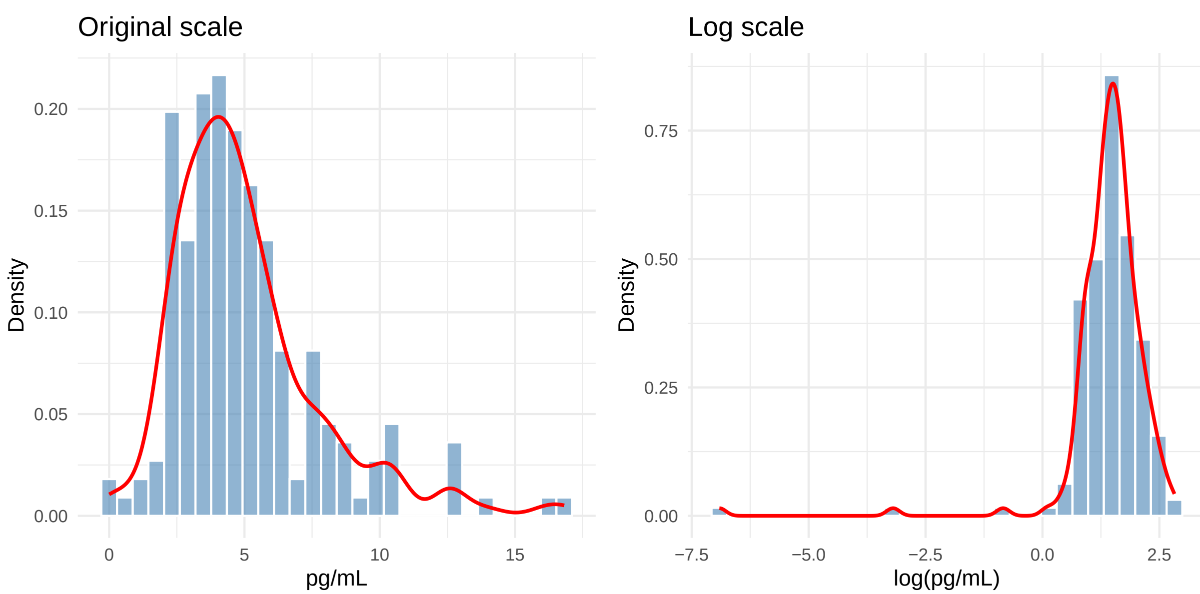
3.16 IL-6 (`IL6`)

Skewness: original scale = 11.36 | log scale = -1.86



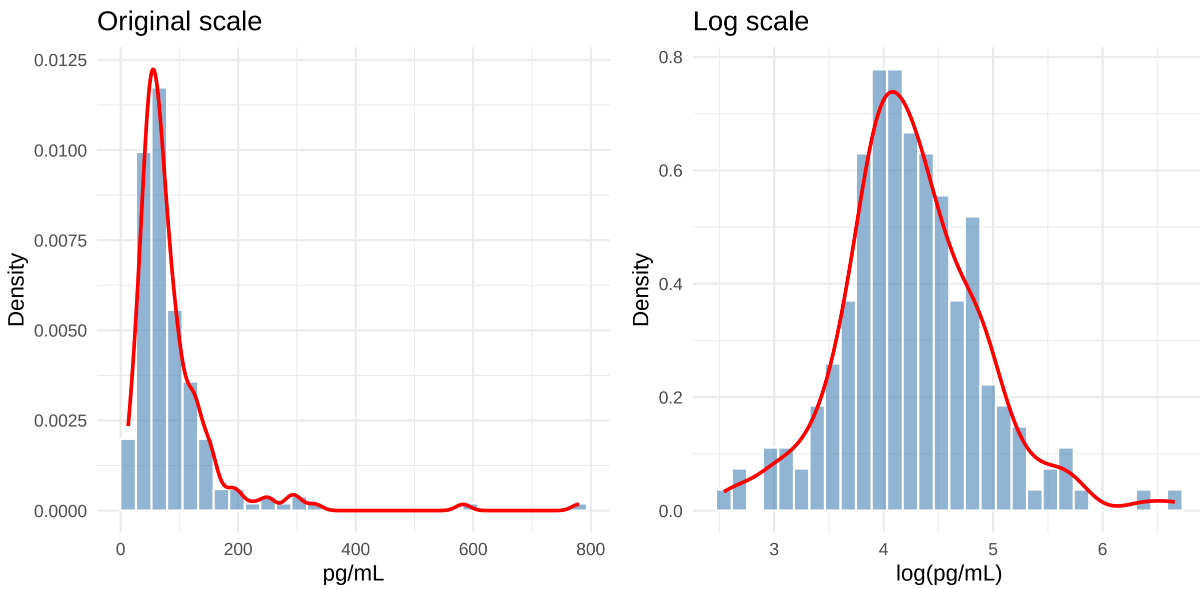
3.17 IL-8 (IL8)

Skewness: original scale = 1.47 | log scale = -5.5



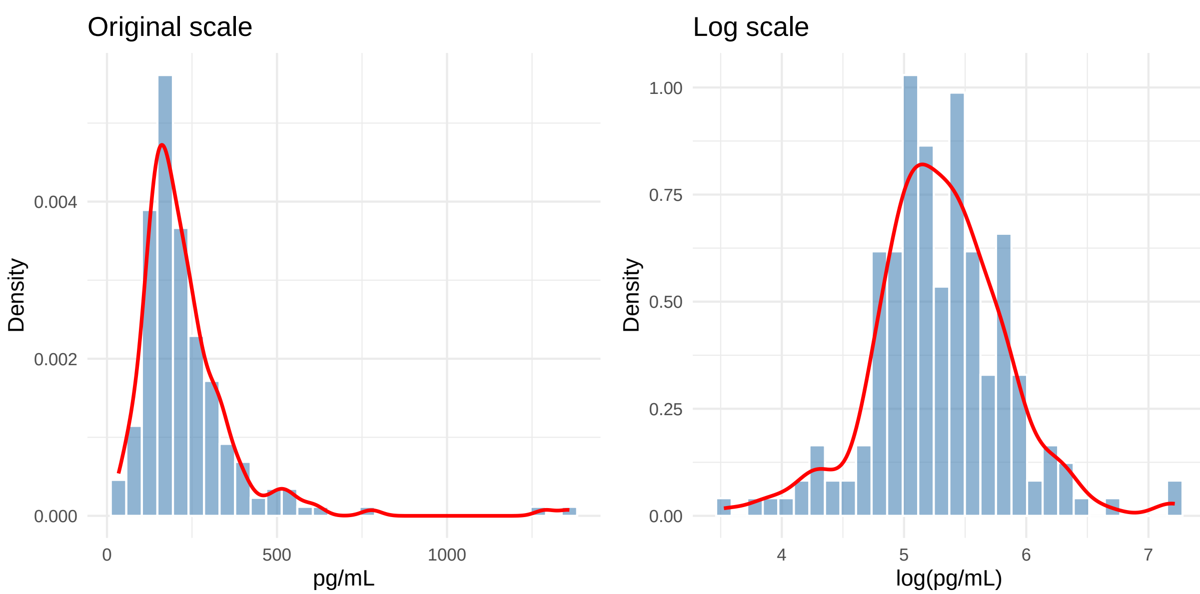
3.18 IP-10 (CXCL10) (IP10)

Skewness: original scale = 4.71 | log scale = 0.4



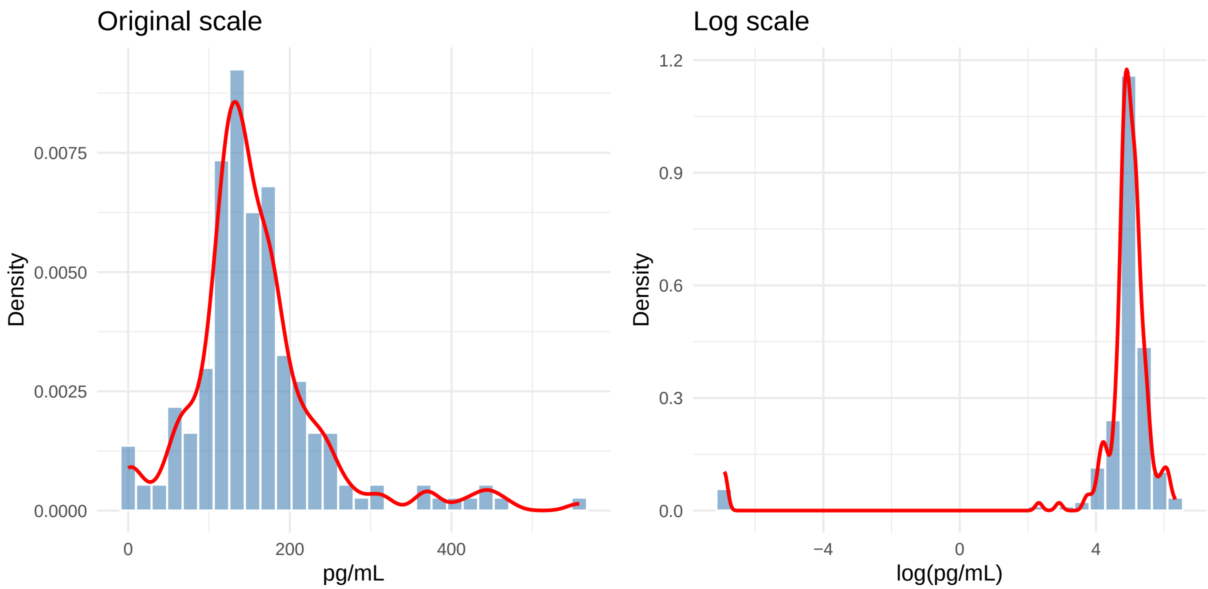
3.19 I-TAC (CXCL11) (`ITAC`)

Skewness: original scale = 3.74 | log scale = 0.11



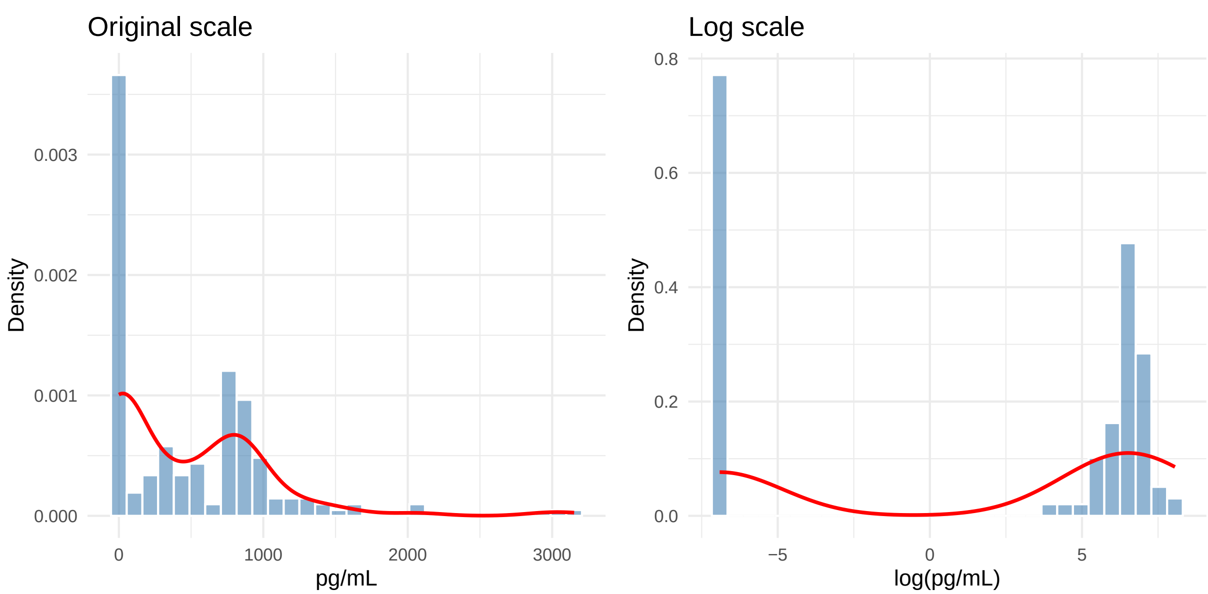
3.20 MCP-1 (`MCP1`)

Skewness: original scale = 1.65 | log scale = -5.34



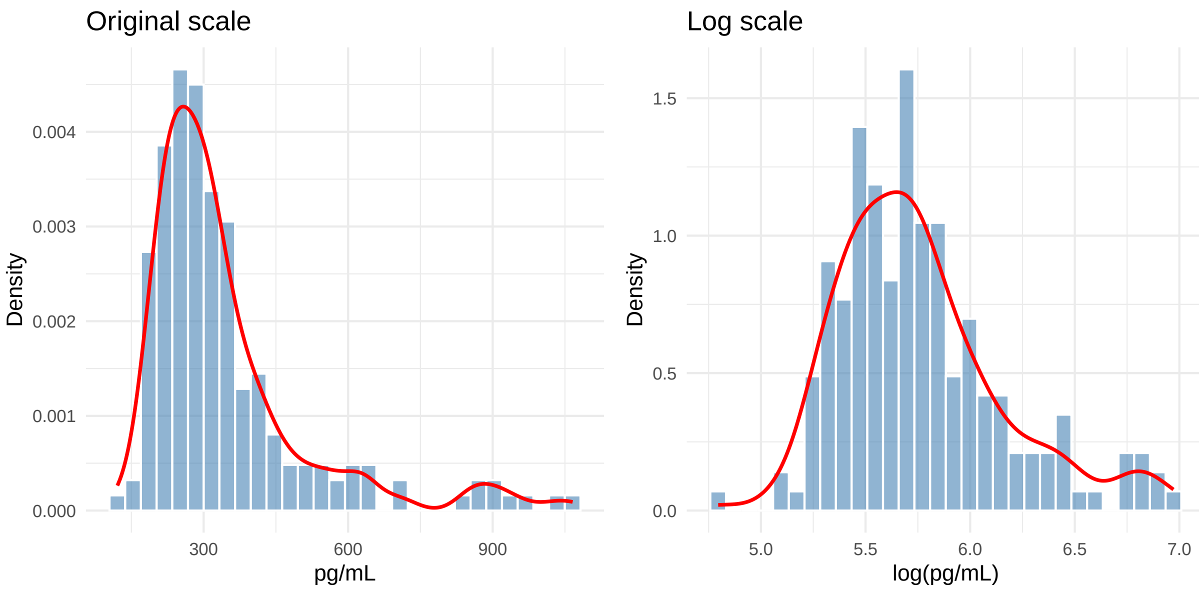
3.21 MIG (CXCL9) (`MIG`)

Skewness: original scale = 1.74 | log scale = -0.39



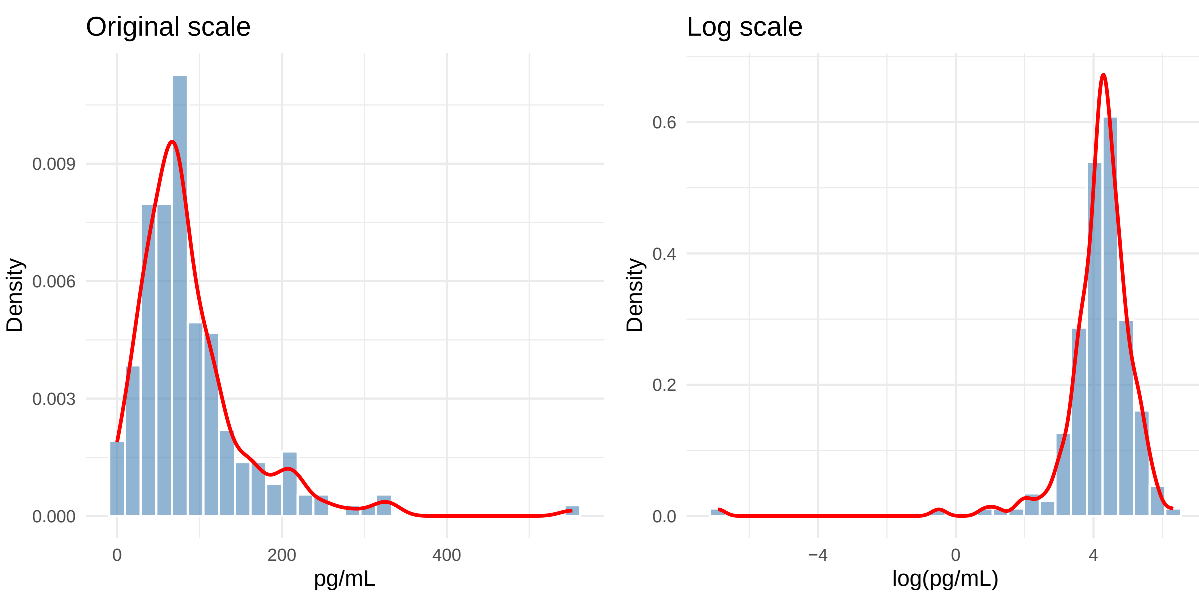
3.22 MMP-10 (`MMP10`)

Skewness: original scale = 2.08 | log scale = 0.93



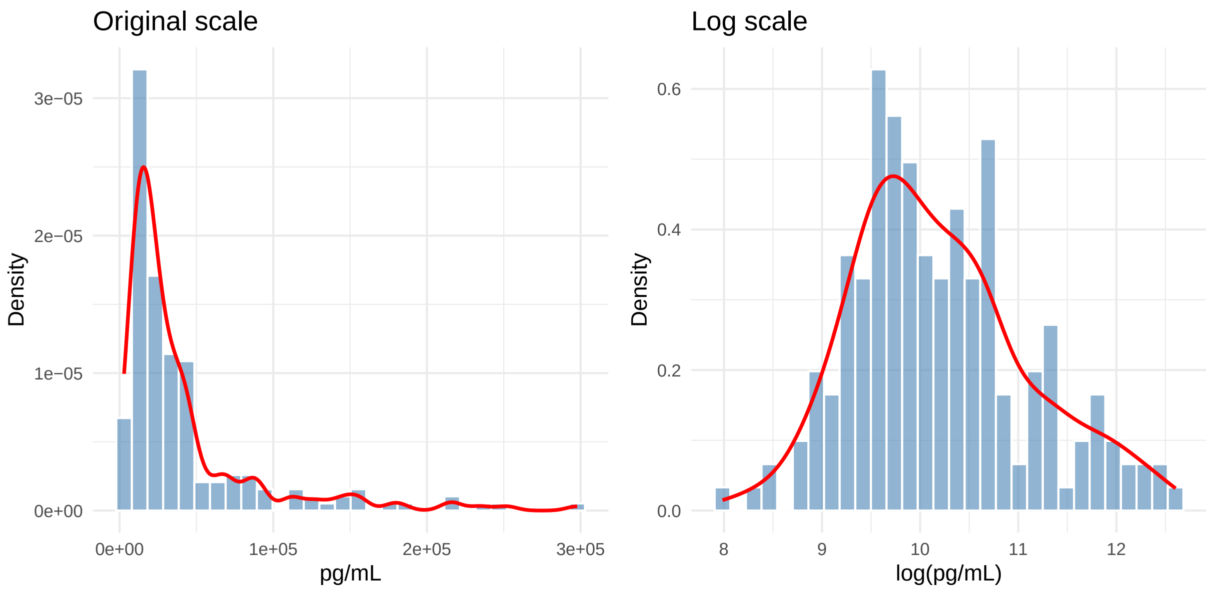
3.23 PDGF-DD (`PDGFDD`)

Skewness: original scale = 2.48 | log scale = -4.88



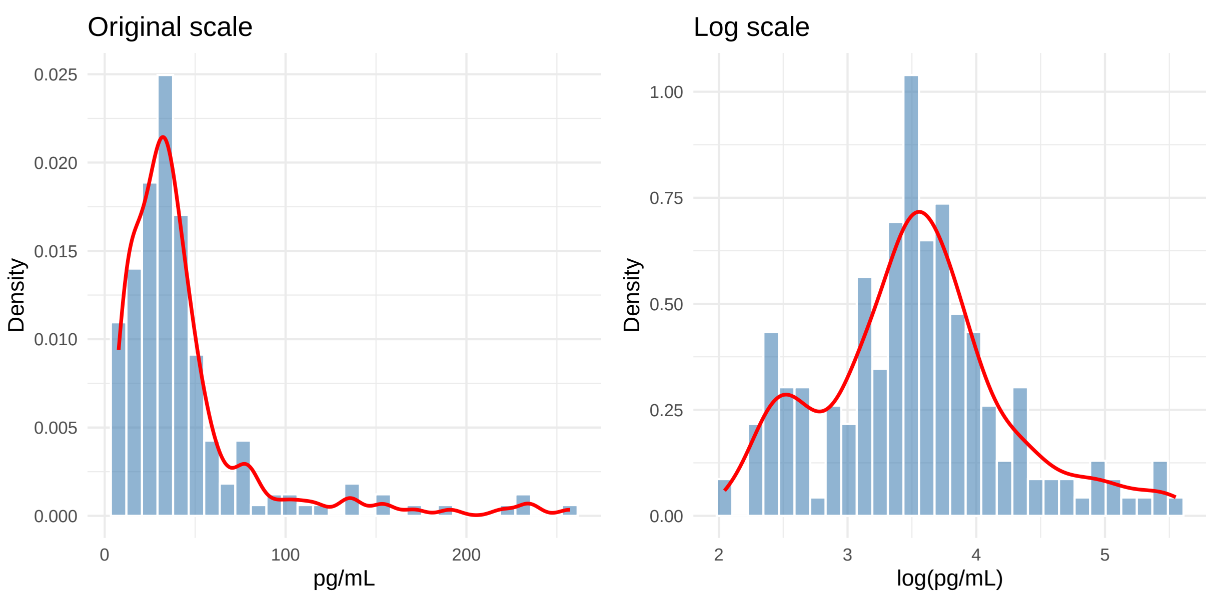
3.24 PECAM-1 (`PECAM1`)

Skewness: original scale = 2.64 | log scale = 0.48



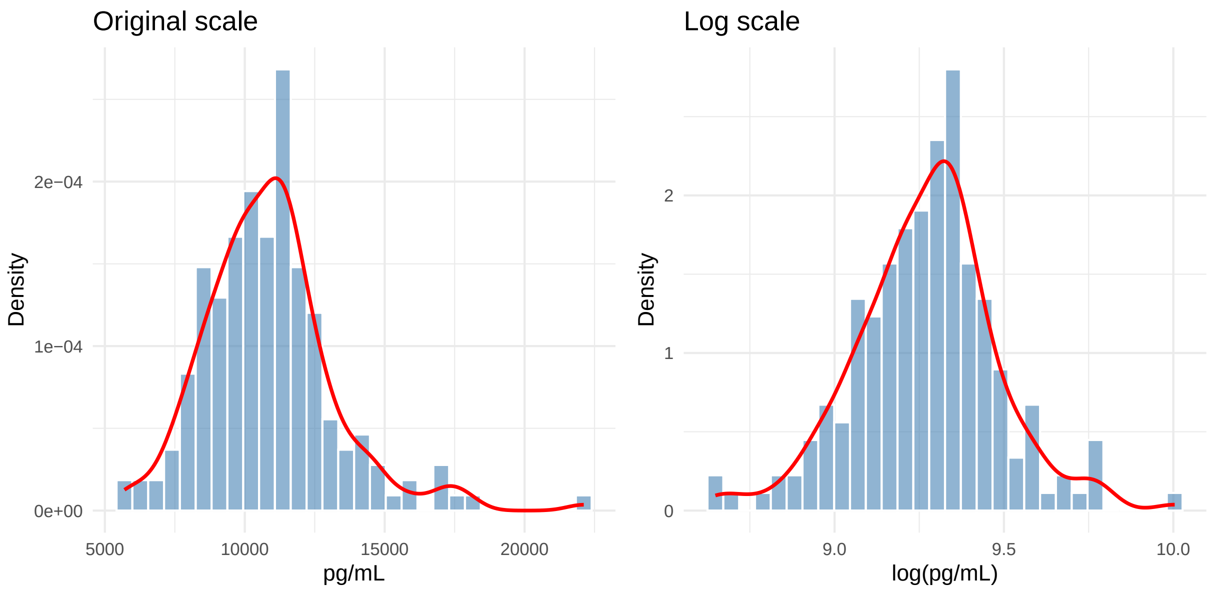
3.25 PIGF (PIGF)

Skewness: original scale = 2.83 | log scale = 0.4



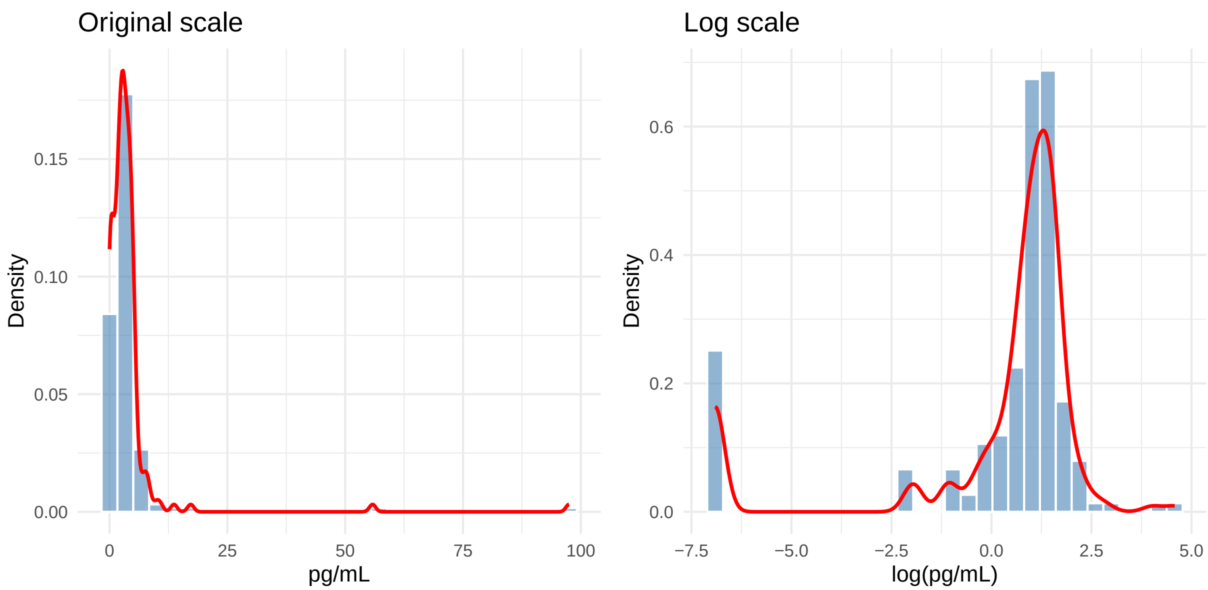
3.26 Tenascin C (TENASC)

Skewness: original scale = 1.01 | log scale = 0.01



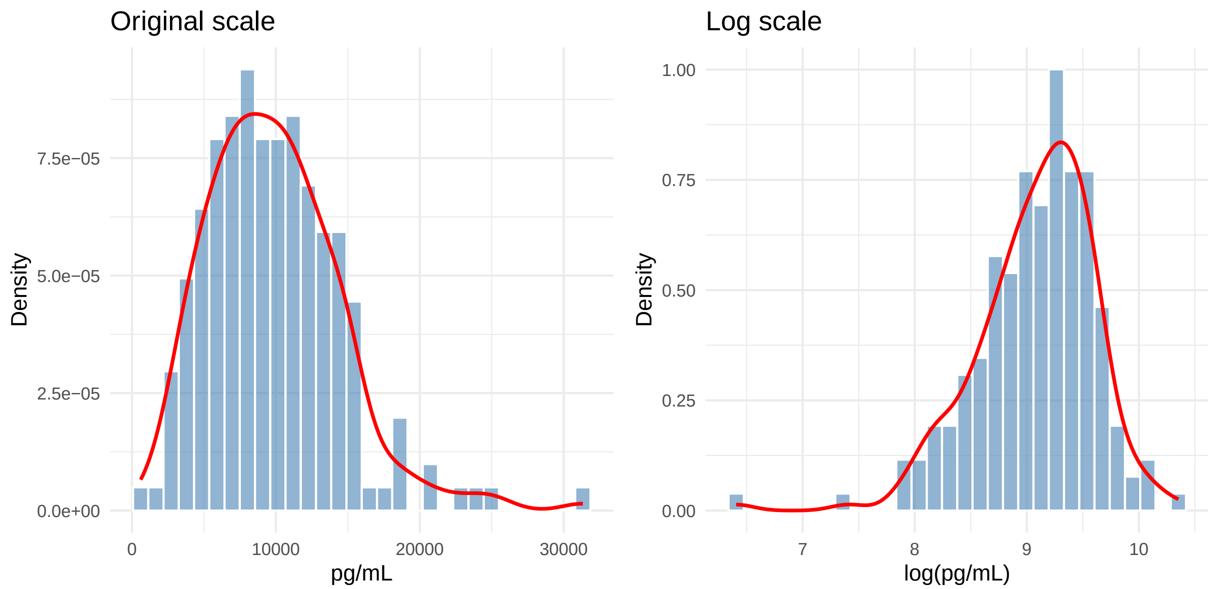
3.27 TGF-alpha (`TGFA`)

Skewness: original scale = 9.27 | log scale = -2.11



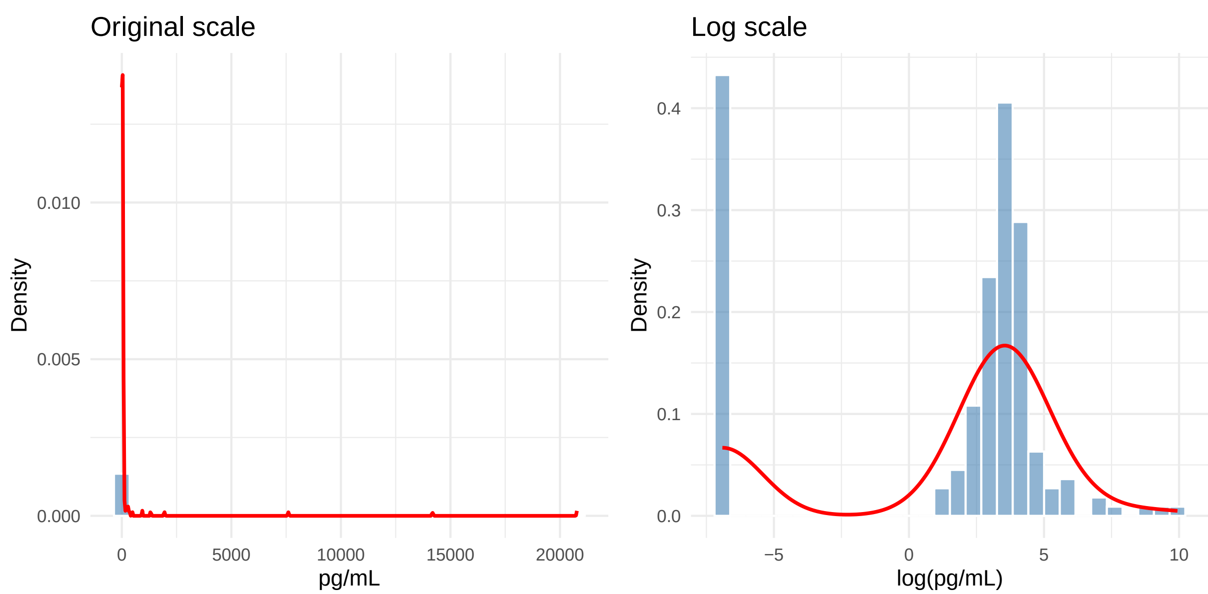
3.28 Thrombospondin-2 (`THBS2`)

Skewness: original scale = 1.02 | log scale = -0.99



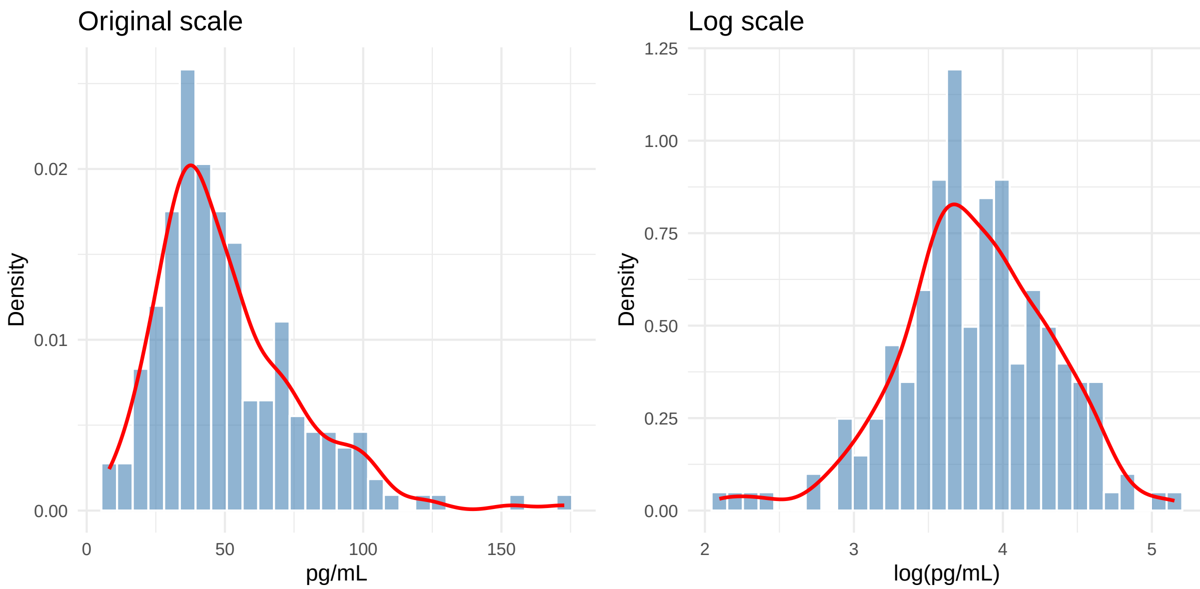
3.29 TNFRSF18 (GITR) (`TNFRSF18`)

Skewness: original scale = 8.99 | log scale = -0.92



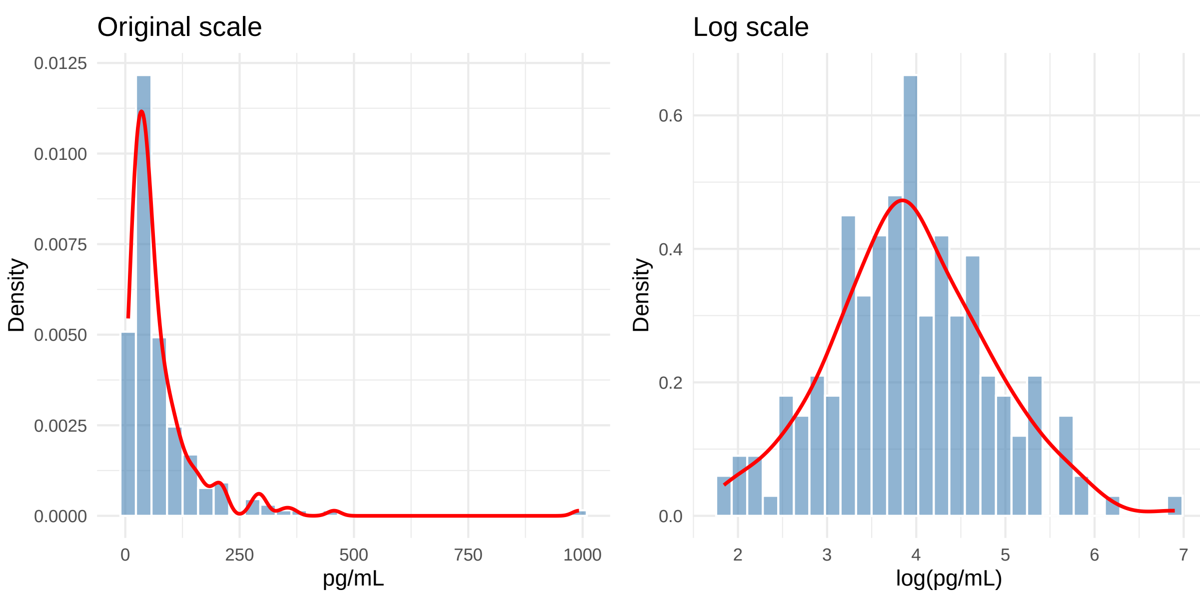
3.30 TNFSF6 (FasL) (`TNFSF6`)

Skewness: original scale = 1.35 | log scale = -0.36



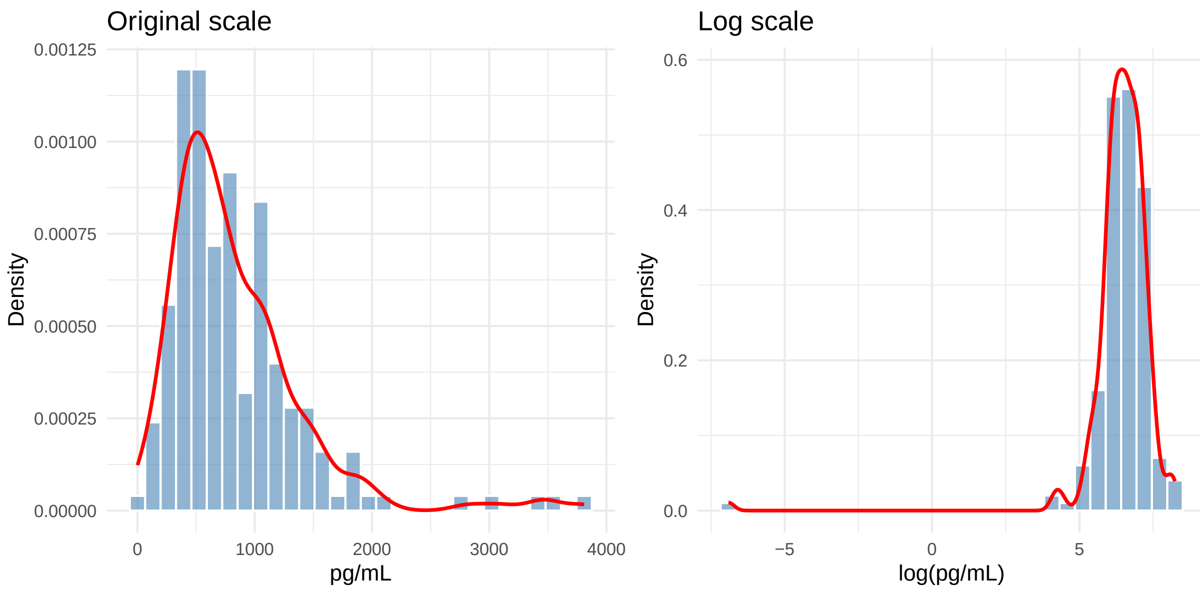
3.31 VEGF-A (VEGFA)

Skewness: original scale = 5.07 | log scale = 0.16



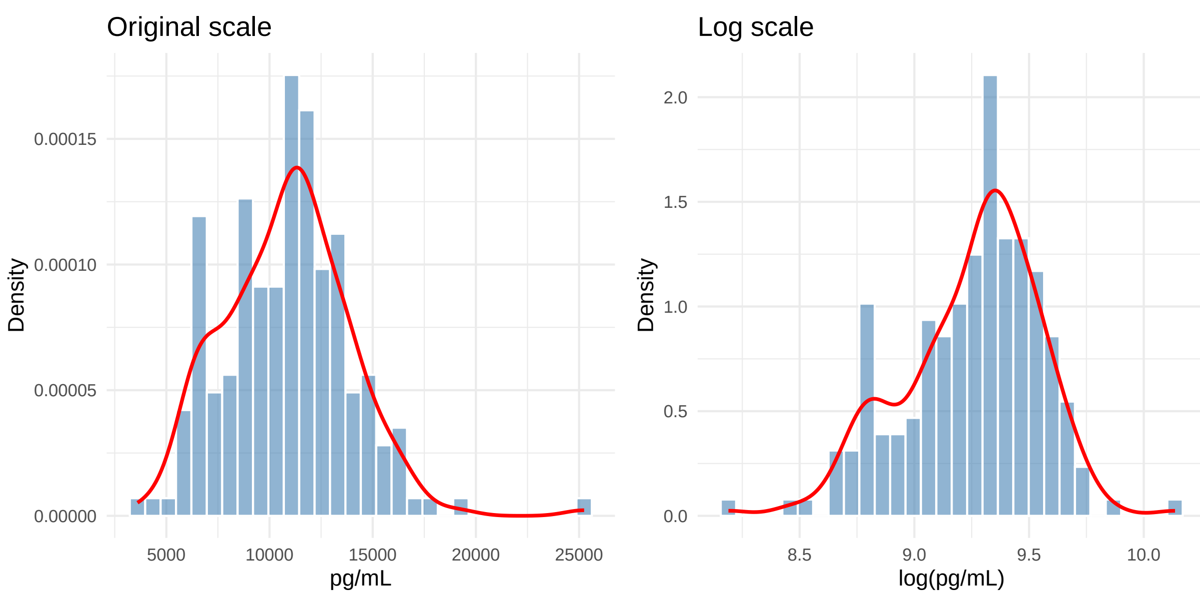
3.32 VEGF-C (VEGFC)

Skewness: original scale = 2.23 | log scale = -7.72



3.33 VEGFR2 (`VEGFR2`)

Skewness: original scale = 0.52 | log scale = -0.52



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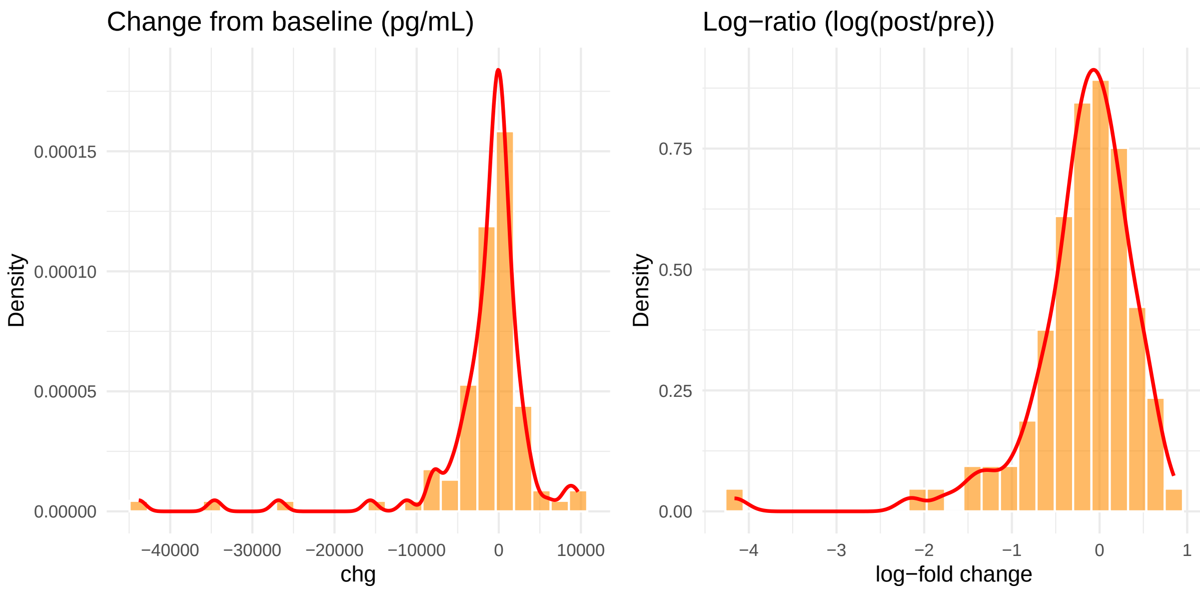
4 Change-score distributions

Change scores (`chg = aval - base`) are the response variable in the linear model. The log-ratio model uses `log(aval/base)` ~ `log(aval + offset) - log(base + offset)`.

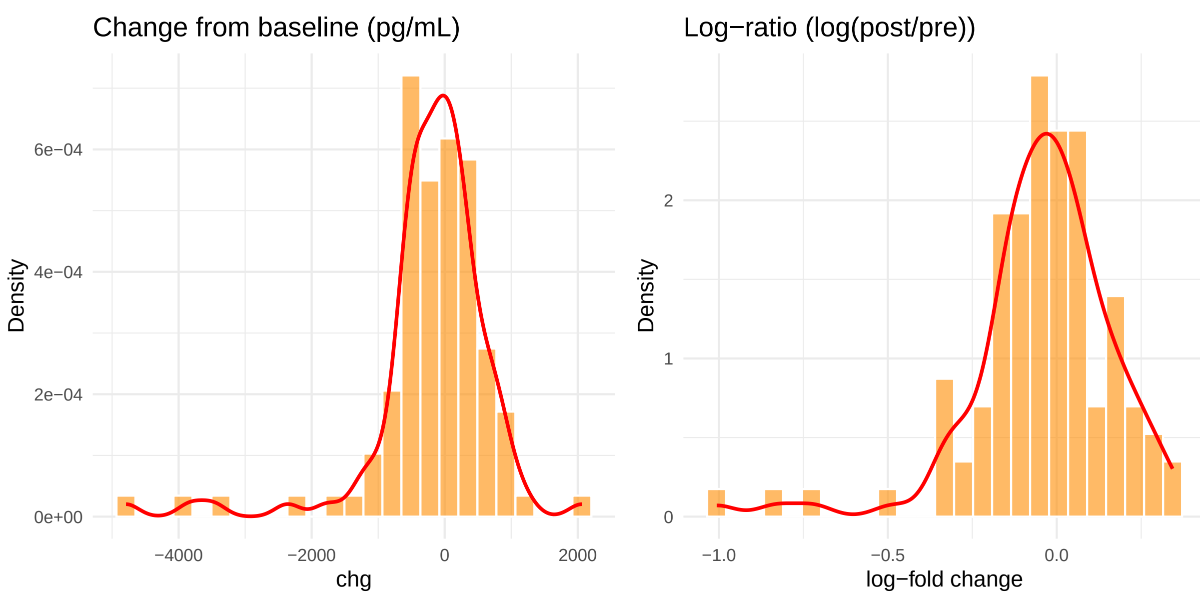
4.1 Angiopoietin-1

Linear chg: skewness = -3.69, excess kurtosis = 17.48 | Log-ratio skewness = -2.62



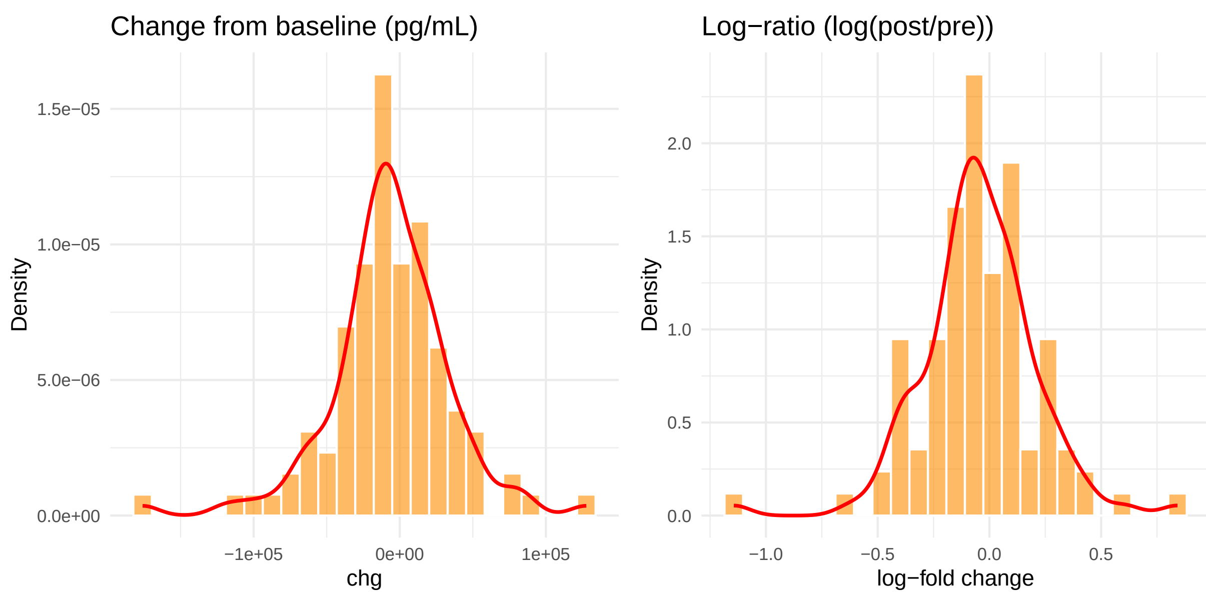
4.2 Angiopoietin-2

Linear chg: skewness = -2.24, excess kurtosis = 8.6 | Log-ratio skewness = -1.53



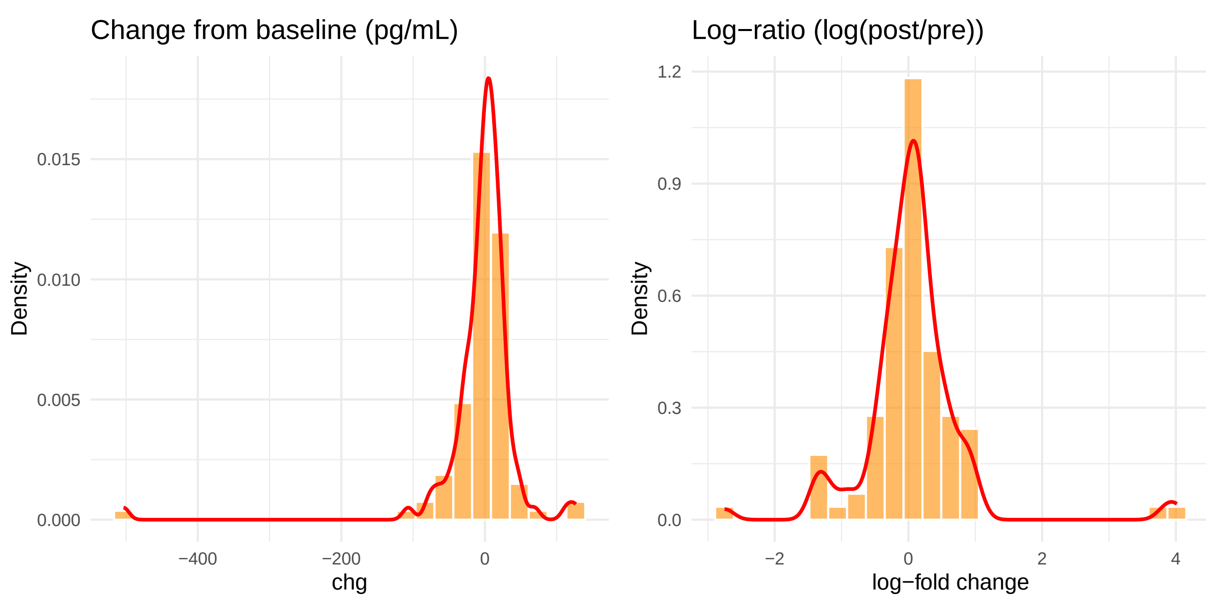
4.3 ANGPTL4

Linear chg: skewness = -0.46, excess kurtosis = 2.89 | Log-ratio skewness = -0.27



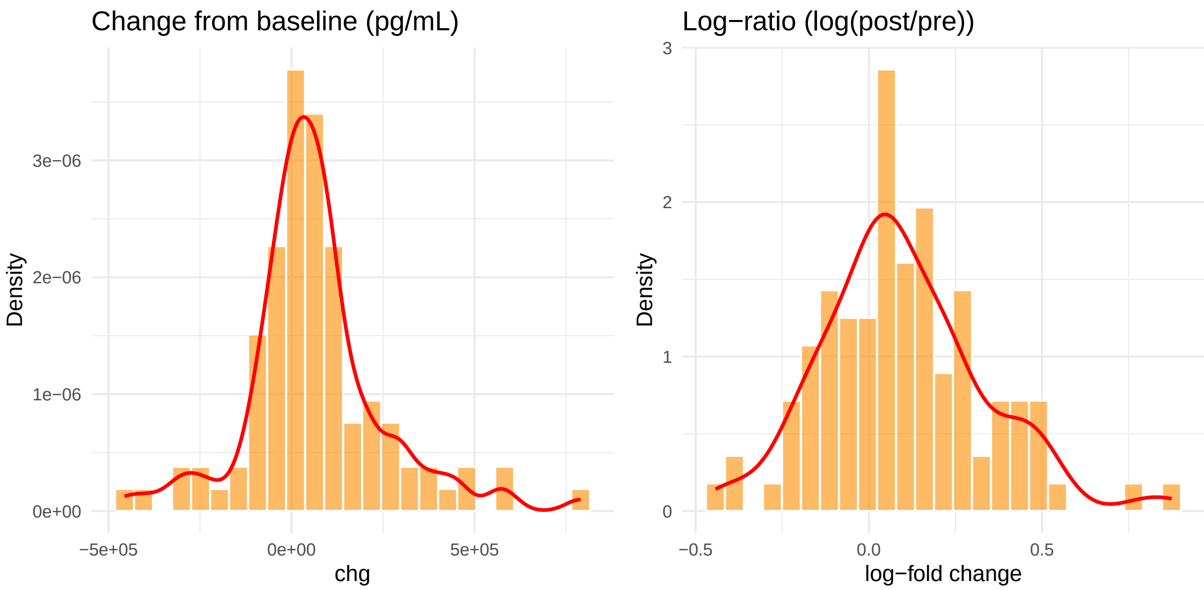
4.4 CA9

Linear chg: skewness = -5.72, excess kurtosis = 46.71 | Log-ratio skewness = 1.65



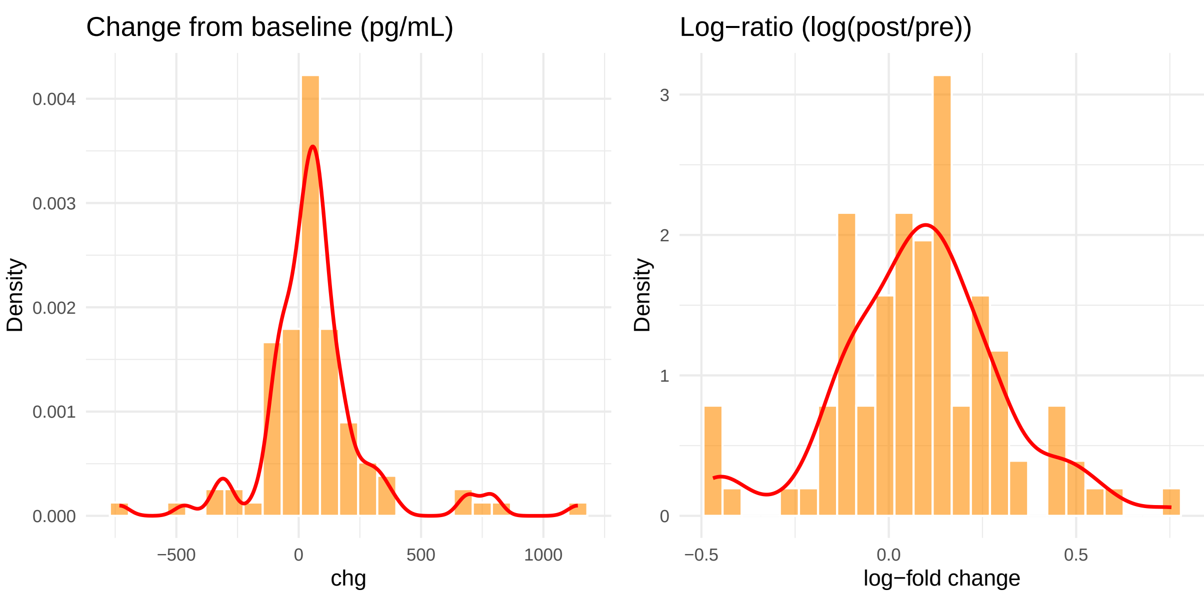
4.5 CD163

Linear chg: skewness = 0.67, excess kurtosis = 2.4 | Log-ratio skewness = 0.52



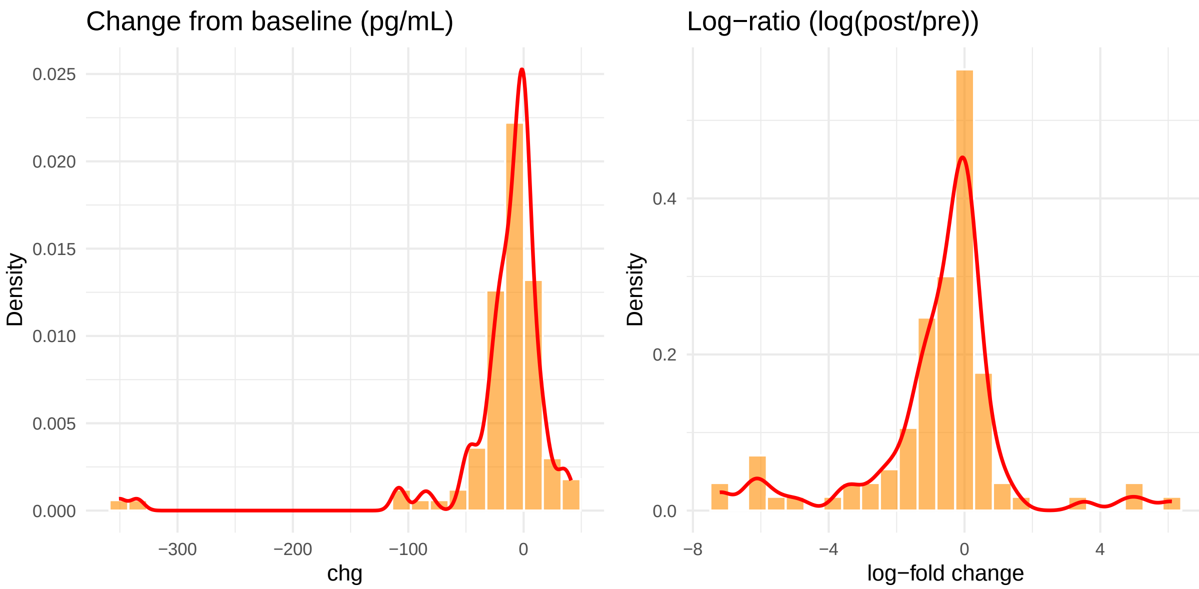
4.6 Collagen IV alpha 1

Linear chg: skewness = 1.19, excess kurtosis = 5.7 | Log-ratio skewness = -0.03



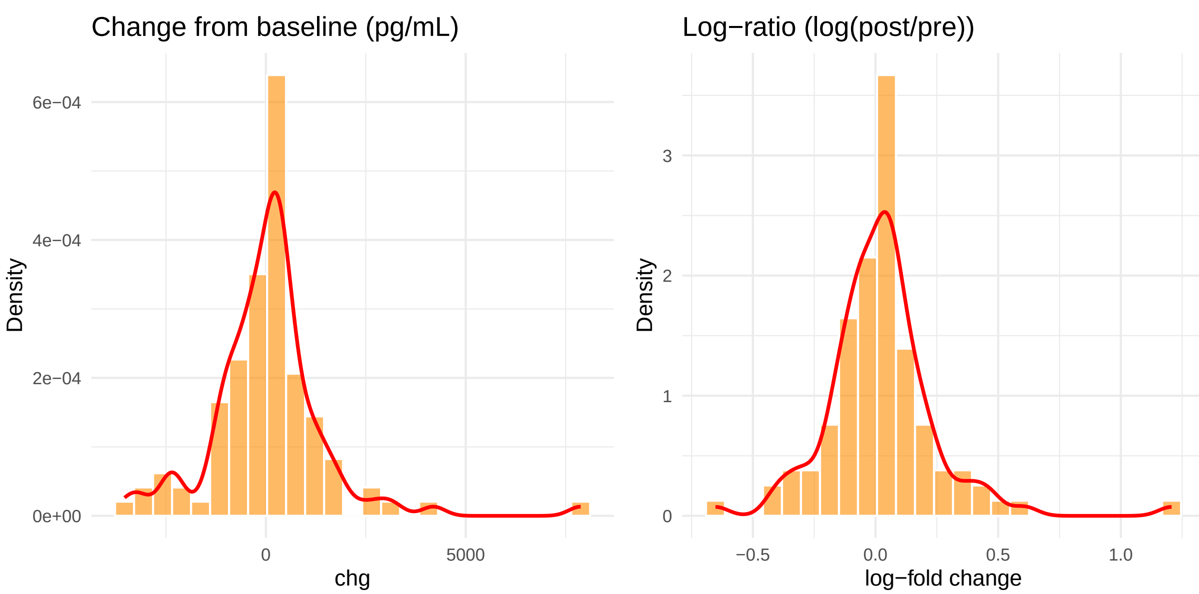
4.7 EGF

Linear chg: skewness = -4.72, excess kurtosis = 26.08 | Log-ratio skewness = -0.58



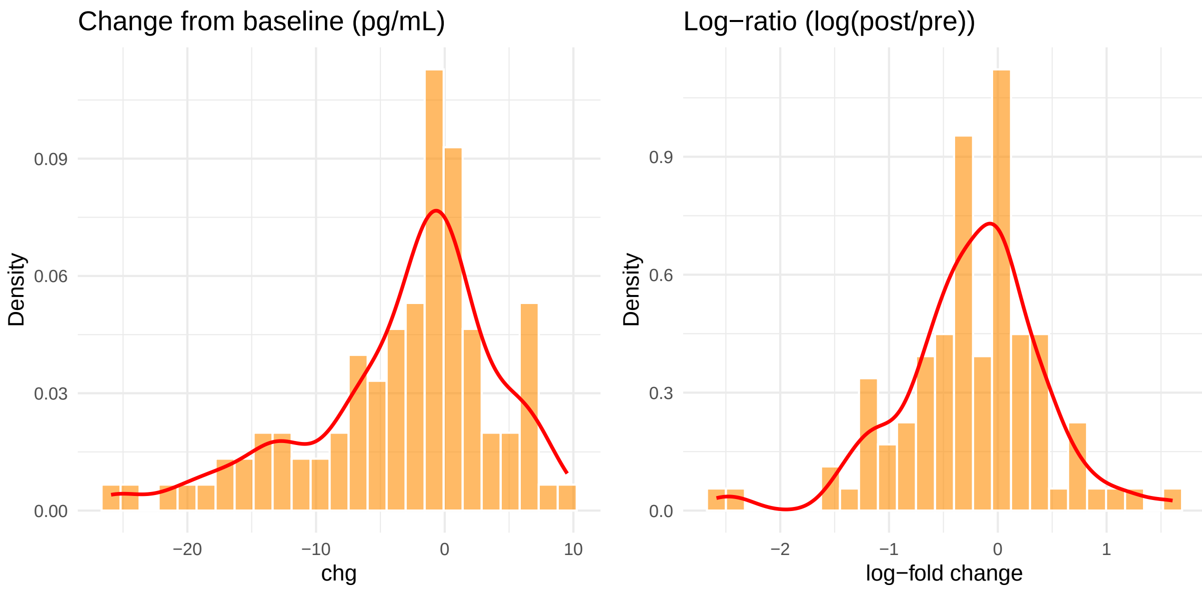
4.8 ErbB2

Linear chg: skewness = 1.37, excess kurtosis = 7.37 | Log-ratio skewness = 1.27



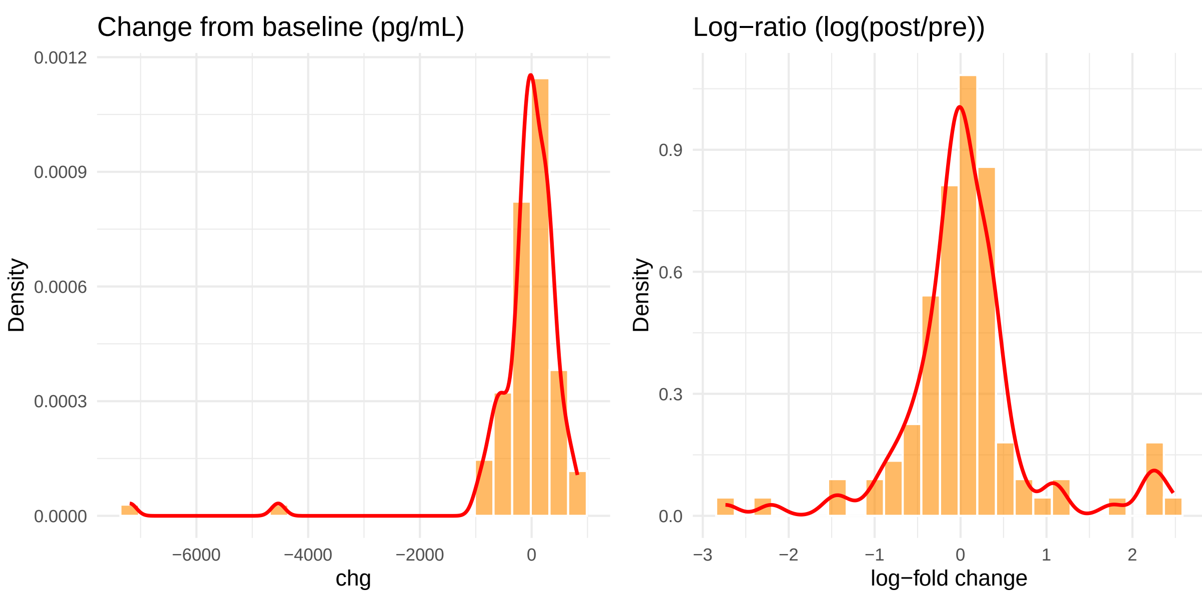
4.9 FGF Basic

Linear chg: skewness = -0.94, excess kurtosis = 0.63 | Log-ratio skewness = -0.52



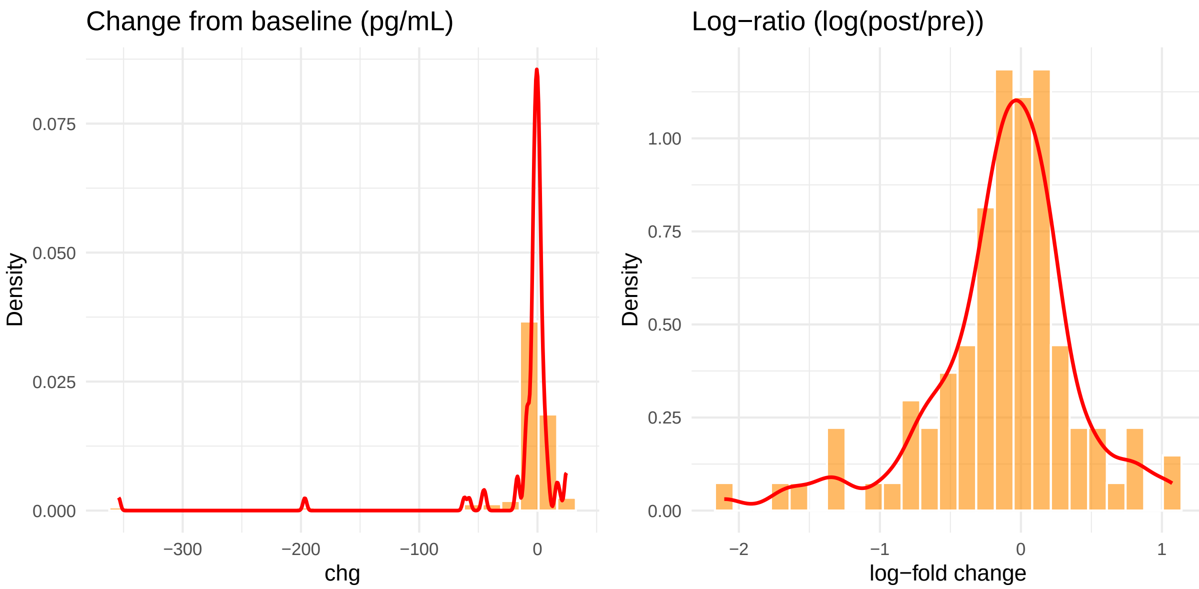
4.10 Fractalkine

Linear chg: skewness = -5.6, excess kurtosis = 37.51 | Log-ratio skewness = 0.39



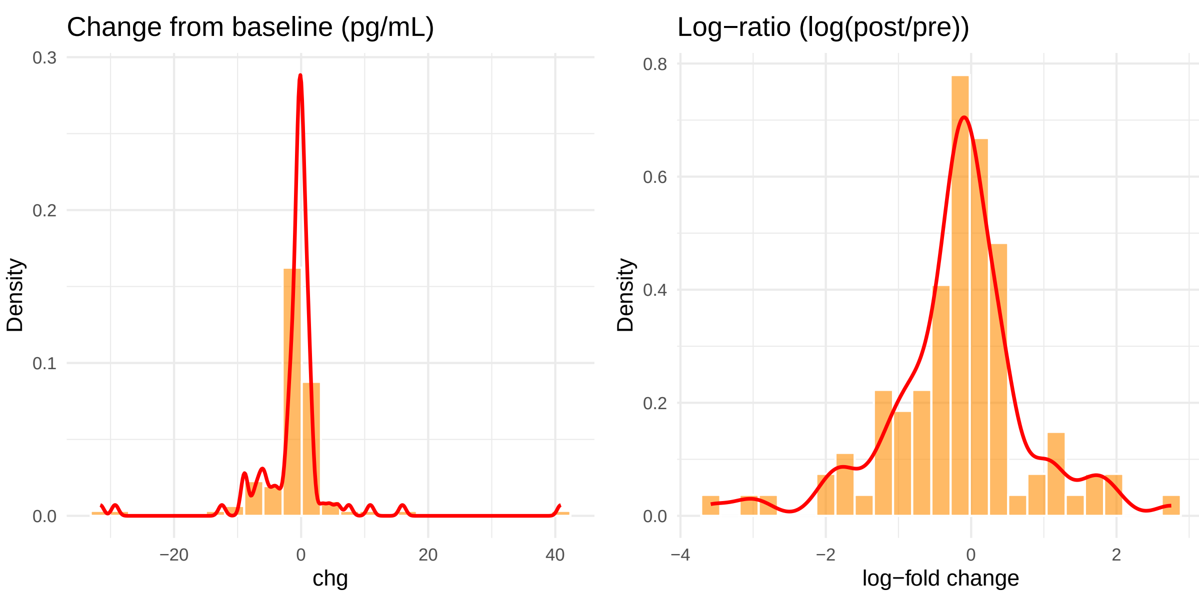
4.11 Granzyme B

Linear chg: skewness = -6.62, excess kurtosis = 48.53 | Log-ratio skewness = -0.92



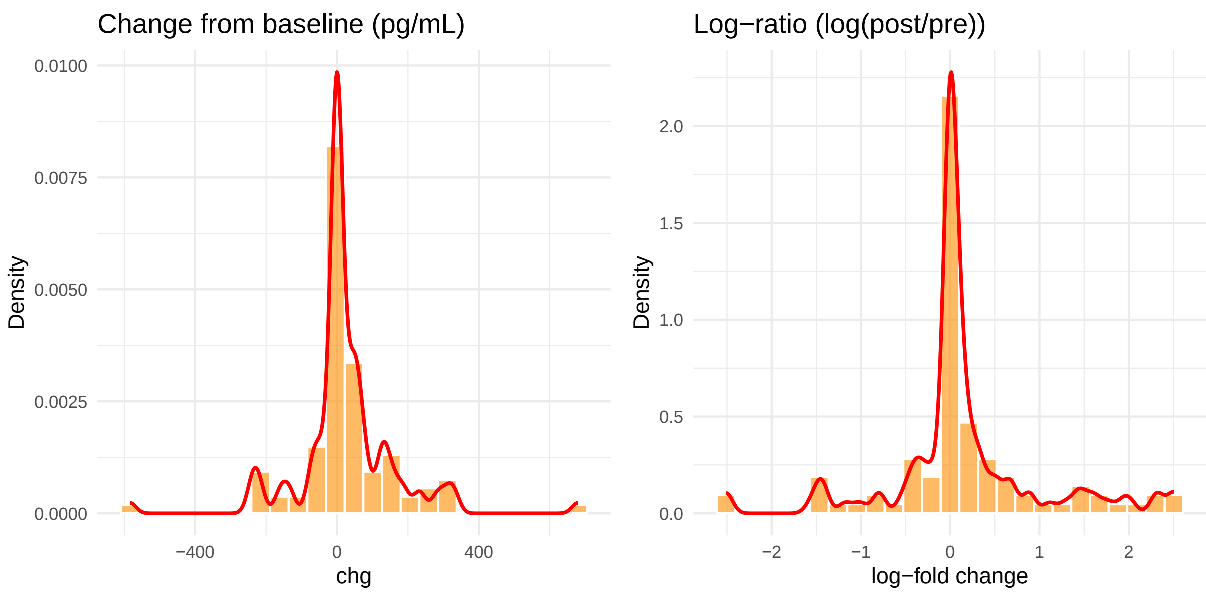
4.12 IFN-g

Linear chg: skewness = 0.72, excess kurtosis = 17.28 | Log-ratio skewness = -0.43



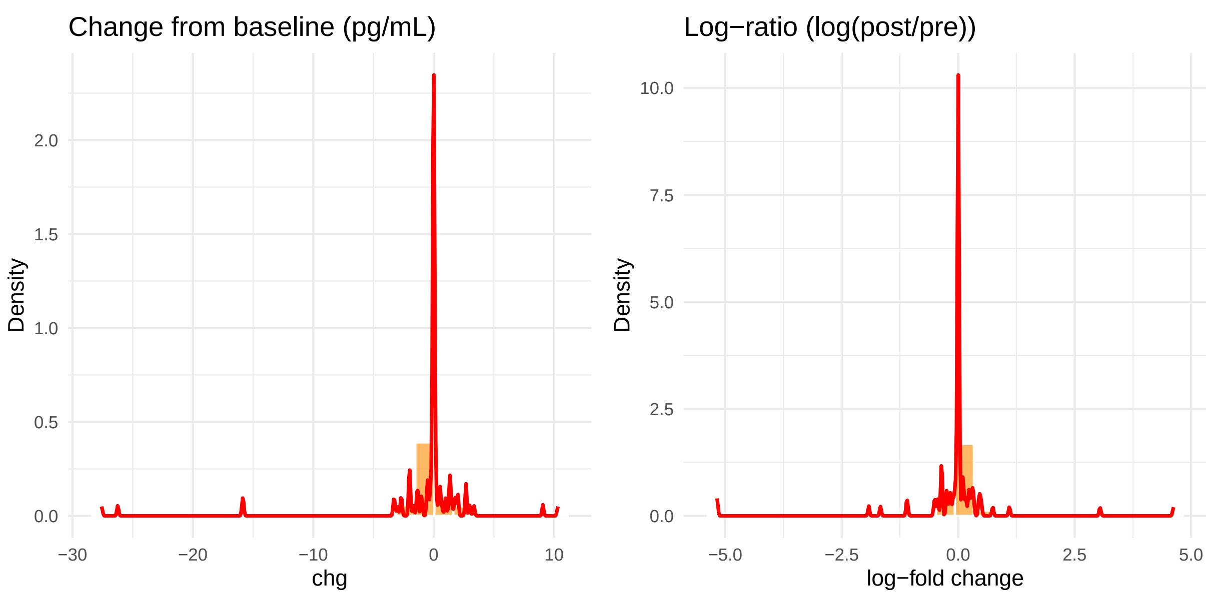
4.13 IL-12 p70

Linear chg: skewness = 0.34, excess kurtosis = 6.49 | Log-ratio skewness = 0.26



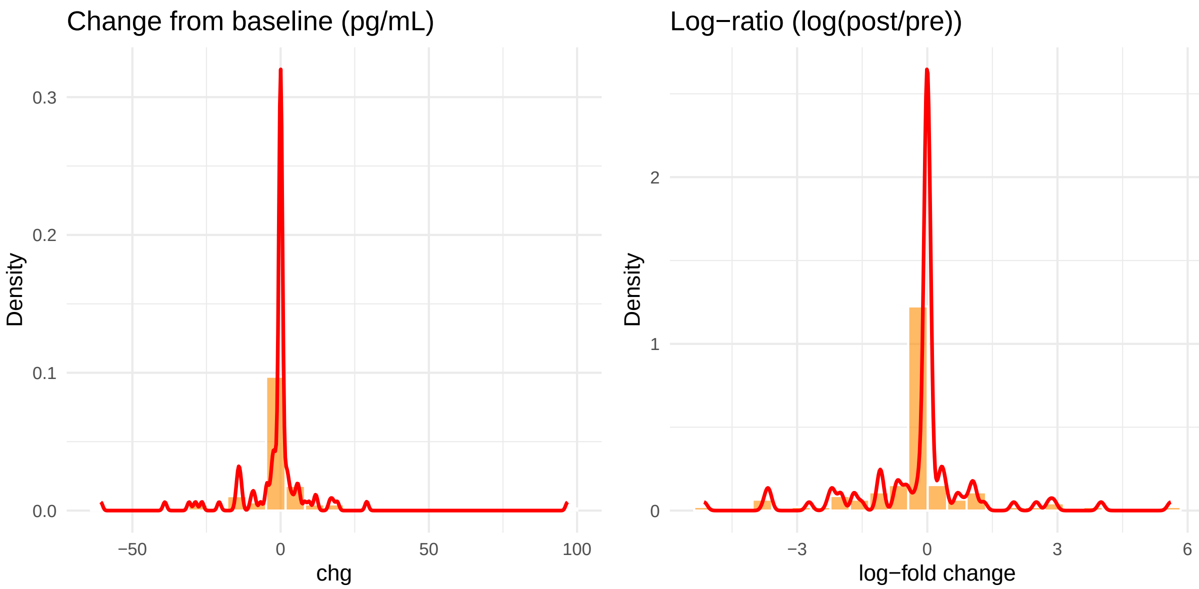
4.14 IL-1b

Linear chg: skewness = -3.82, excess kurtosis = 18.59 | Log-ratio skewness = -1.46



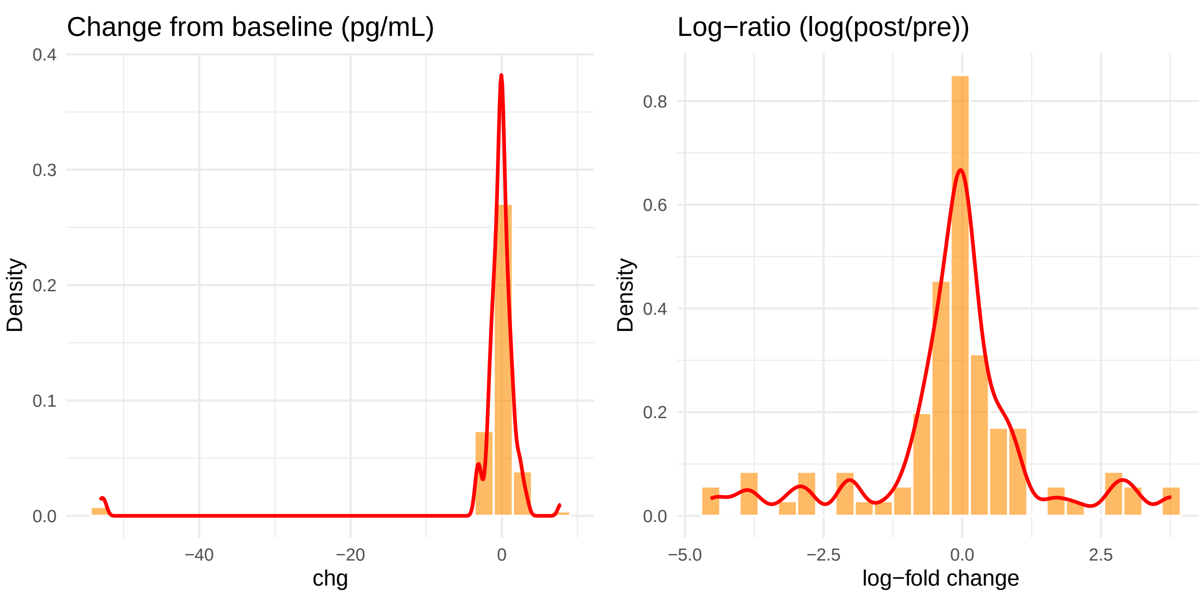
4.15 IL-2

Linear chg: skewness = 2.01, excess kurtosis = 20.61 | Log-ratio skewness = 0.11



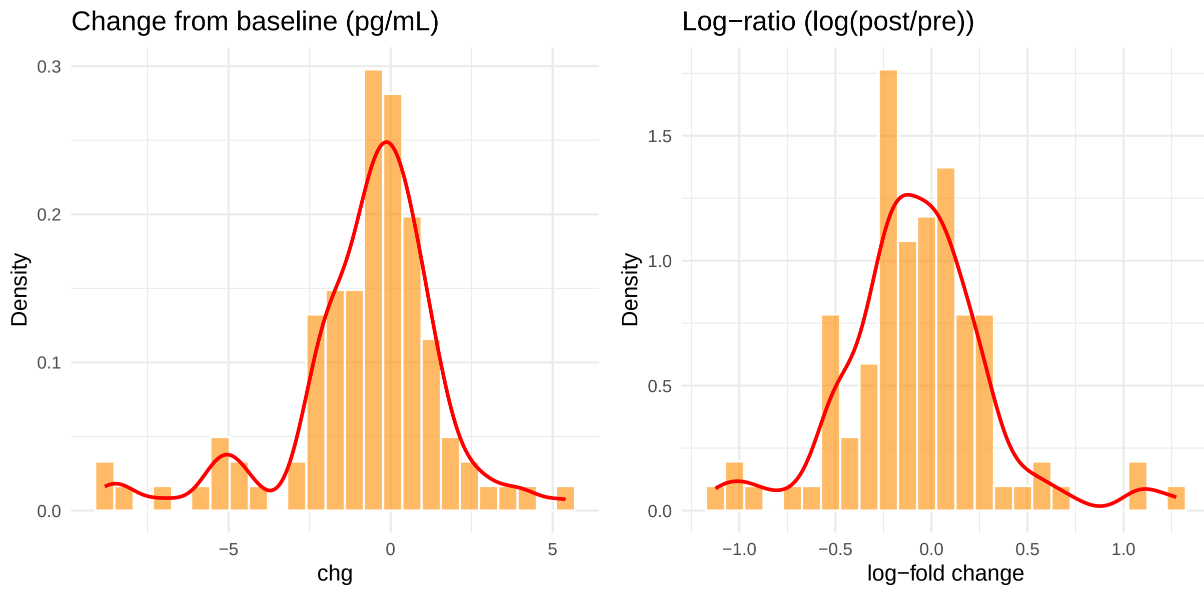
4.16 IL-6

Linear chg: skewness = -6.41, excess kurtosis = 41.45 | Log-ratio skewness = -0.36



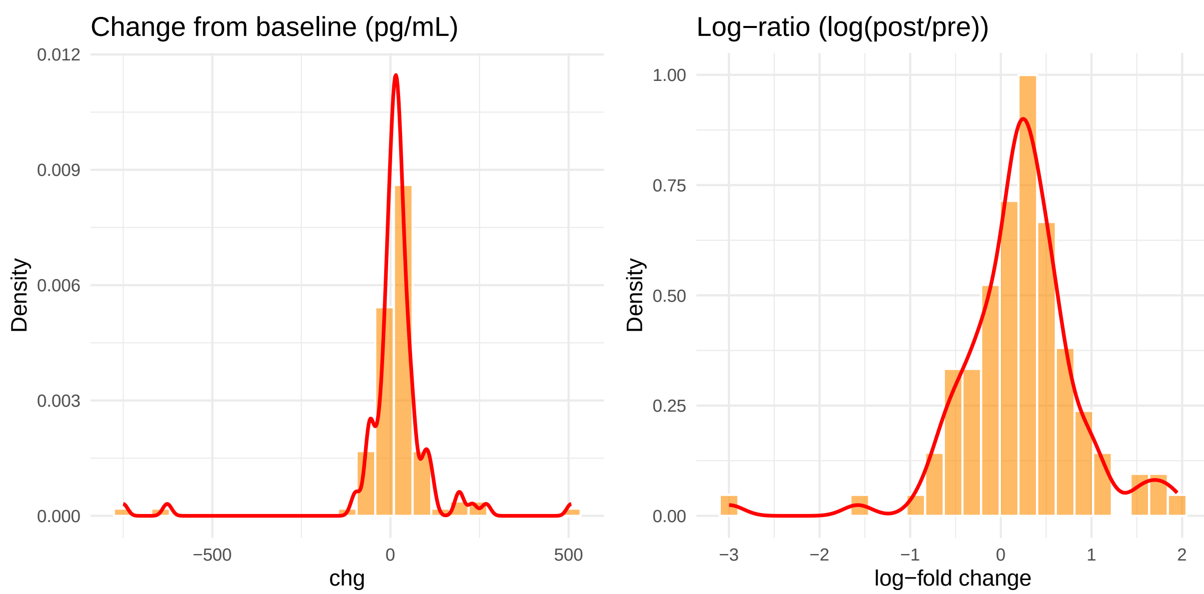
4.17 IL-8

Linear chg: skewness = -1.04, excess kurtosis = 2.2 | Log-ratio skewness = 0.42



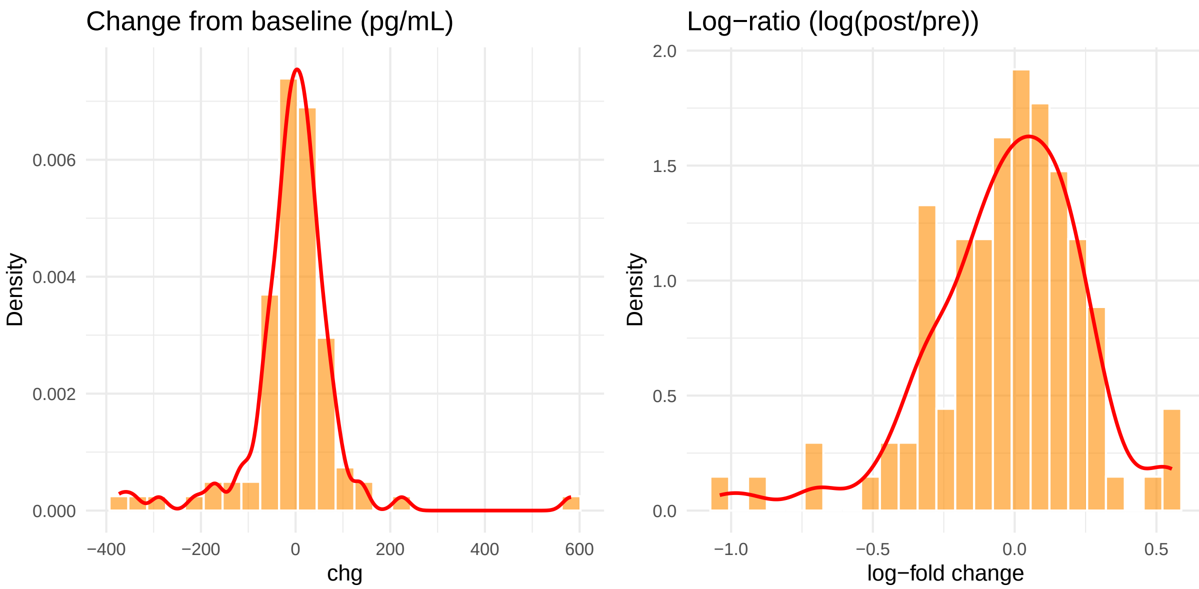
4.18 IP-10 (CXCL10)

Linear chg: skewness = -2.74, excess kurtosis = 20.04 | Log-ratio skewness = -0.85



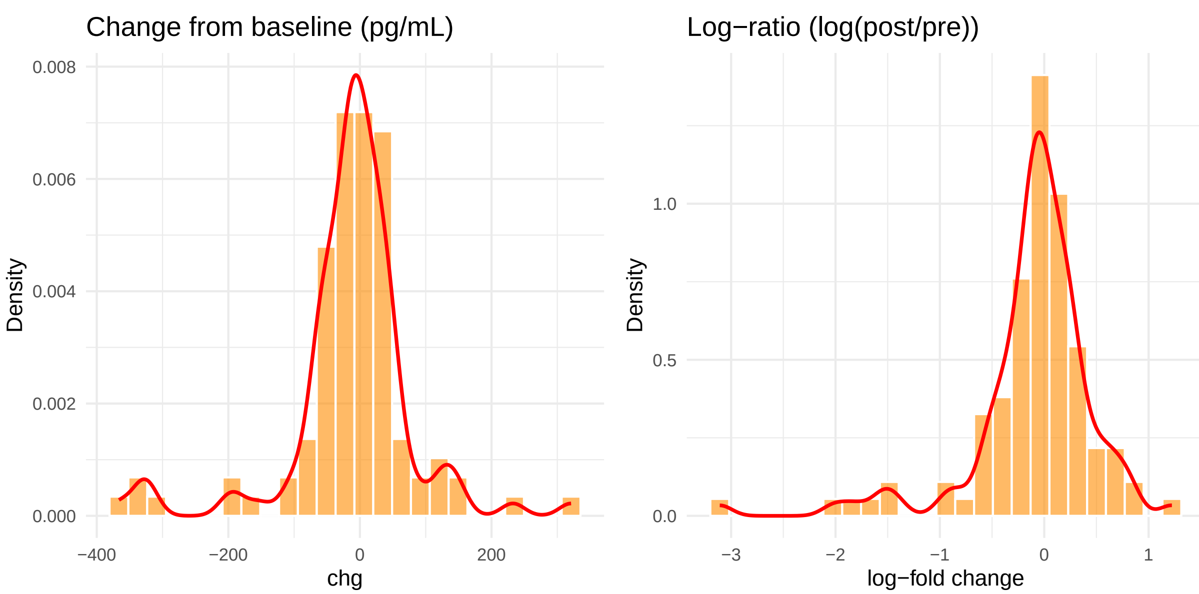
4.19 I-TAC (CXCL11)

Linear chg: skewness = 0.81, excess kurtosis = 11.55 | Log-ratio skewness = -0.84



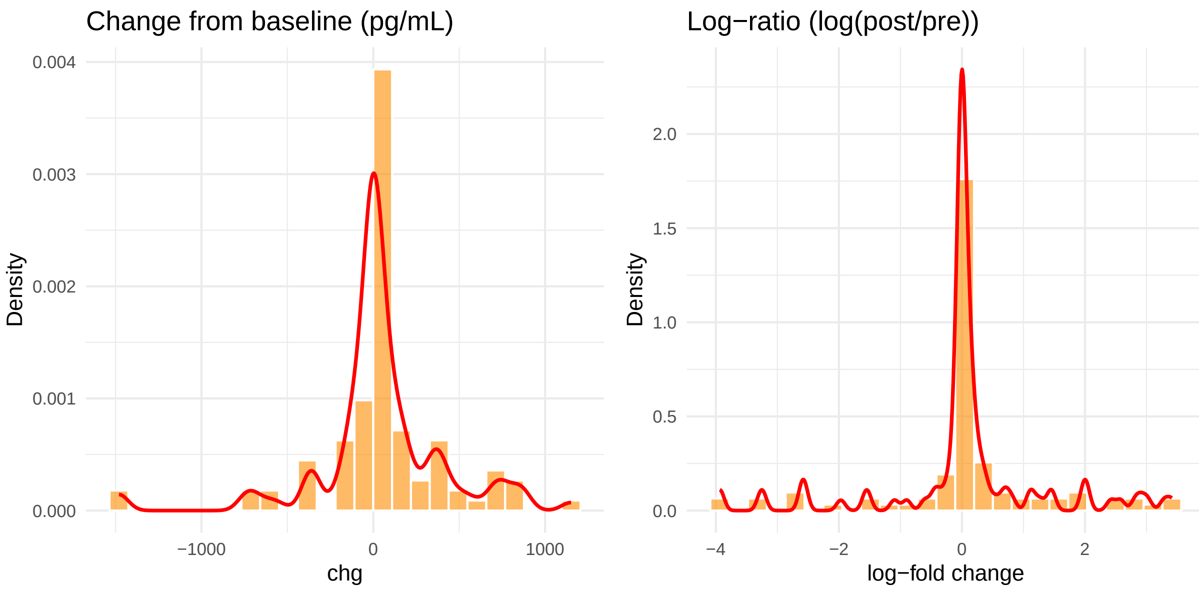
4.20 MCP-1

Linear chg: skewness = -0.91, excess kurtosis = 4.65 | Log-ratio skewness = -1.94



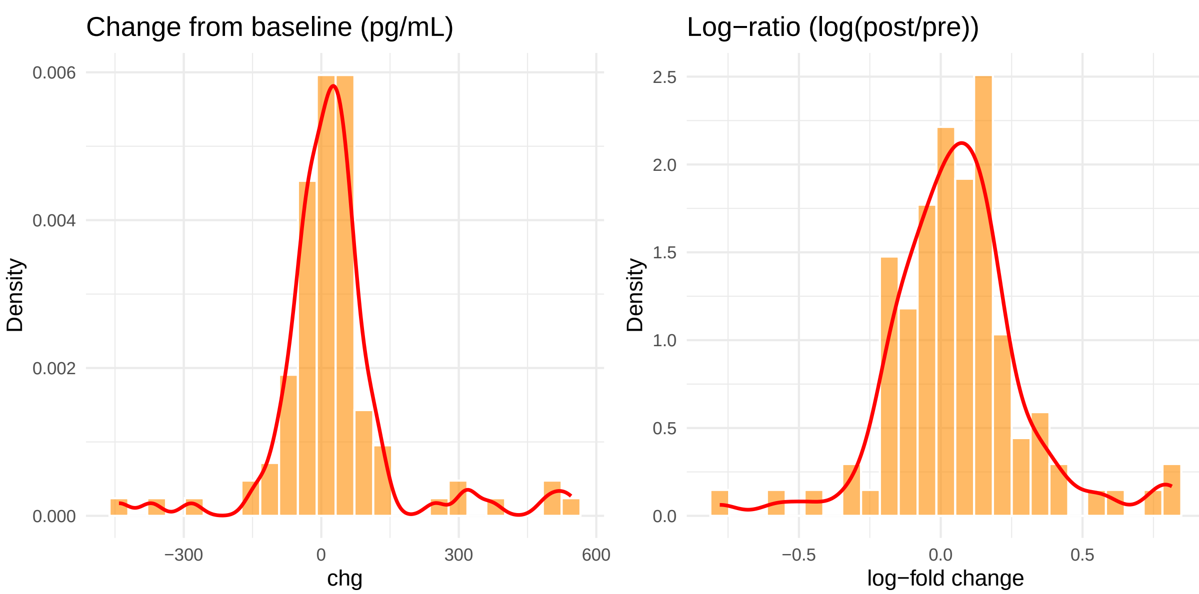
4.21 MIG (CXCL9)

Linear chg: skewness = -0.73, excess kurtosis = 4.29 | Log-ratio skewness = -0.36



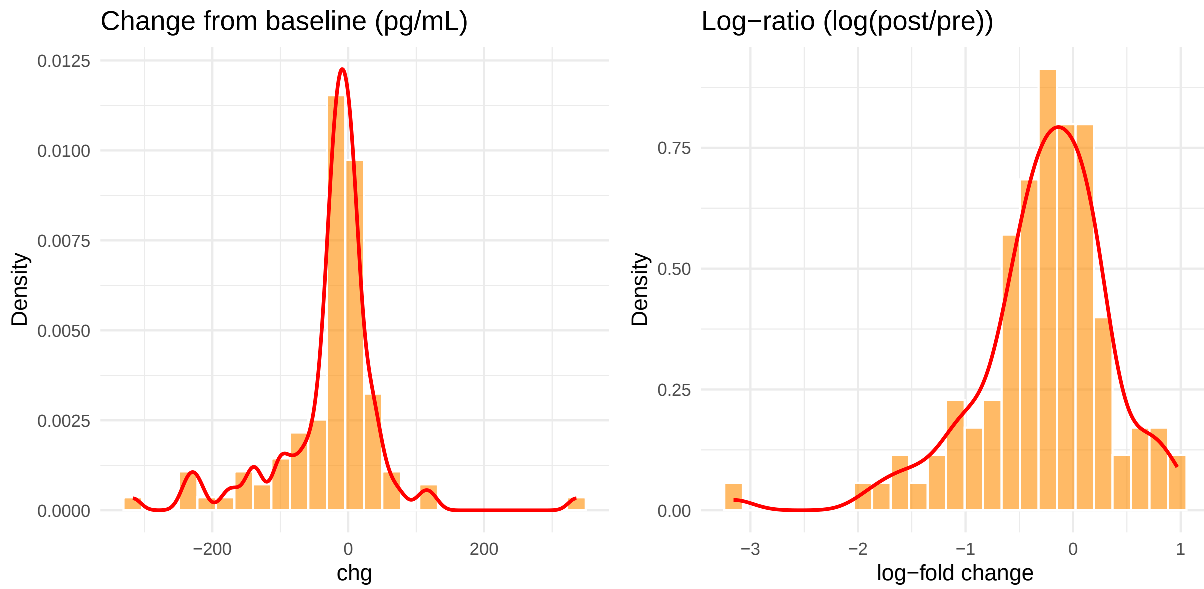
4.22 MMP-10

Linear chg: skewness = 0.95, excess kurtosis = 5.38 | Log-ratio skewness = 0.29



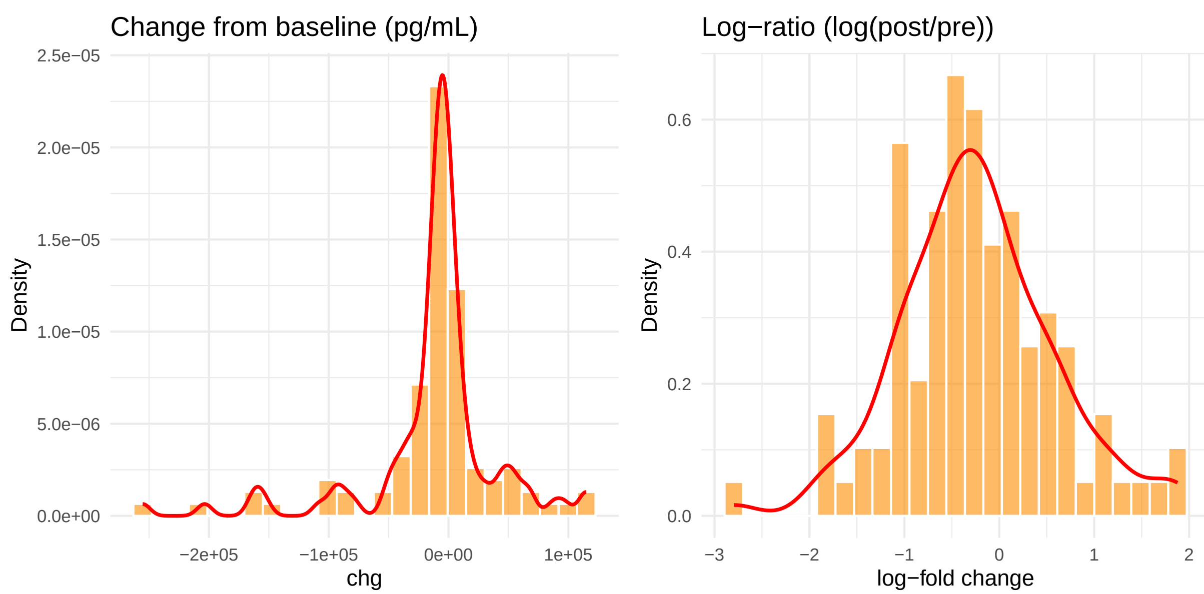
4.23 PDGF-DD

Linear chg: skewness = -0.3, excess kurtosis = 5.36 | Log-ratio skewness = -1.2



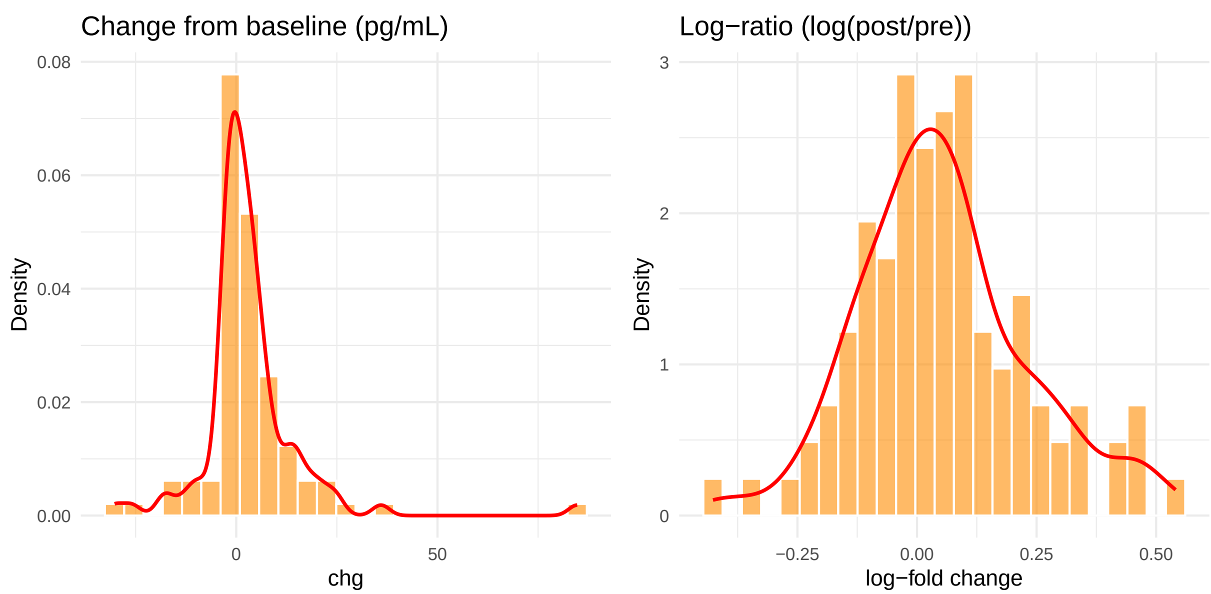
4.24 PECAM-1

Linear chg: skewness = -1.49, excess kurtosis = 4.93 | Log-ratio skewness = 0.11



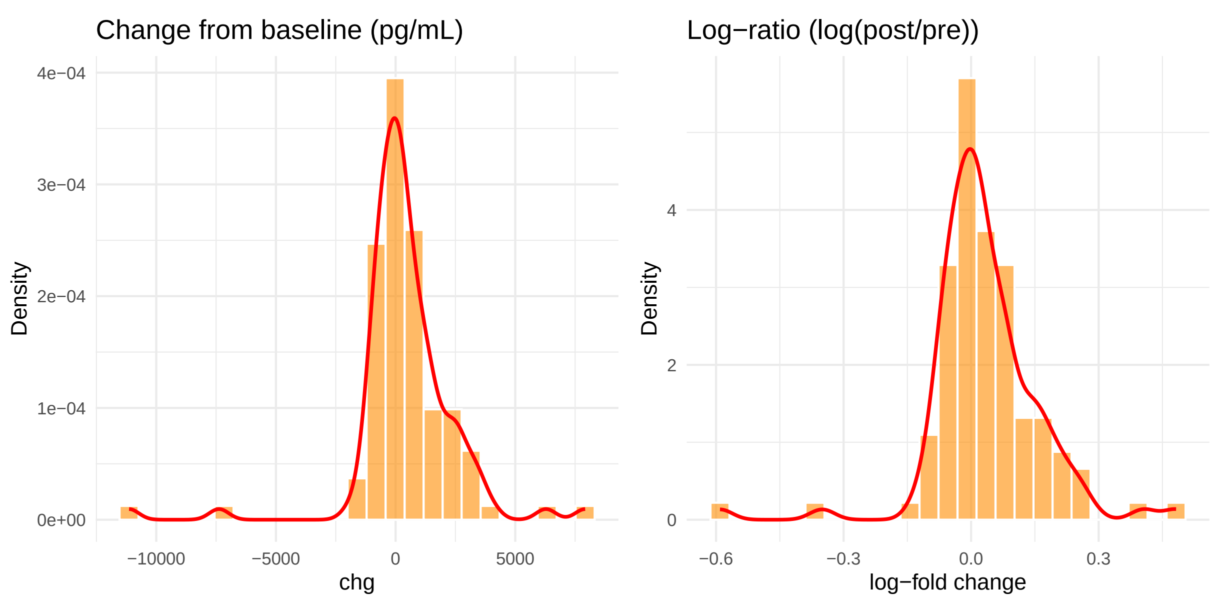
4.25 PIGF

Linear chg: skewness = 2.71, excess kurtosis = 17.16 | Log-ratio skewness = 0.36



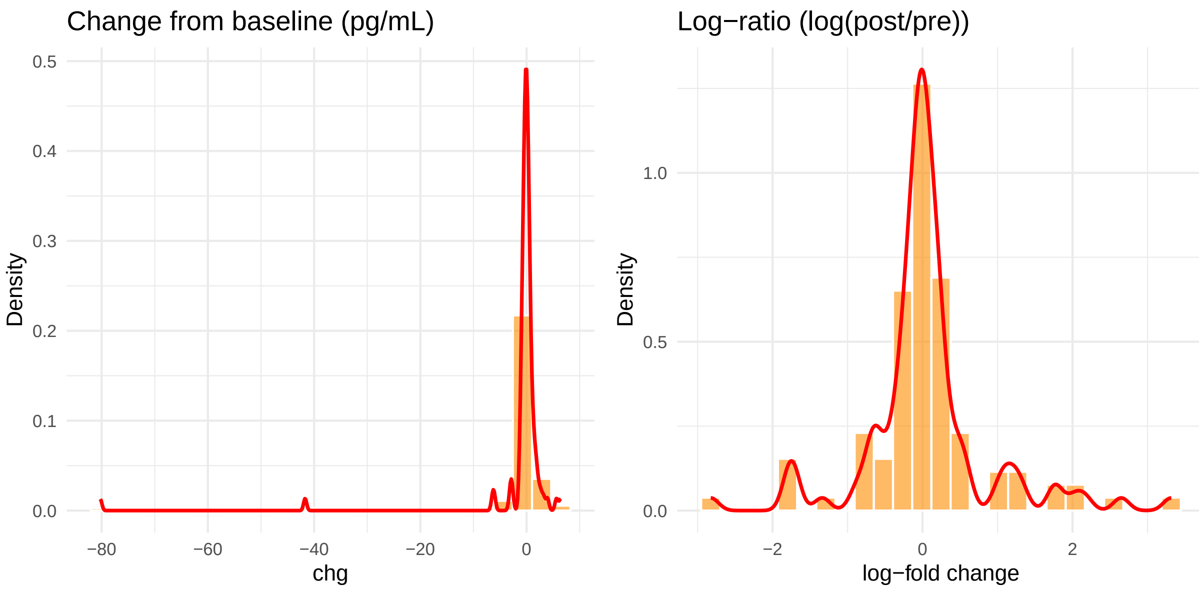
4.26 Tenascin C

Linear chg: skewness = -1.36, excess kurtosis = 11.25 | Log-ratio skewness = -0.5



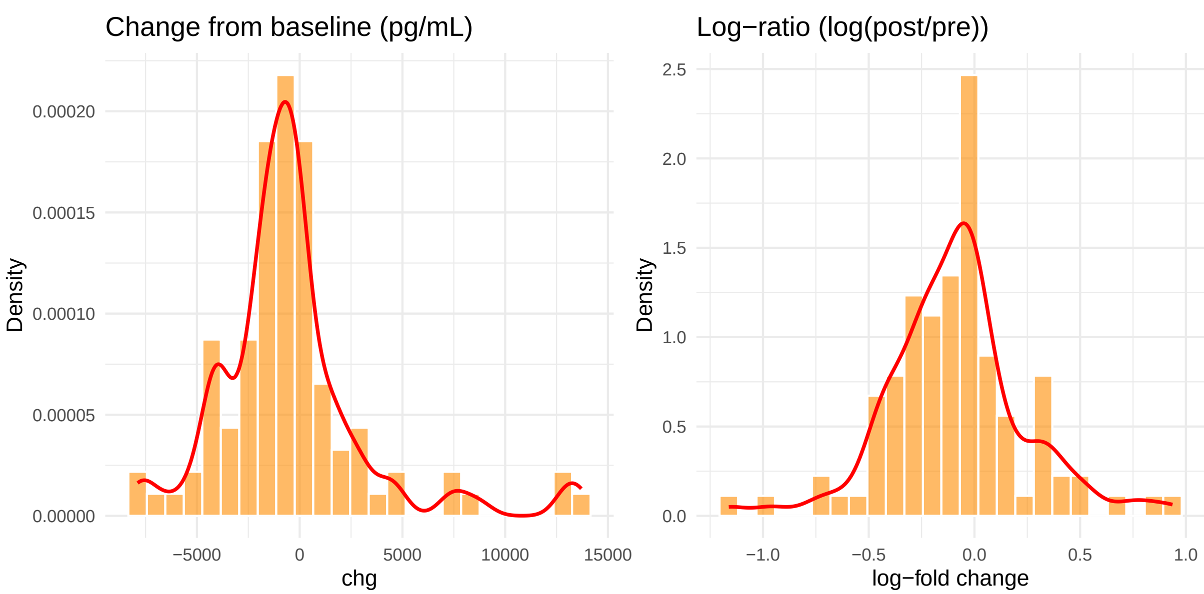
4.27 TGF-alpha

Linear chg: skewness = -7.37, excess kurtosis = 57.72 | Log-ratio skewness = 0.58



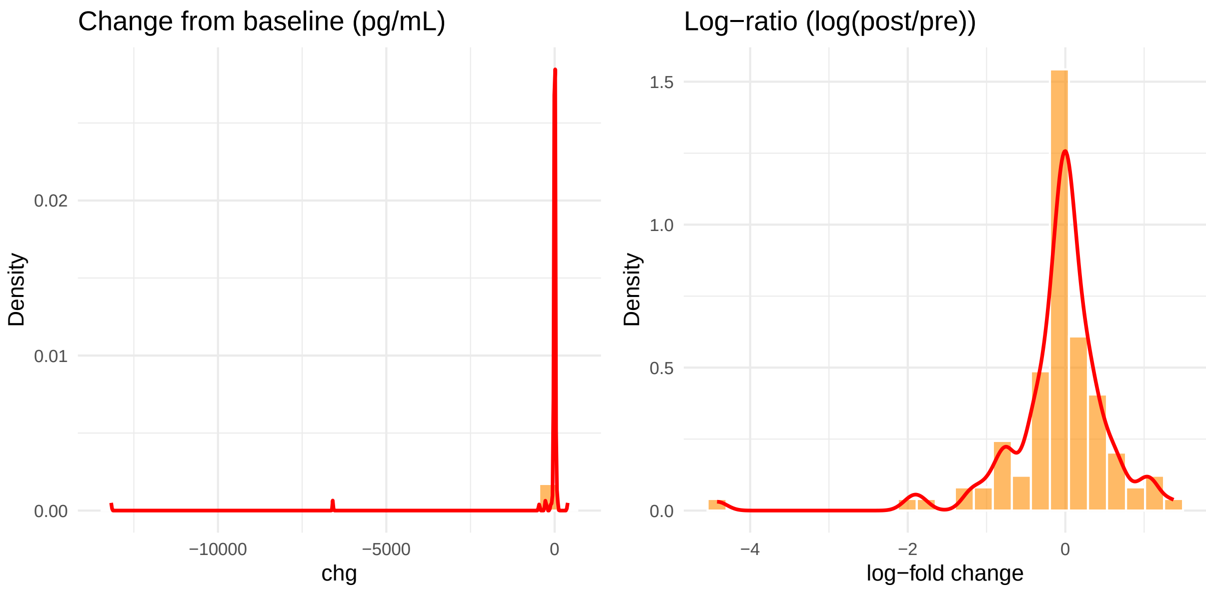
4.28 Thrombospondin-2

Linear chg: skewness = 1.53, excess kurtosis = 4.13 | Log-ratio skewness = 0.15



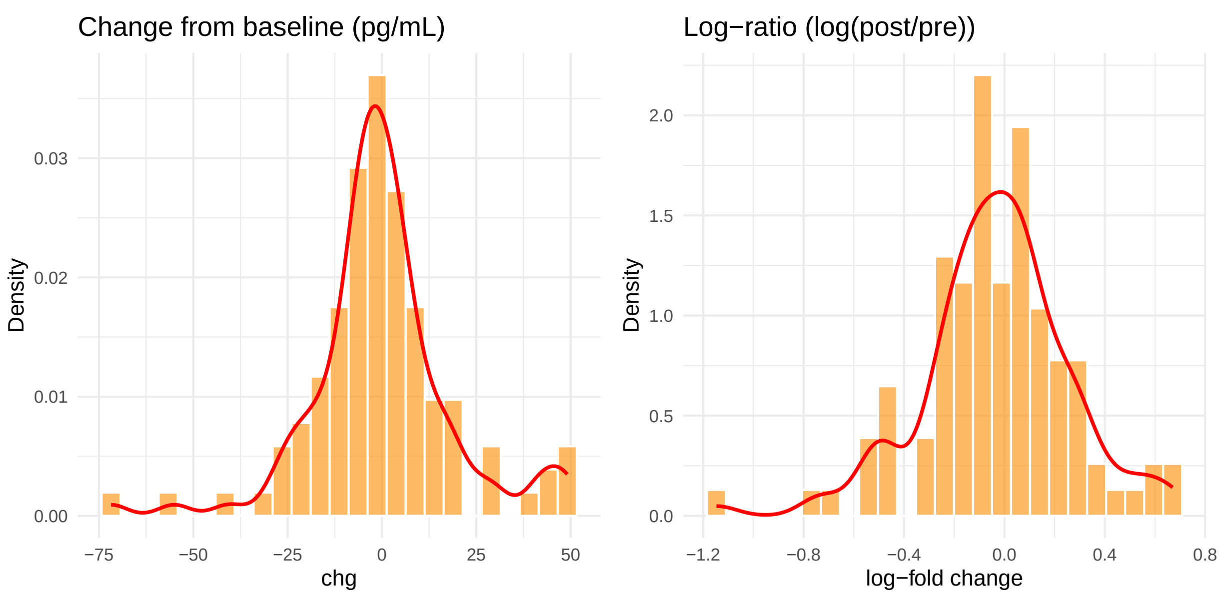
4.29 TNFRSF18 (GITR)

Linear chg: skewness = -7.8, excess kurtosis = 62.85 | Log-ratio skewness = -2.65



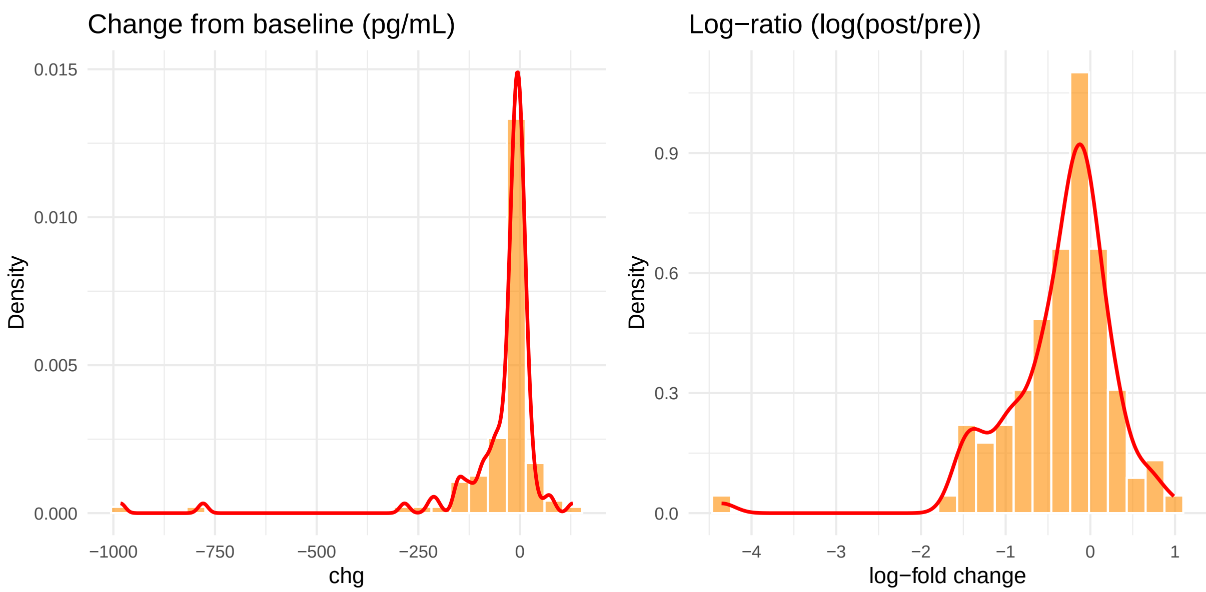
4.30 TNFSF6 (FasL)

Linear chg: skewness = -0.08, excess kurtosis = 2.51 | Log-ratio skewness = -0.45



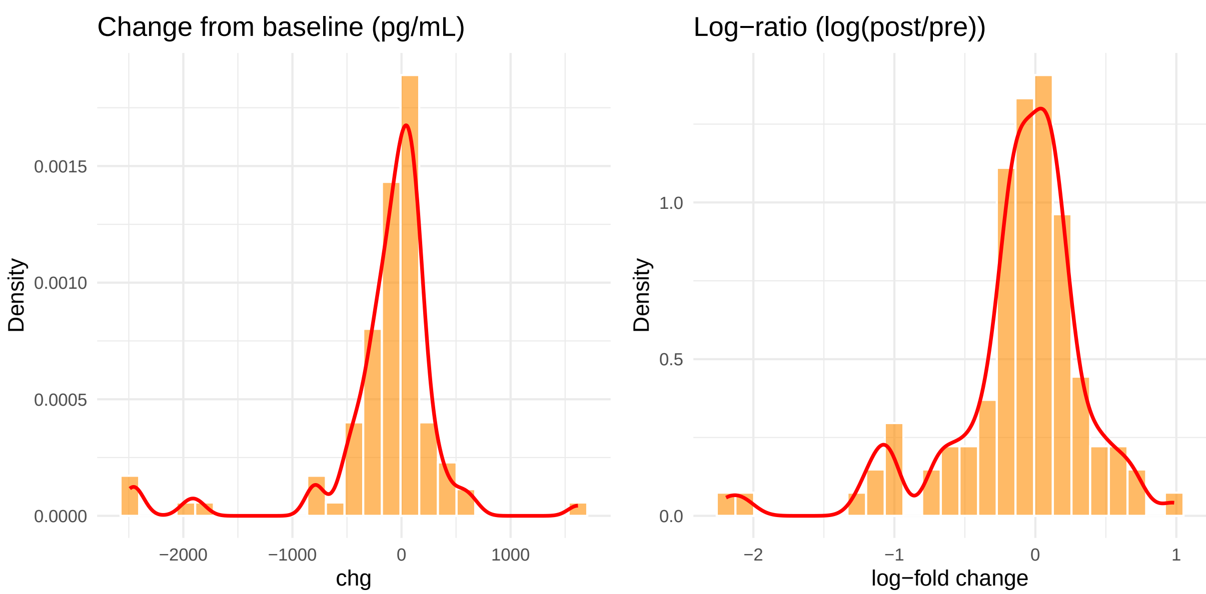
4.31 VEGF-A

Linear chg: skewness = -5.14, excess kurtosis = 30.45 | Log-ratio skewness = -2.17



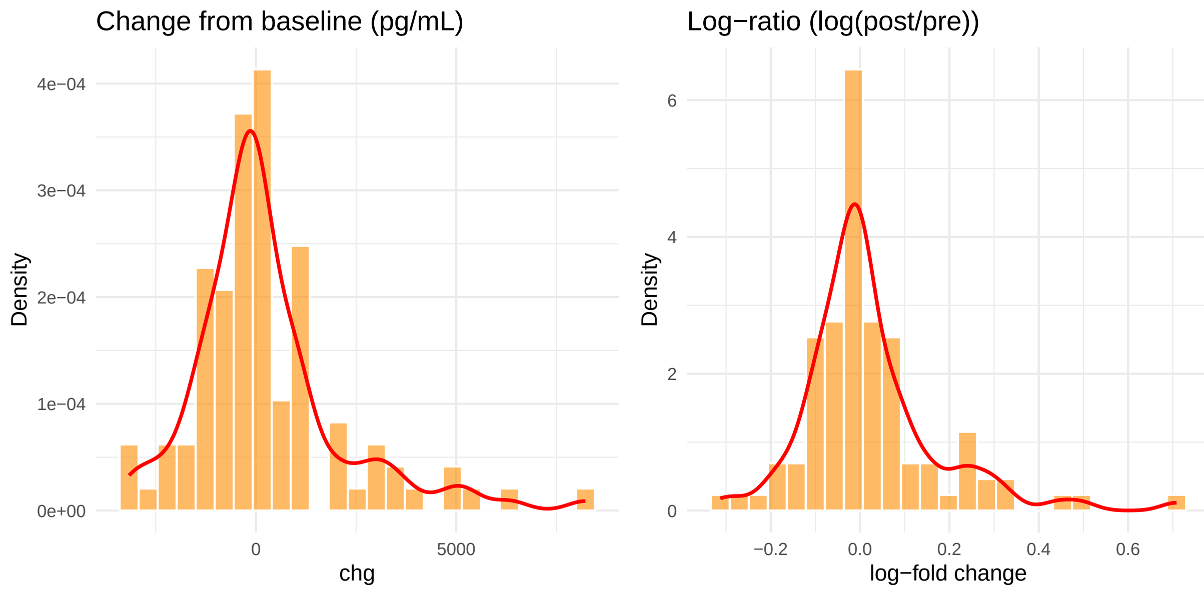
4.32 VEGF-C

Linear chg: skewness = -2.21, excess kurtosis = 7.88 | Log-ratio skewness = -1.49



4.33 VEGFR2

Linear chg: skewness = 1.44, excess kurtosis = 3.2 | Log-ratio skewness = 1.46



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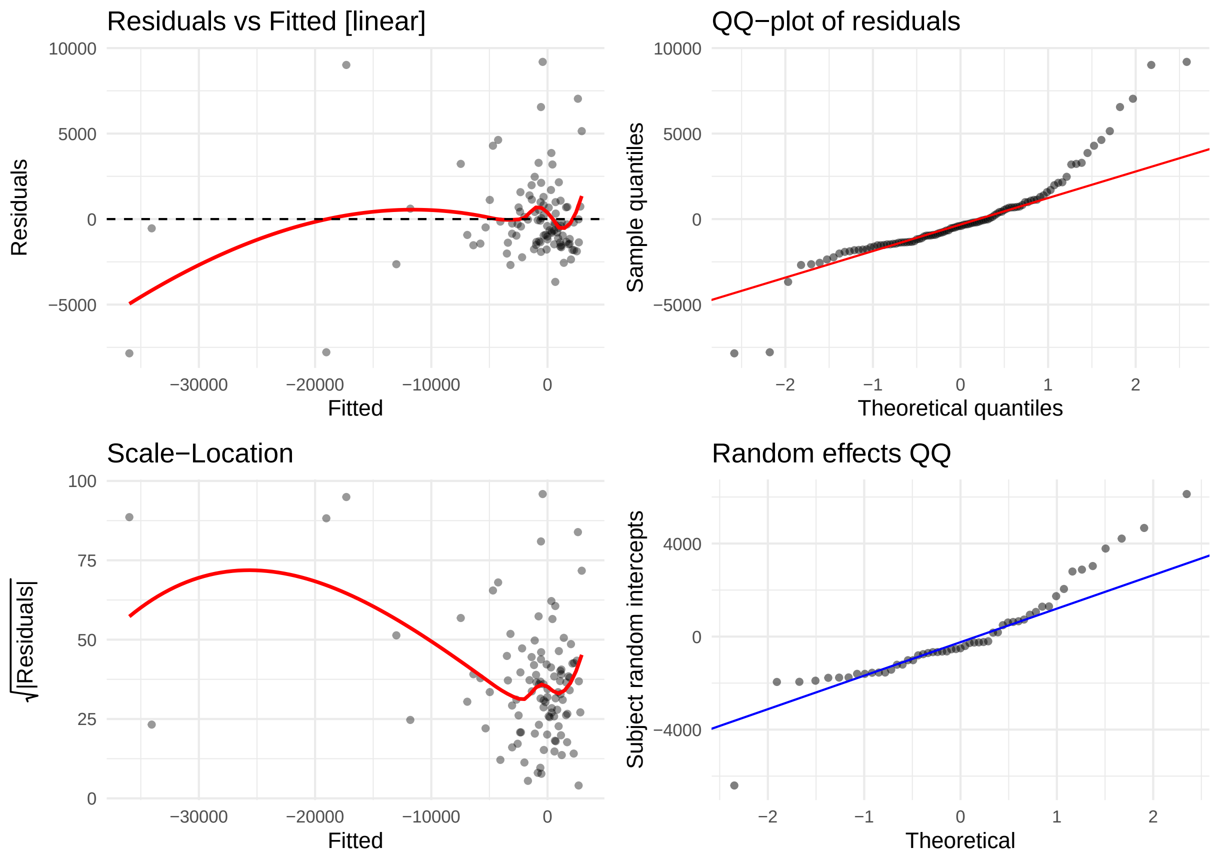
5 Residual diagnostics per analyte

For each analyte both LMMs are fitted on early visits (Days 1–43). Four diagnostic panels are shown per model: **Residuals vs Fitted** (homoscedasticity), **QQ-plot** (normality), **Scale-Location** (trend in residual variance), and **Random effects QQ** (normality of subject-level intercepts). A Shapiro-Wilk test is reported when the residual sample size is between 5 and 5000.

5.1 Angiopoietin-1 (`ANGPT1`)

AIC linear: 1922 | **AIC log-ratio:** 199.9 | **Preferred: log**


Residuals vs Fitted [linear]
Residuals
Fitted
10000
5000
0
−5000
−30000
−20000
−10000
0
QQ−plot of residuals
Sample quantiles
Theoretical quantiles
10000
5000
0
−5000
−2
−1
0
1
2
Scale−Location
√|Residuals|
Fitted
100
75
50
25
0
−30000
−20000
−10000
0
Random effects QQ
Subject random intercepts
Theoretical
4000
0
−4000
−2
−1
0
1
2

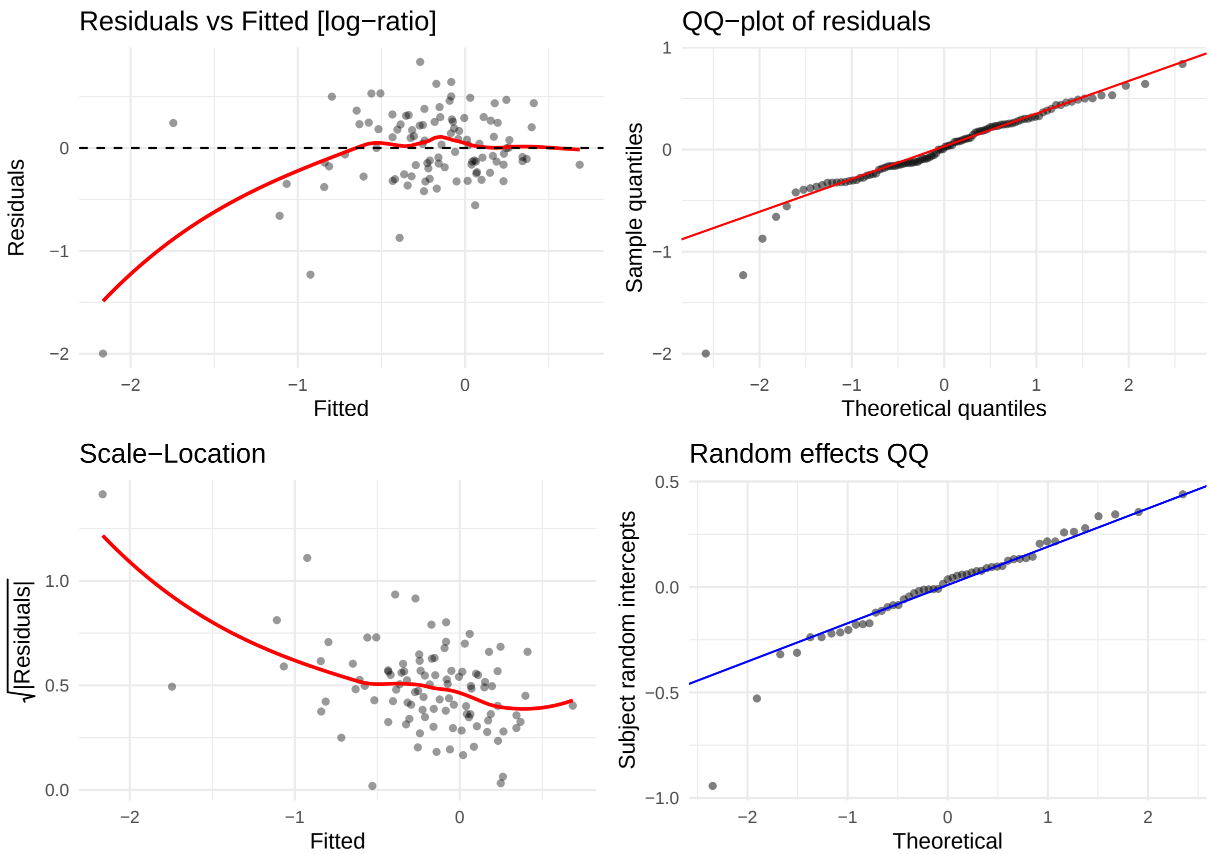
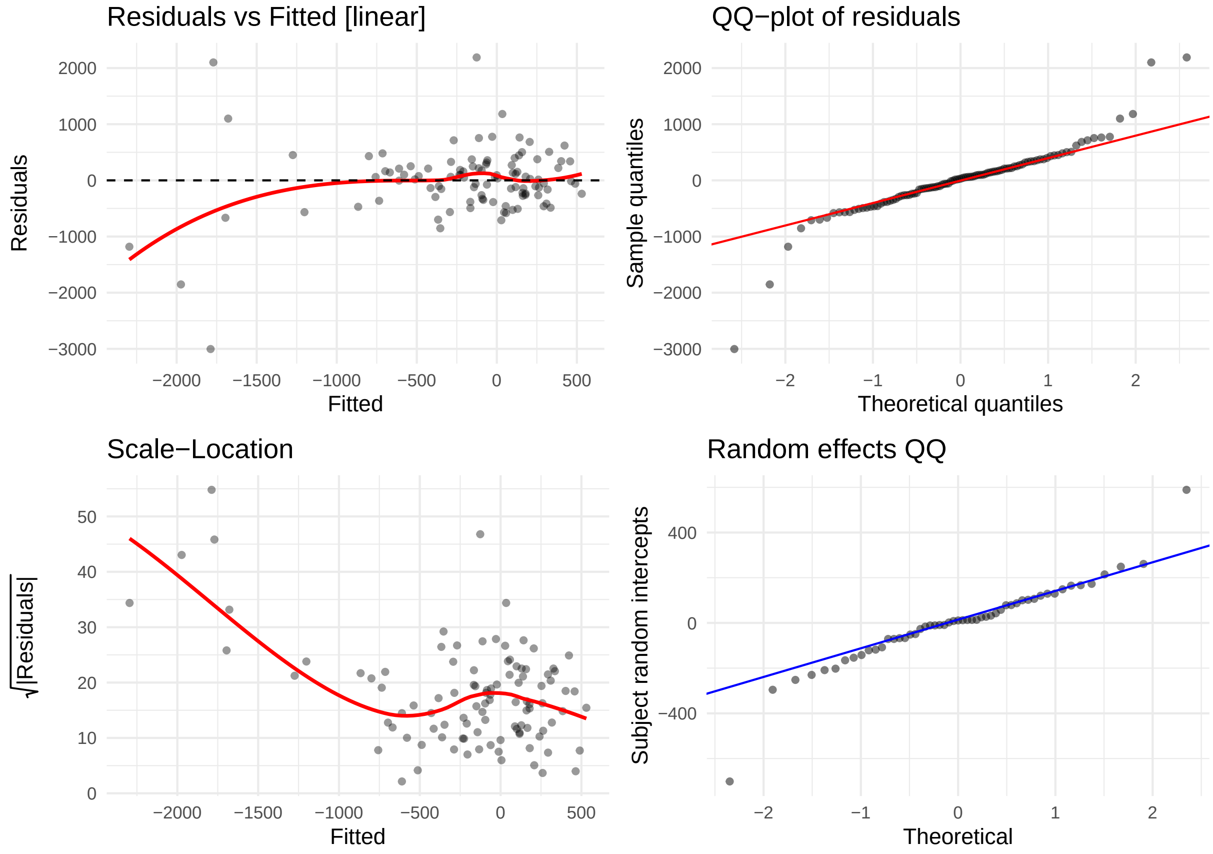


Table 3: Shapiro-Wilk — Angiopoietin-1

Scale	W	p-value	Interpretation
linear	0.8684	0	Strong departure from normality
log-ratio	0.8960	0	Strong departure from normality

5.2 Angiopoietin-2 (ANGPT2)

AIC linear: 1602.8 | **AIC log-ratio:** -7.9 | **Preferred: log**


Residuals vs Fitted [linear]
Residuals
Fitted
QQ−plot of residuals
Sample quantiles
Theoretical quantiles
Scale−Location
√|Residuals|
Fitted
Random effects QQ
Subject random intercepts
Theoretical

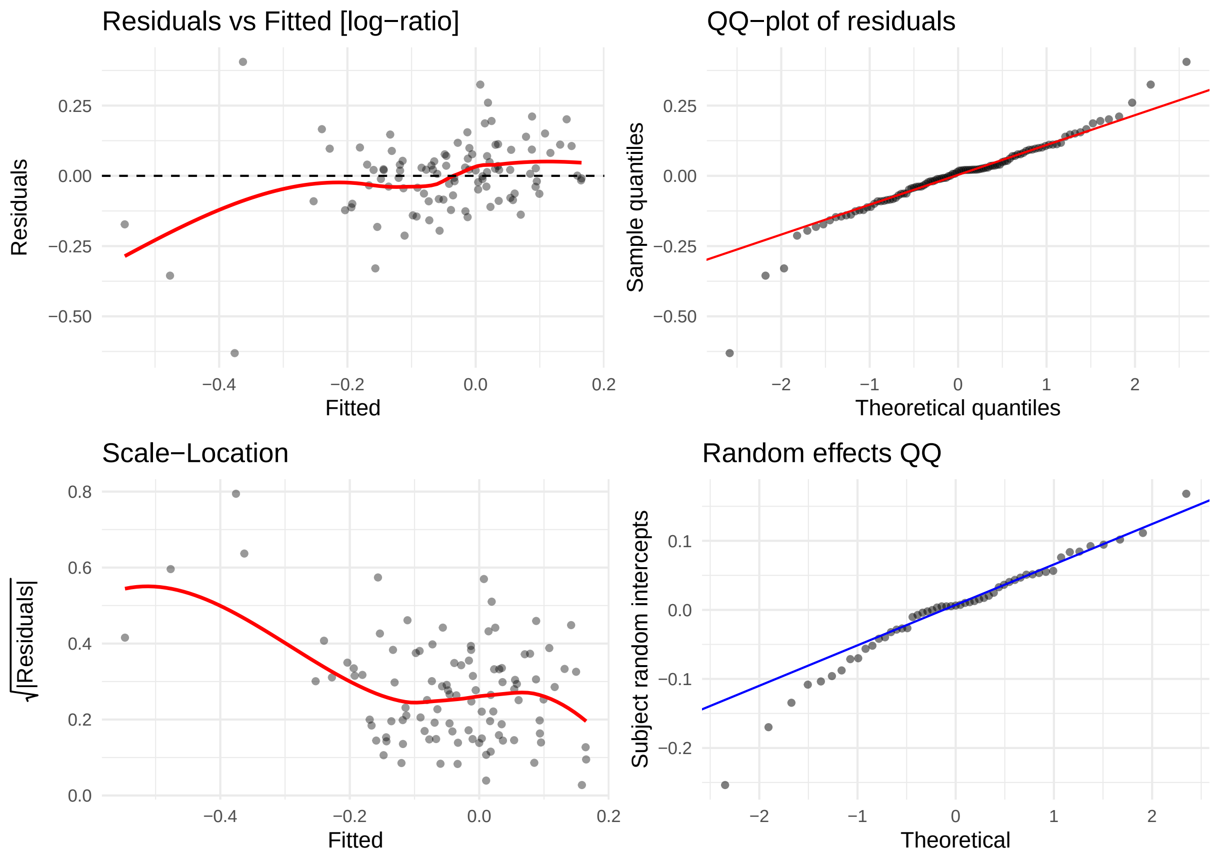
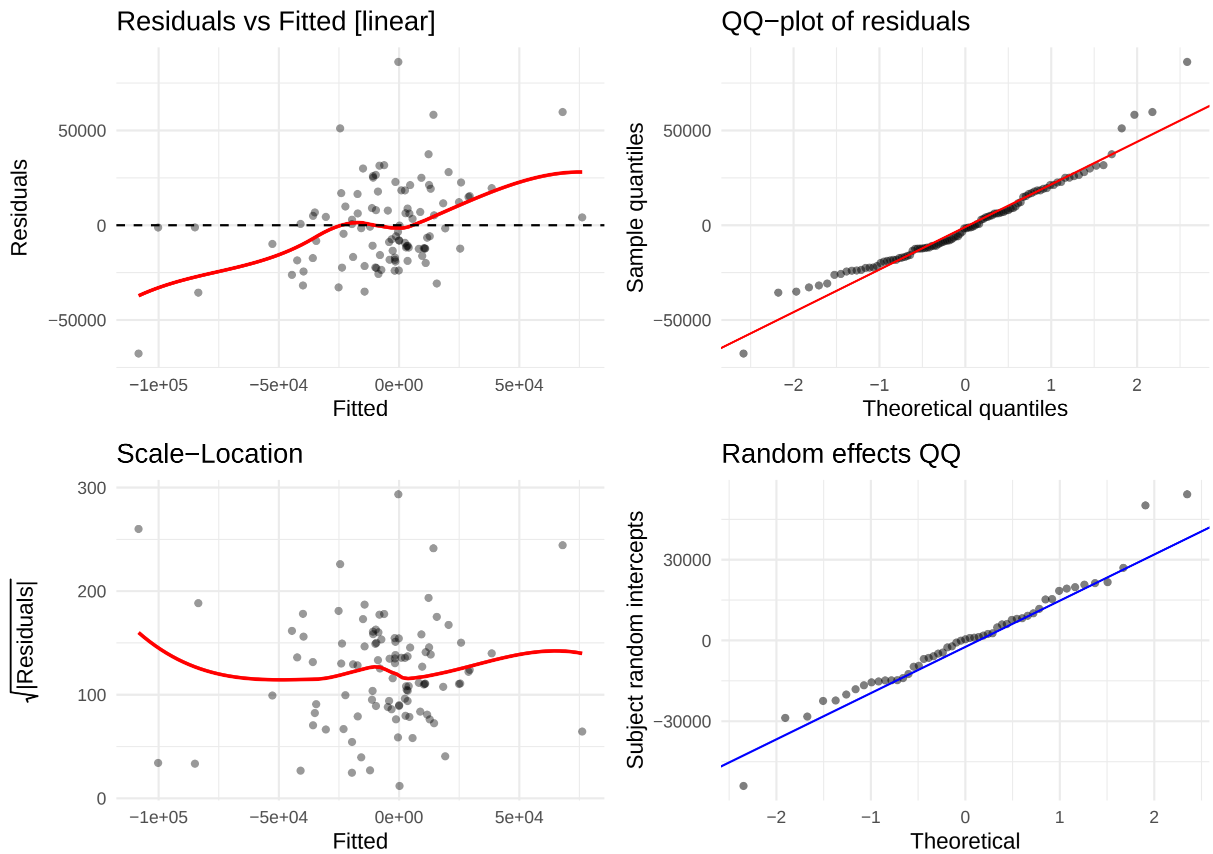


Table 4: Shapiro-Wilk — Angiopoietin-2

Scale	W	p-value	Interpretation
linear	0.8745	0e+00	Strong departure from normality
log-ratio	0.9322	1e-04	Strong departure from normality

5.3 ANGPTL4 (`ANGPTL4`)

AIC linear: 2347.3 | **AIC log-ratio:** 28.2 | **Preferred: log**



Residuals vs Fitted [linear]
Residuals
50000
0
−50000
−1e+05
−5e+04
0e+00
5e+04
Fitted
QQ−plot of residuals
Sample quantiles
50000
0
−50000
−2
−1
0
1
2
Theoretical quantiles
Scale−Location
$\sqrt{|\text{Residuals}|}$
300
200
100
0
−1e+05
−5e+04
0e+00
5e+04
Fitted
Random effects QQ
Subject random intercepts
30000
0
−30000
−2
−1
0
1
2
Theoretical

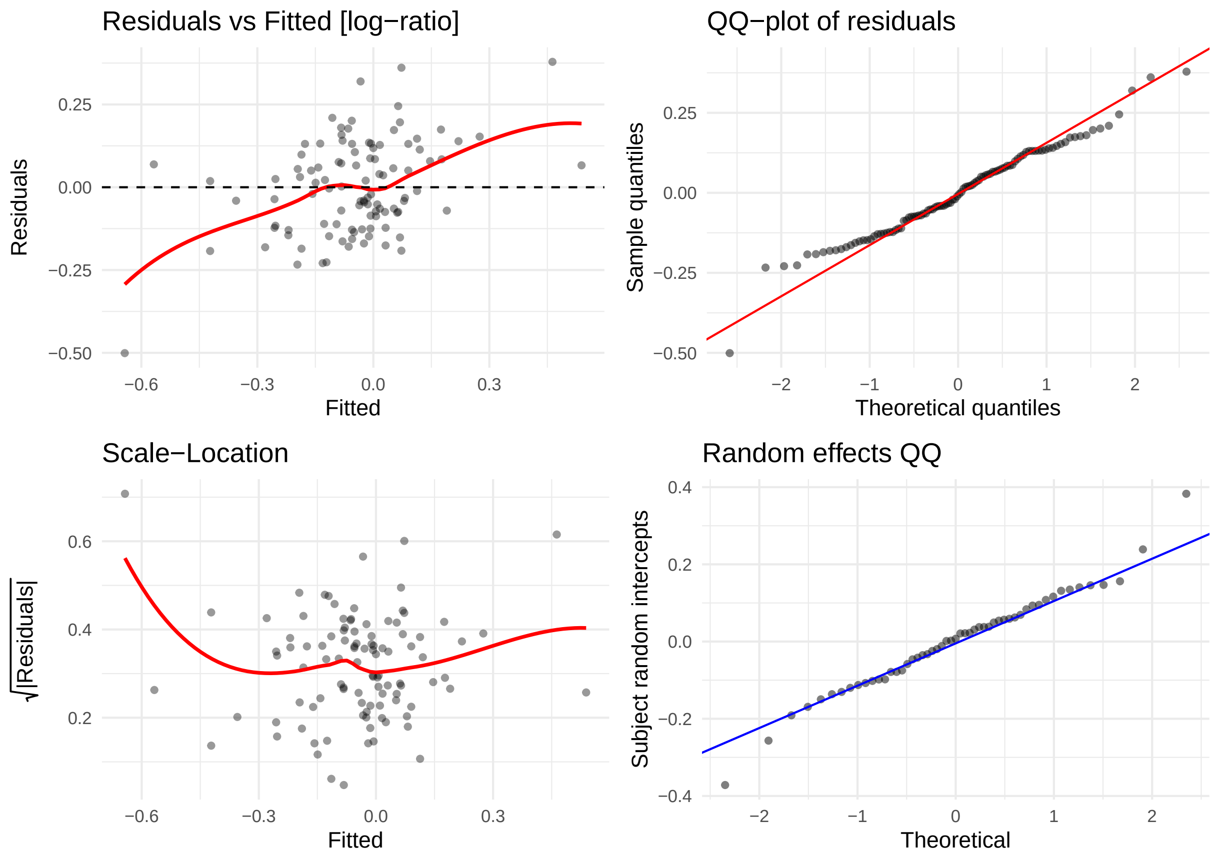
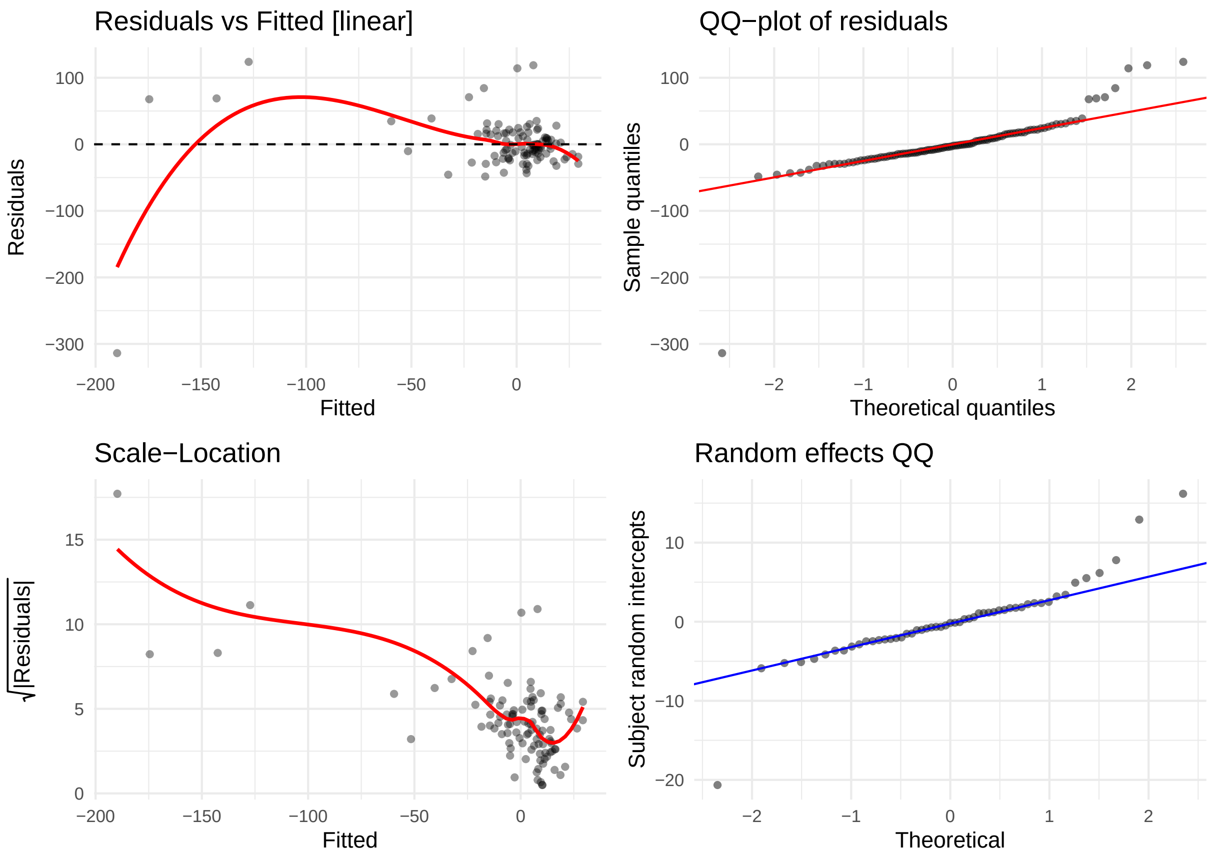


Table 5: Shapiro-Wilk — ANGPTL4

Scale	W	p-value	Interpretation
linear	0.9593	0.0032	Strong departure from normality
log-ratio	0.9791	0.1060	No evidence of non-normality

5.4 CA9 (`CA9`)

AIC linear: 1073.9 | **AIC log-ratio:** 233.3 | **Preferred: log**


Residuals vs Fitted [linear]
Residuals
Fitted
QQ−plot of residuals
Sample quantiles
Theoretical quantiles
Scale−Location
√|Residuals|
Fitted
Random effects QQ
Subject random intercepts
Theoretical

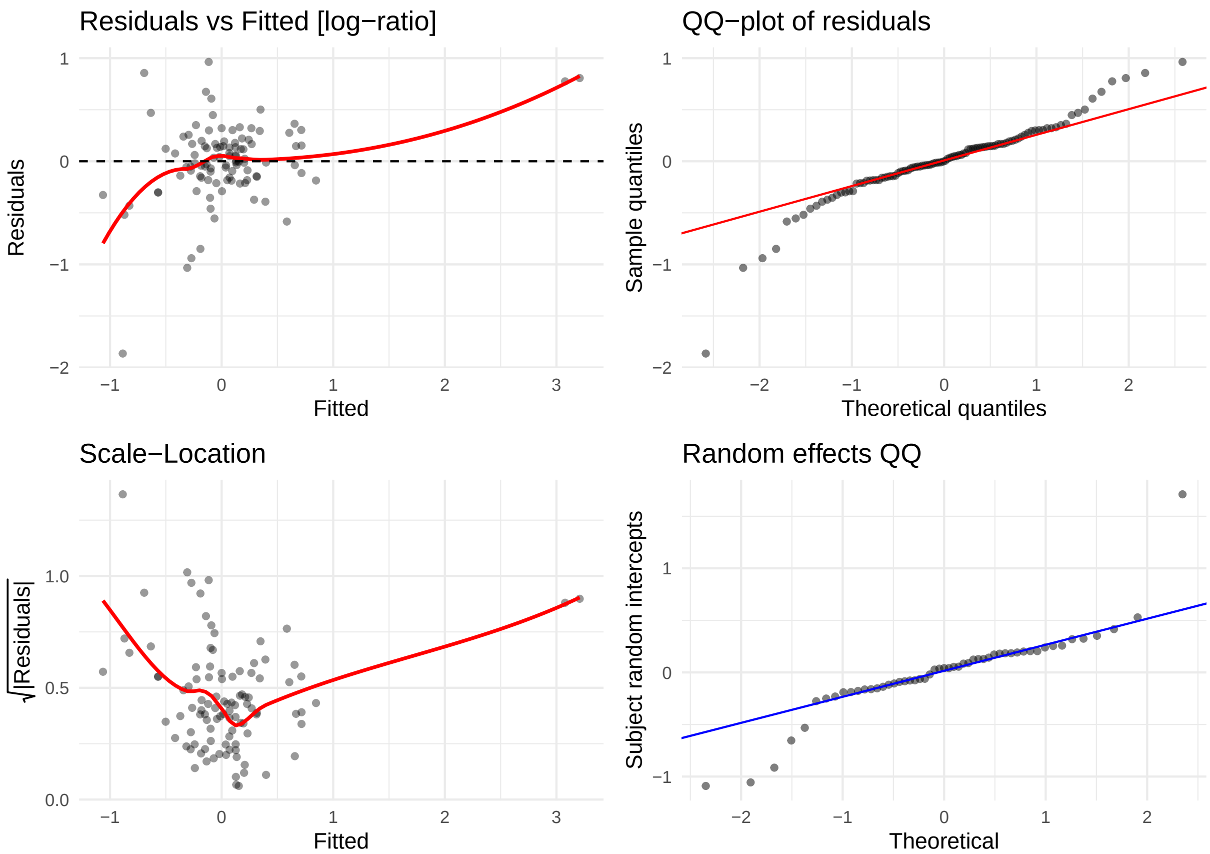
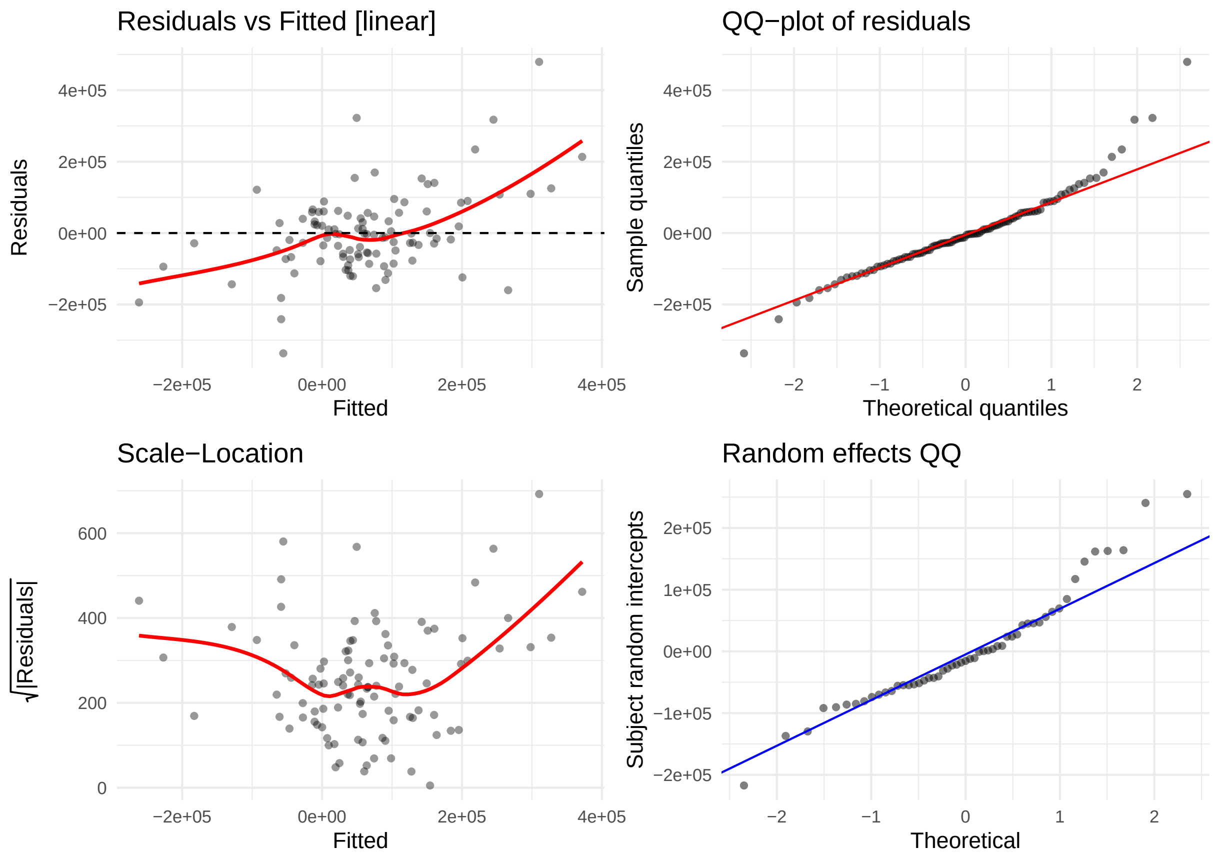


Table 6: Shapiro-Wilk — CA9

Scale	W	p-value	Interpretation
linear	0.6961	0	Strong departure from normality
log-ratio	0.9127	0	Strong departure from normality

5.5 CD163 (`CD163`)

AIC linear: 2660.3 | **AIC log-ratio:** 21.8 | **Preferred: log**


Residuals vs Fitted [linear]
QQ−plot of residuals
Scale−Location
Random effects QQ
Residuals
Fitted
Sample quantiles
Theoretical quantiles
√|Residuals|
Subject random intercepts
Theoretical
4e+05
2e+05
0e+00
−2e+05
−2e+05
0e+00
2e+05
4e+05
−2
−1
0
1
2
600
400
200
0
1e+05
−1e+05

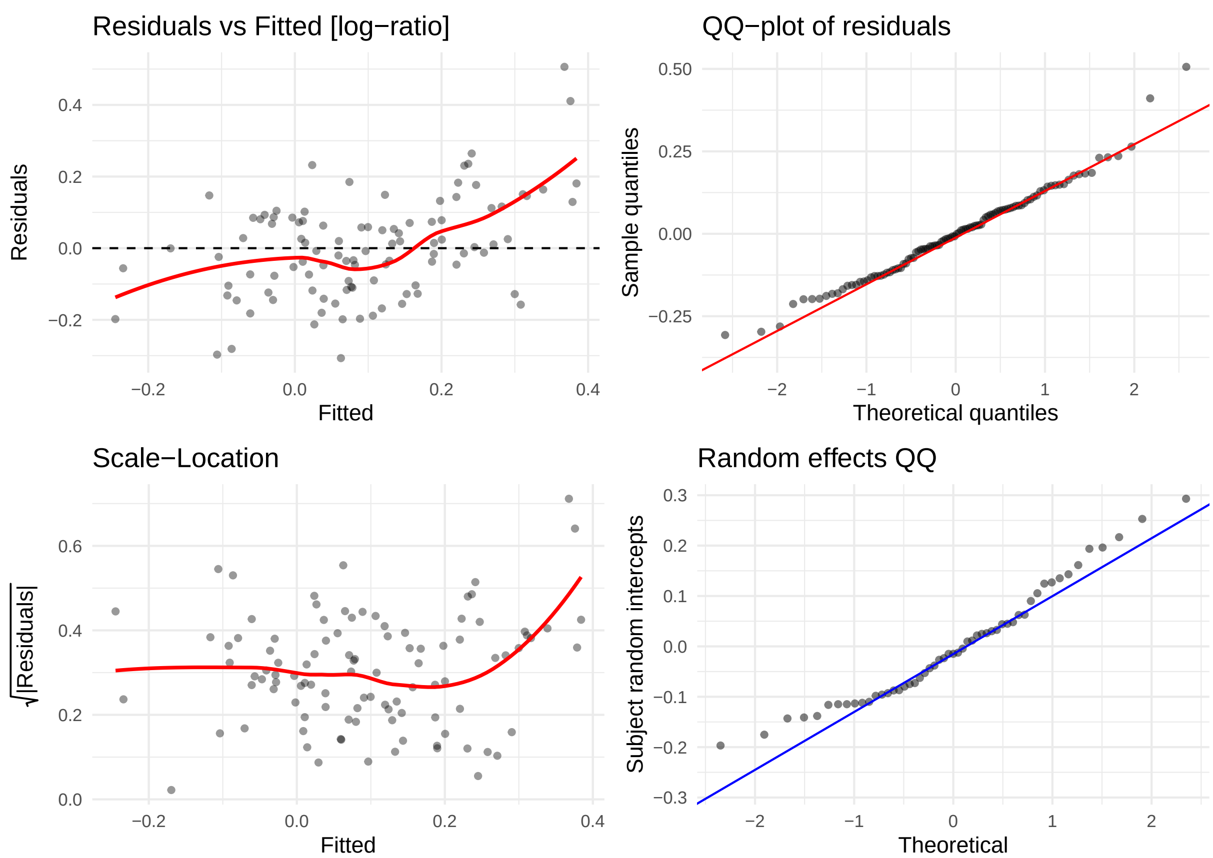
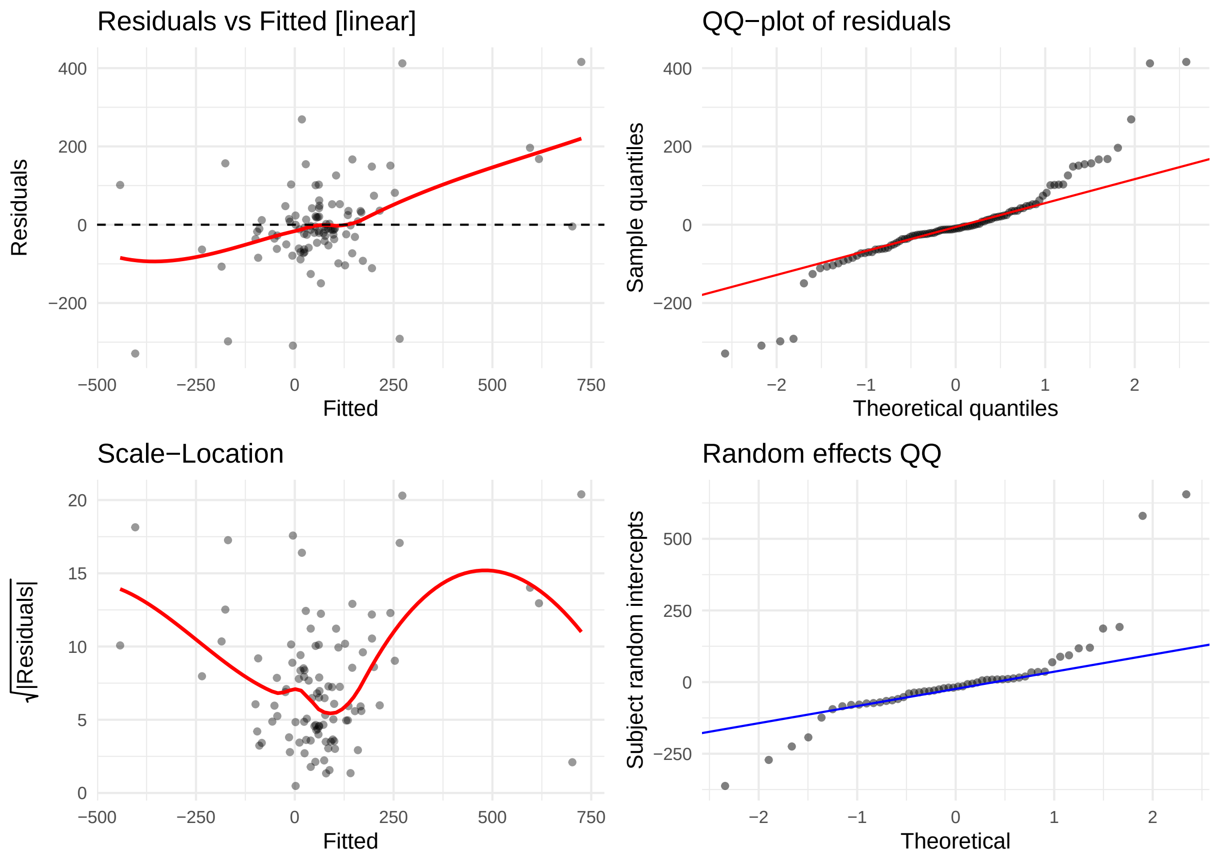


Table 7: Shapiro-Wilk — CD163

Scale	W	p-value	Interpretation
linear	0.9429	0.0003	Strong departure from normality
log-ratio	0.9783	0.0913	Borderline

5.6 Collagen IV alpha 1 (`COLIVA1`)

AIC linear: 1327.4 | **AIC log-ratio:** 10.8 | **Preferred: log**


Residuals vs Fitted [linear]
Residuals
Fitted
QQ−plot of residuals
Sample quantiles
Theoretical quantiles
Scale−Location
$\sqrt{|\text{Residuals}|}$
Fitted
Random effects QQ
Subject random intercepts
Theoretical

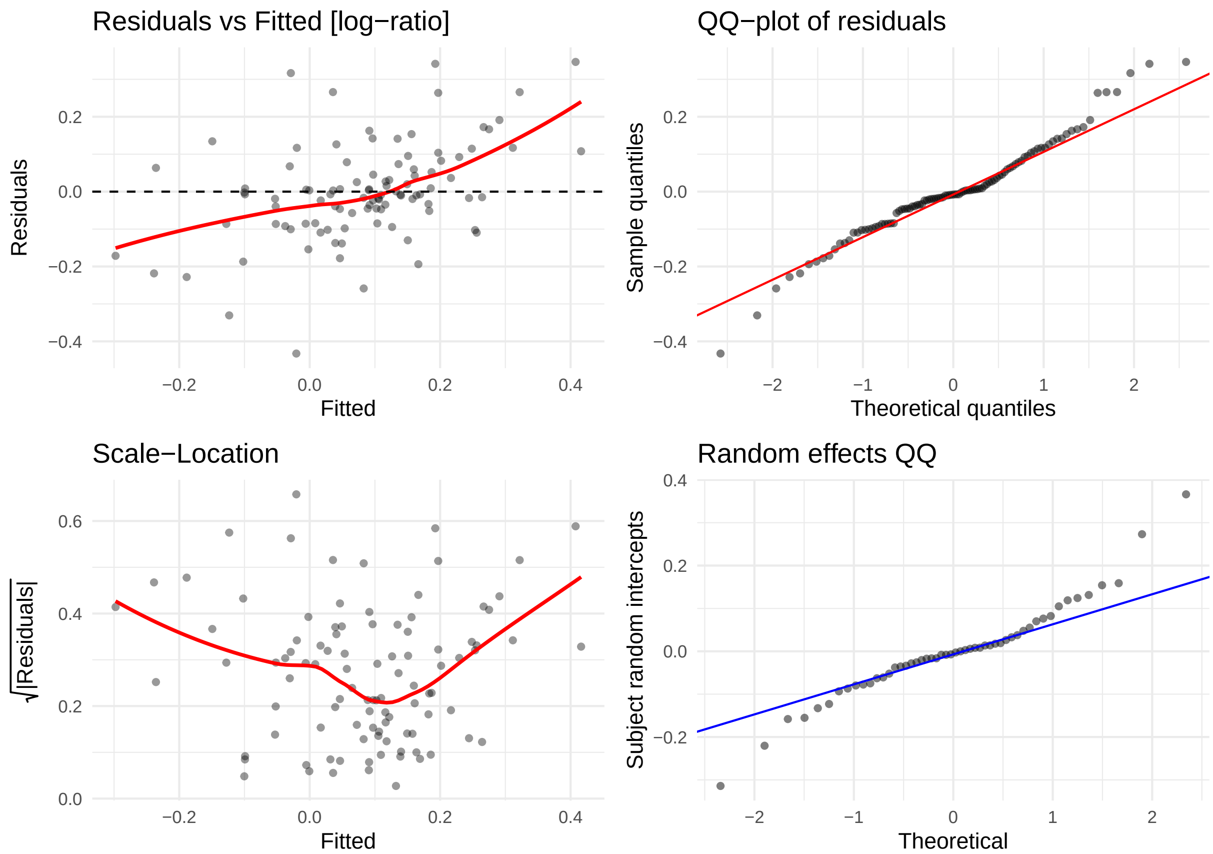
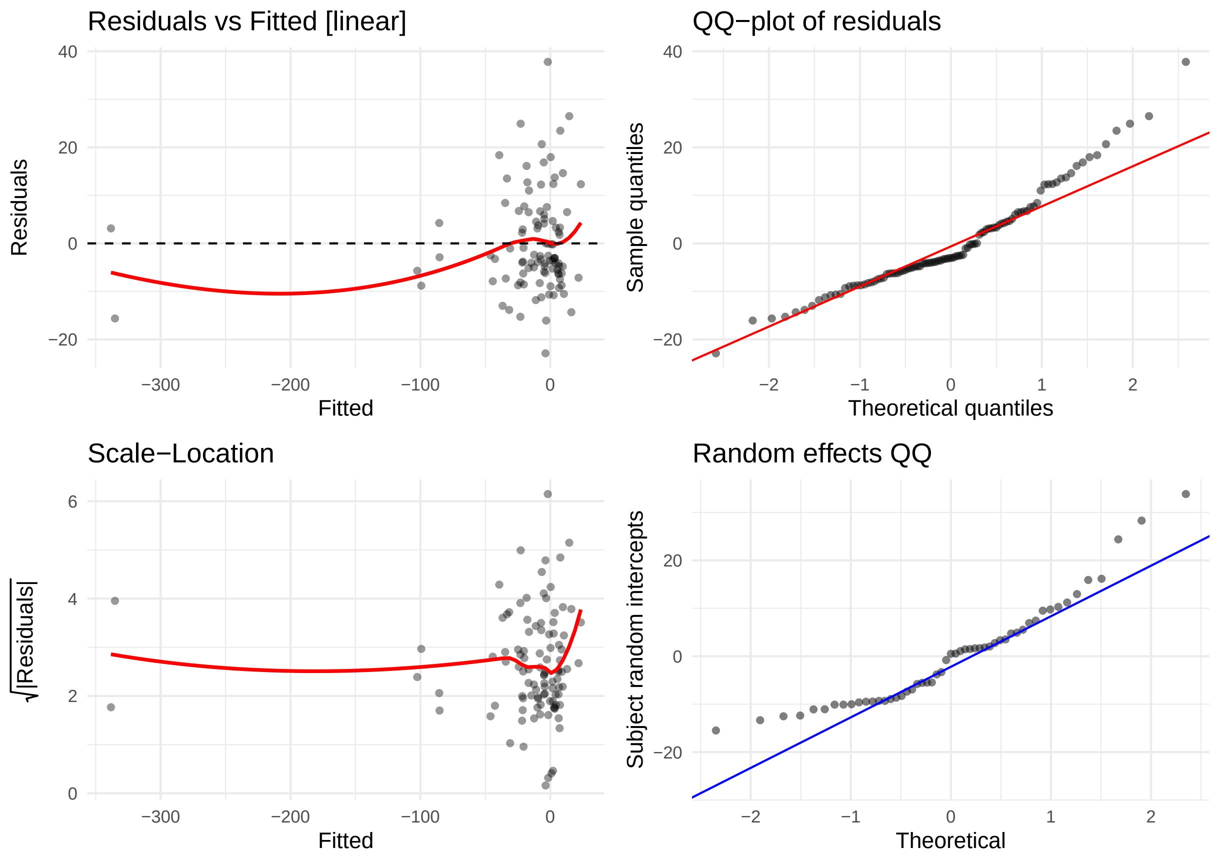


Table 8: Shapiro-Wilk — Collagen IV alpha 1

Scale	W	p-value	Interpretation
linear	0.8866	0.0000	Strong departure from normality
log-ratio	0.9706	0.0244	Some departure from normality

5.7 EGF (`EGF`)

AIC linear: 871.8 | **AIC log-ratio:** 432.7 | **Preferred: log**



Residuals vs Fitted [linear]
Residuals
Fitted
QQ−plot of residuals
Sample quantiles
Theoretical quantiles
Scale−Location
√|Residuals|
Random effects QQ
Subject random intercepts
Theoretical

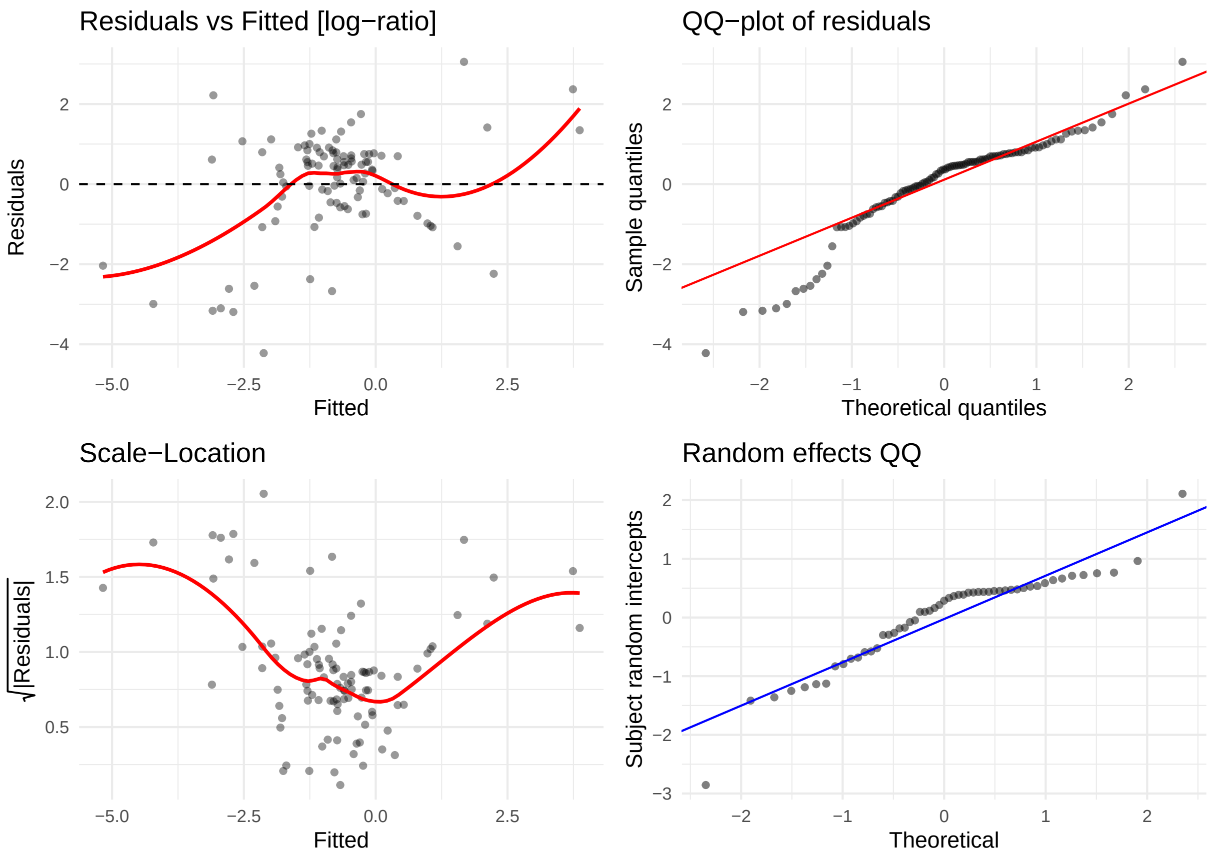
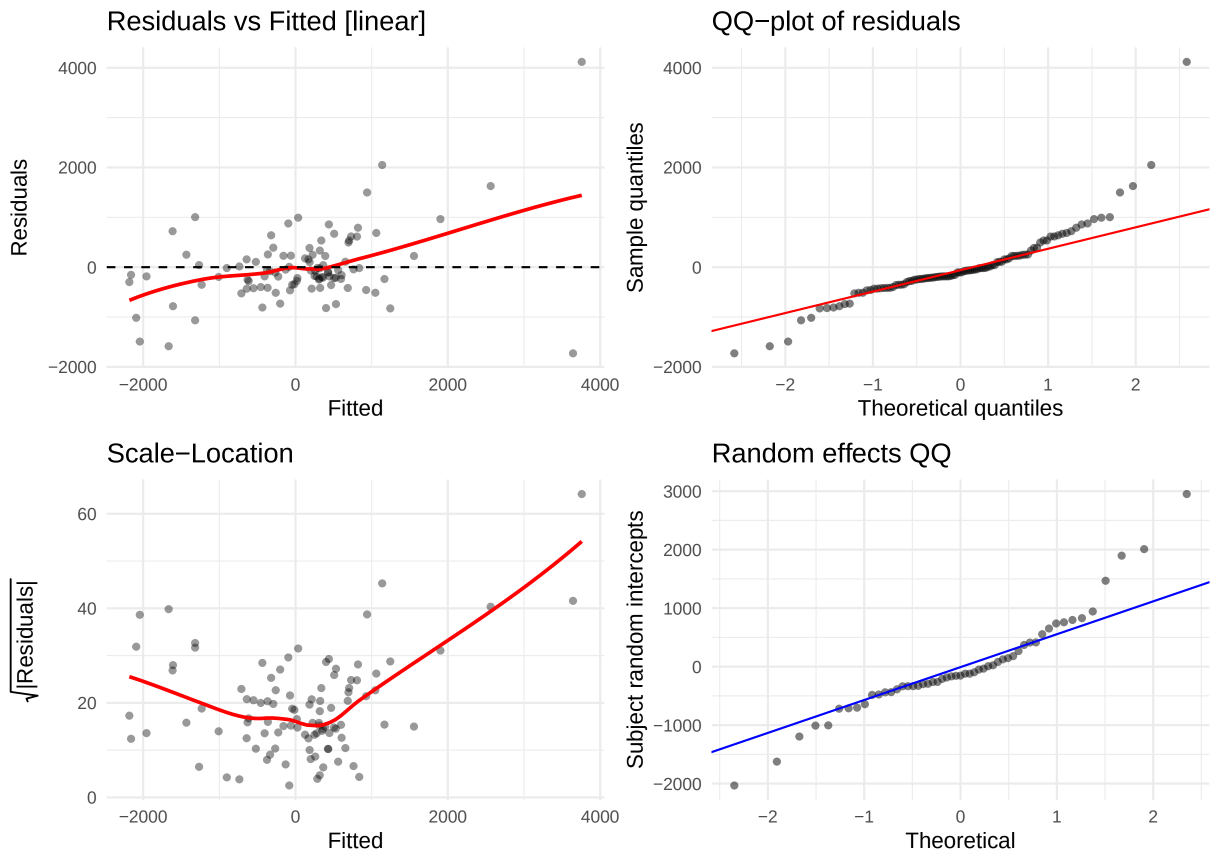


Table 9: Shapiro-Wilk — EGF

Scale	W	p-value	Interpretation
linear	0.9441	3e-04	Strong departure from normality
log-ratio	0.9149	0e+00	Strong departure from normality

5.8 ErbB2 (`ERBB2`)

AIC linear: 1703.6 | **AIC log-ratio:** -3.4 | **Preferred: log**



Residuals vs Fitted [linear]
Residuals
Fitted
QQ−plot of residuals
Sample quantiles
Theoretical quantiles
Scale−Location
√|Residuals|
Fitted
Random effects QQ
Subject random intercepts
Theoretical

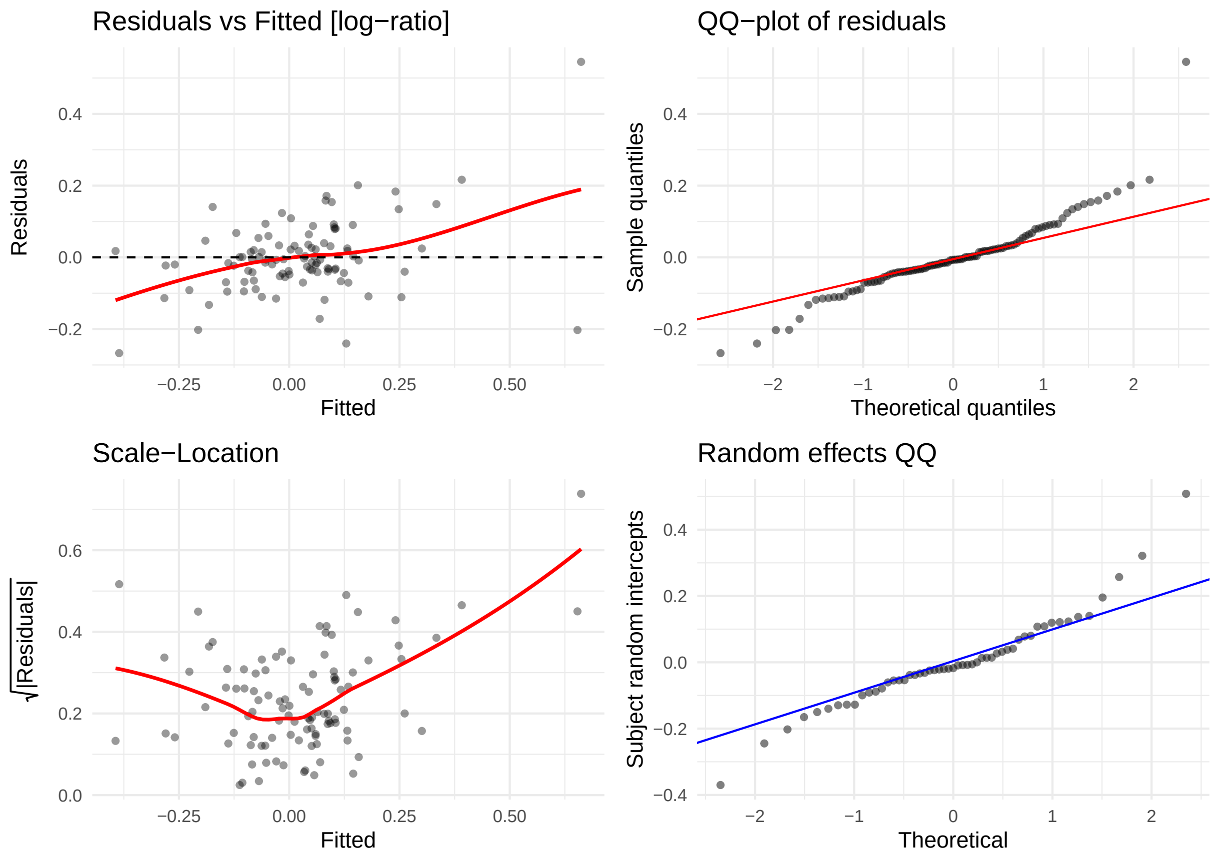
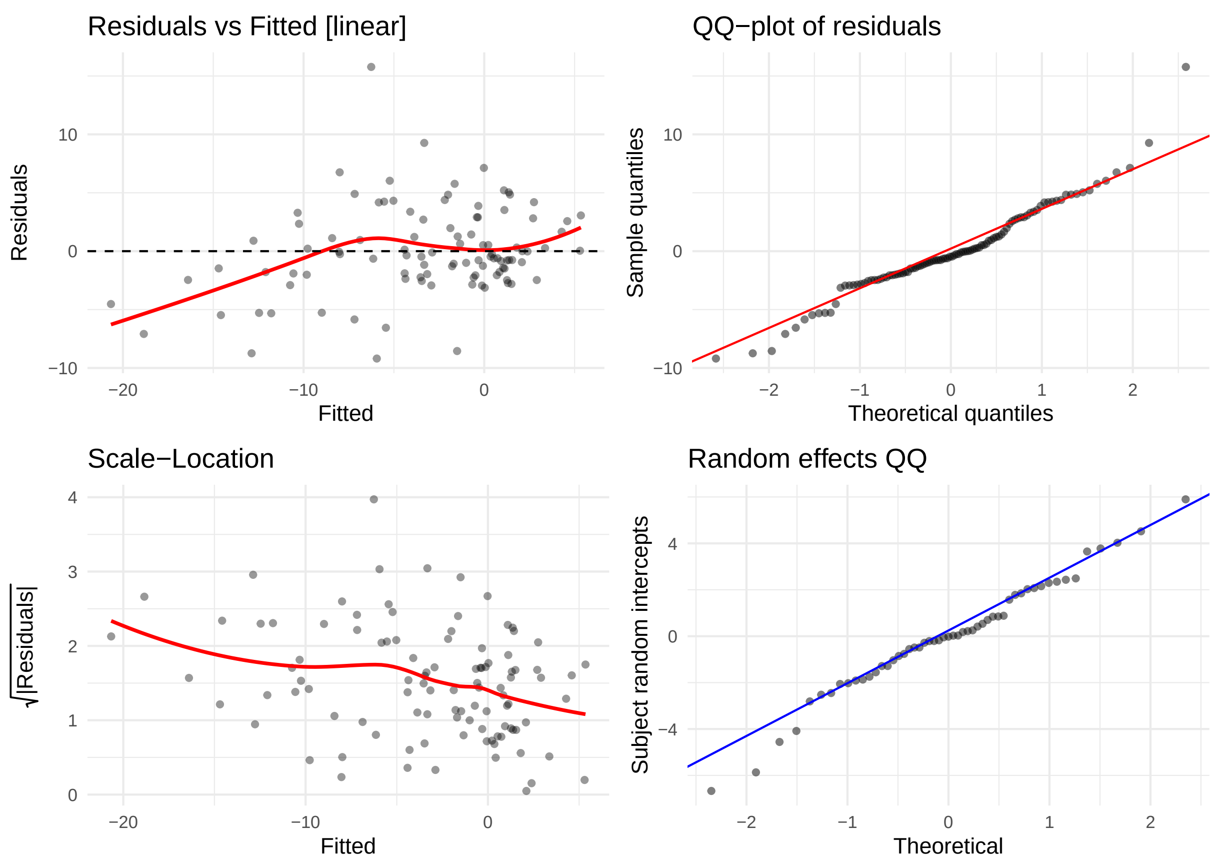


Table 10: Shapiro-Wilk — ErbB2

Scale	W	p-value	Interpretation
linear	0.8490	0	Strong departure from normality
log-ratio	0.9049	0	Strong departure from normality

5.9 FGF Basic (`FGFB`)

AIC linear: 648.8 | **AIC log-ratio:** 209.3 | **Preferred: log**


Residuals vs Fitted [linear]
Residuals
Fitted
QQ−plot of residuals
Sample quantiles
Theoretical quantiles
Scale−Location
√|Residuals|
Fitted
Random effects QQ
Subject random intercepts
Theoretical

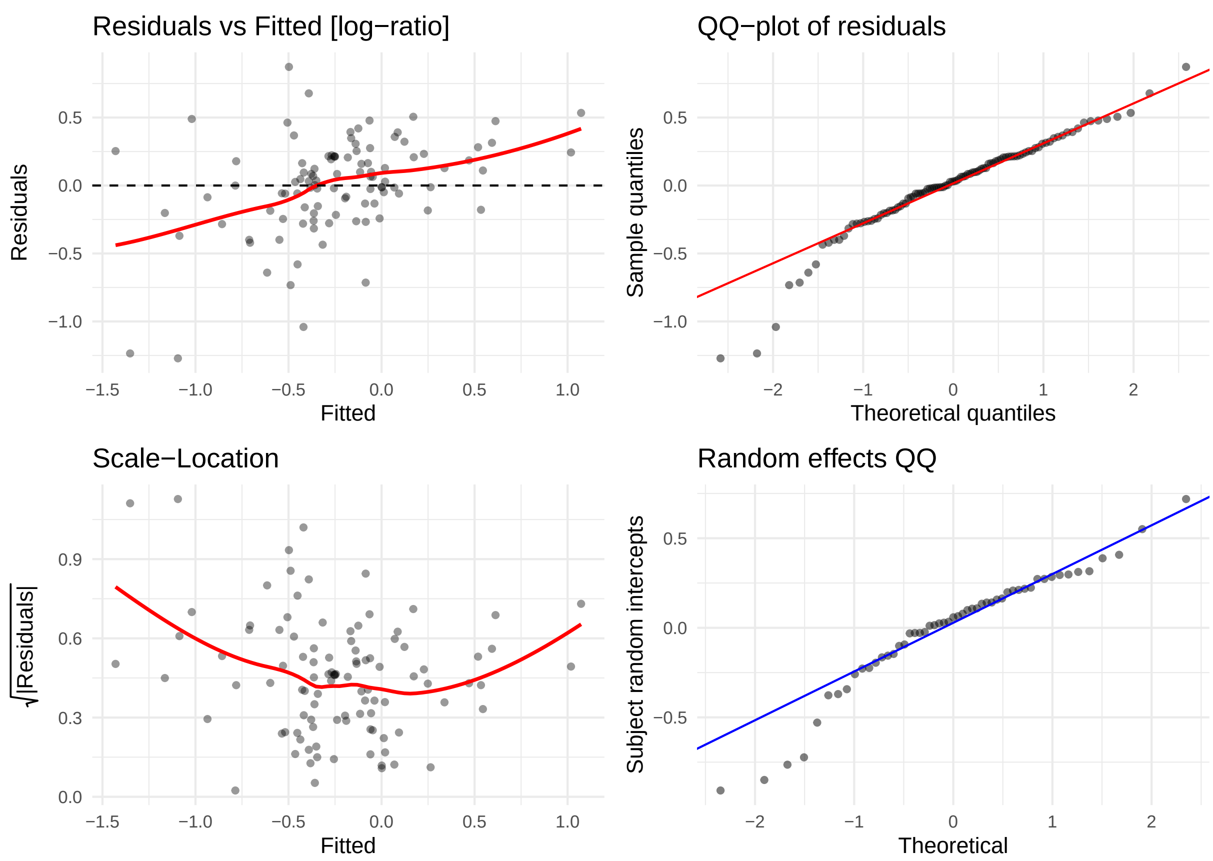
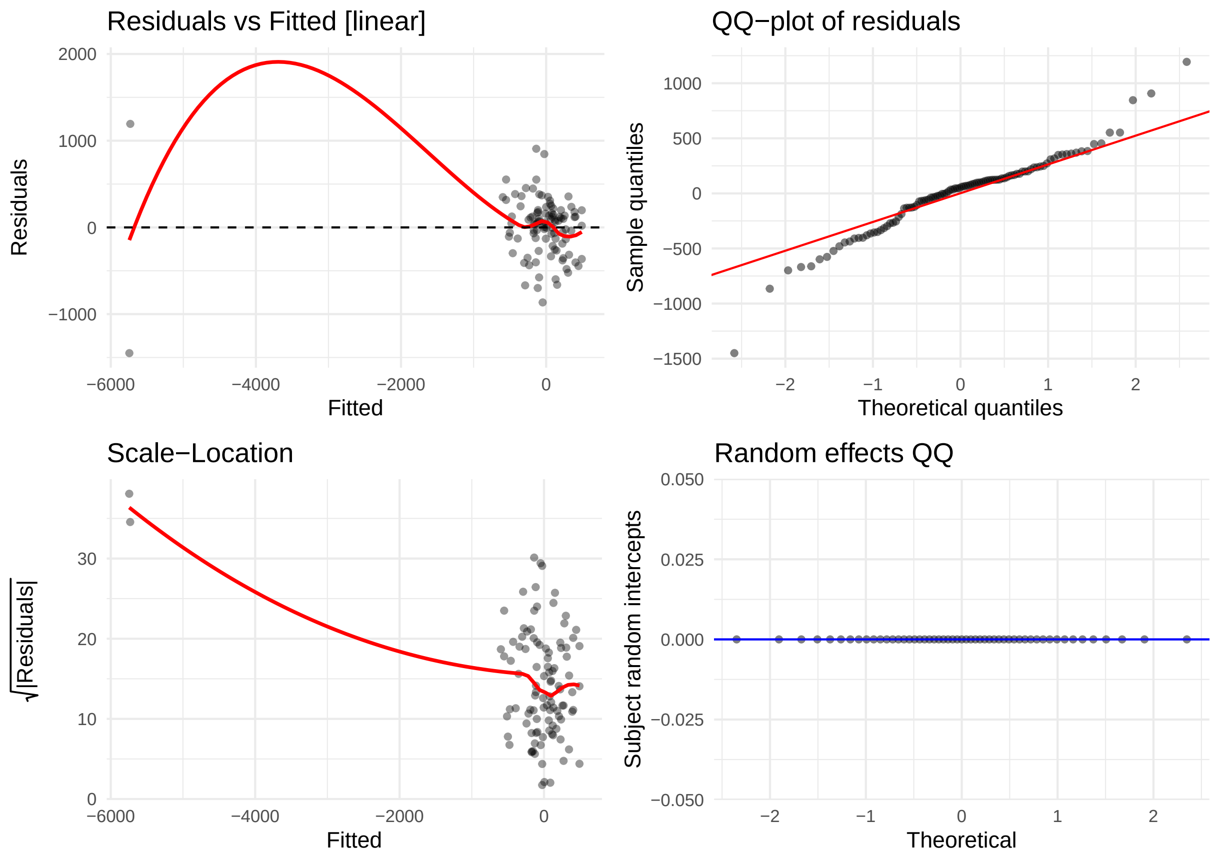


Table 11: Shapiro-Wilk — FGF Basic

Scale	W	p-value	Interpretation
linear	0.9594	0.0032	Strong departure from normality
log-ratio	0.9391	0.0001	Strong departure from normality

5.10 Fractalkine (`FRACTKN`)

AIC linear: 1468.3 | **AIC log-ratio:** 223.3 | **Preferred: log**


Residuals vs Fitted [linear]
2000
1000
0
−1000
Residuals
−6000
−4000
−2000
0
Fitted
QQ−plot of residuals
1000
500
0
−500
−1000
−1500
Sample quantiles
−2
−1
0
1
2
Theoretical quantiles
Scale−Location
30
20
10
0
√|Residuals|
−6000
−4000
−2000
0
Fitted
Random effects QQ
0.050
0.025
0.000
−0.025
−0.050
Subject random intercepts
−2
−1
0
1
2
Theoretical

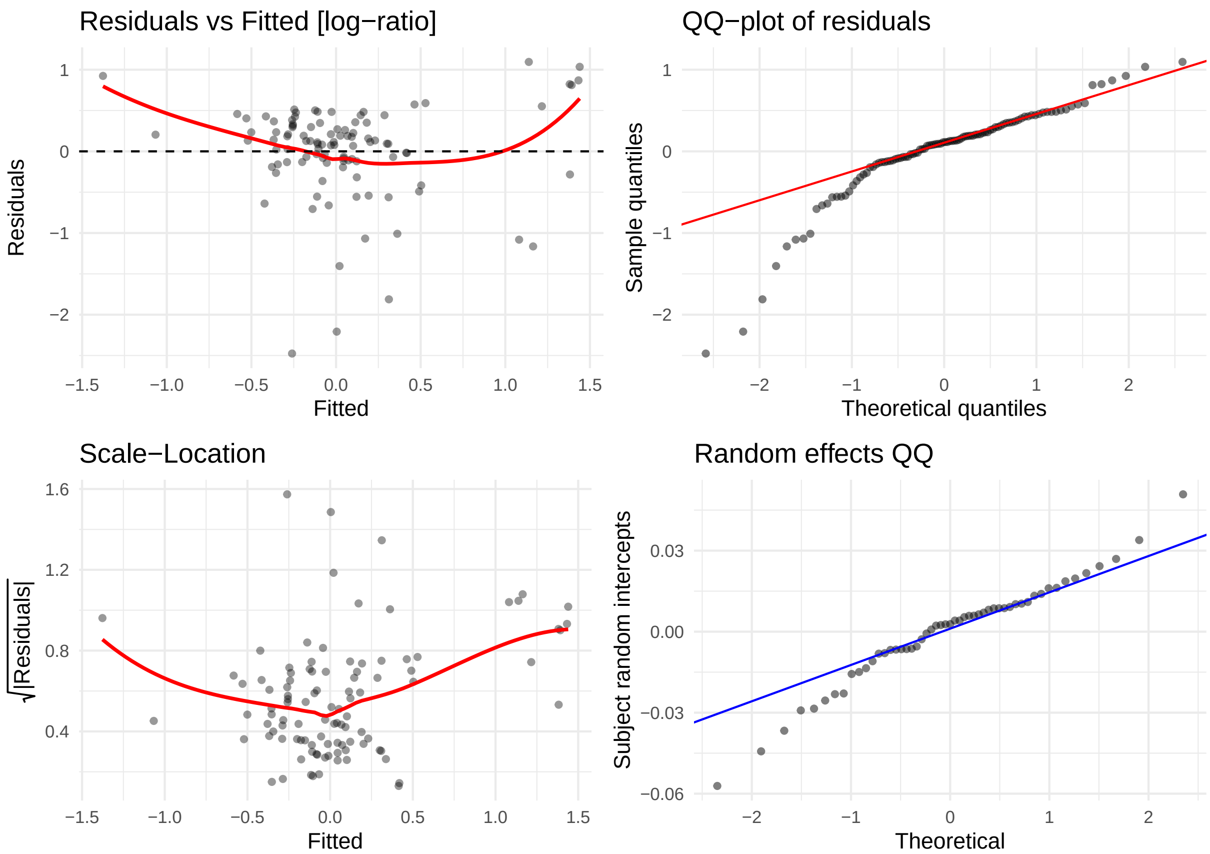
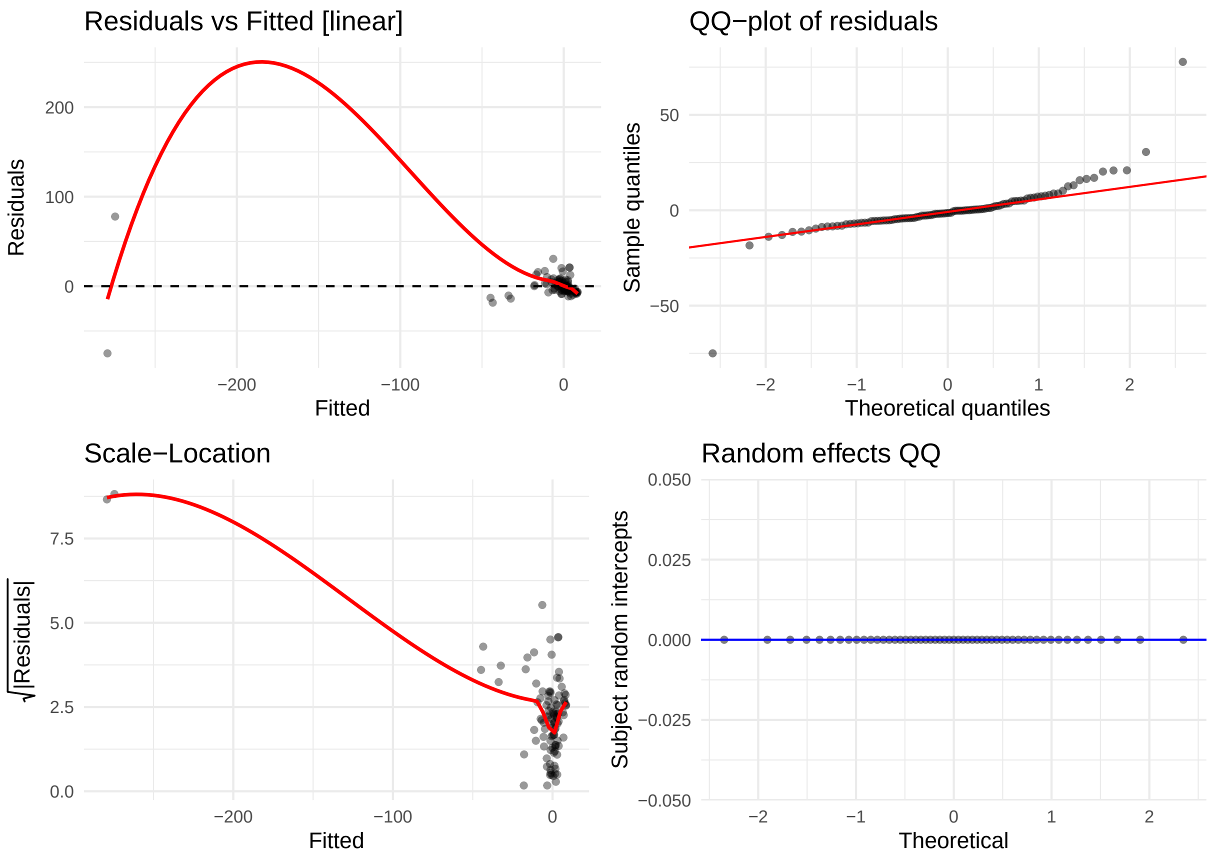


Table 12: Shapiro-Wilk — Fractalkine

Scale	W	p-value	Interpretation
linear	0.9518	0.001	Strong departure from normality
log-ratio	0.8659	0.000	Strong departure from normality

5.11 Granzyme B (GZMB)

AIC linear: 826.6 | **AIC log-ratio:** 161.7 | **Preferred: log**



Residuals vs Fitted [linear]
Residuals
Fitted
200
100
0
−200
−100
0
QQ−plot of residuals
Sample quantiles
Theoretical quantiles
50
0
−50
−2
−1
0
1
2
Scale−Location
√|Residuals|
Fitted
7.5
5.0
2.5
0.0
−200
−100
0
Random effects QQ
Subject random intercepts
Theoretical
0.050
0.025
0.000
−0.025
−0.050
−2
−1
0
1
2

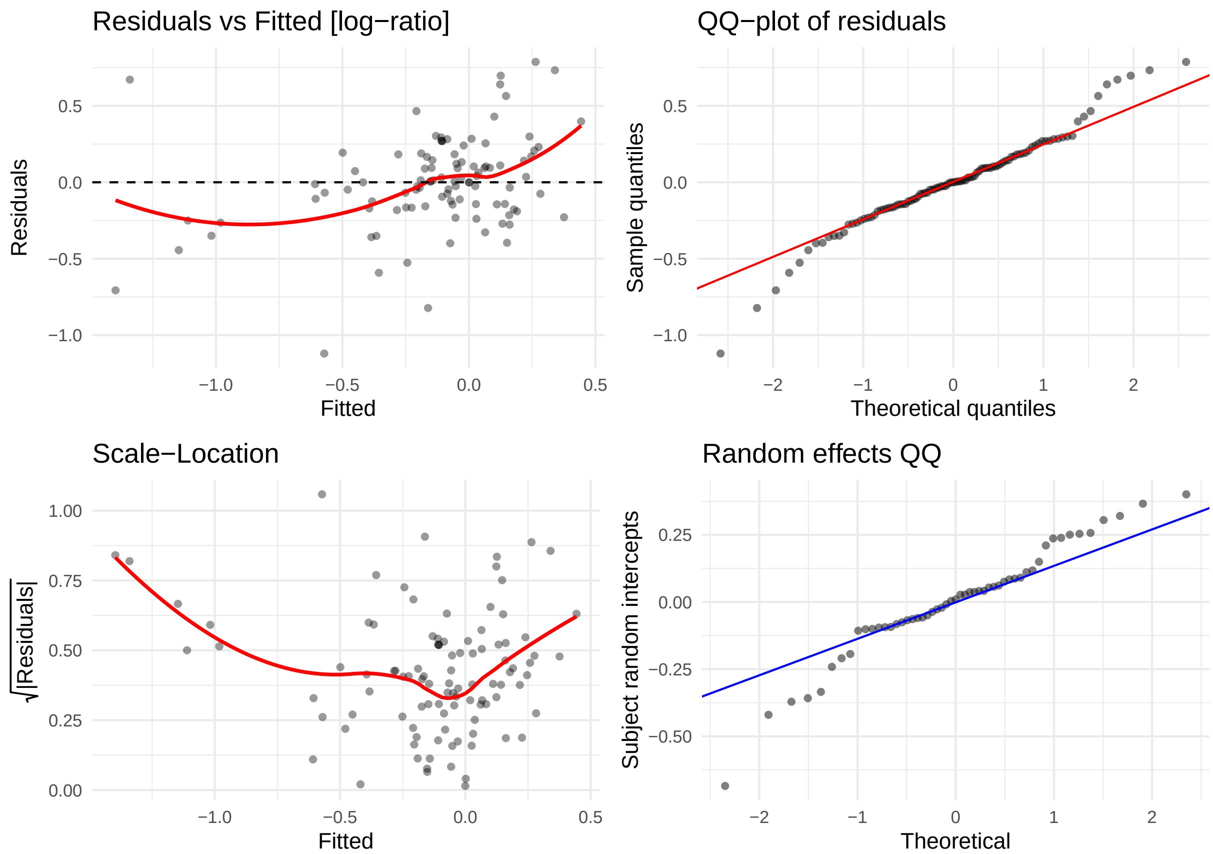
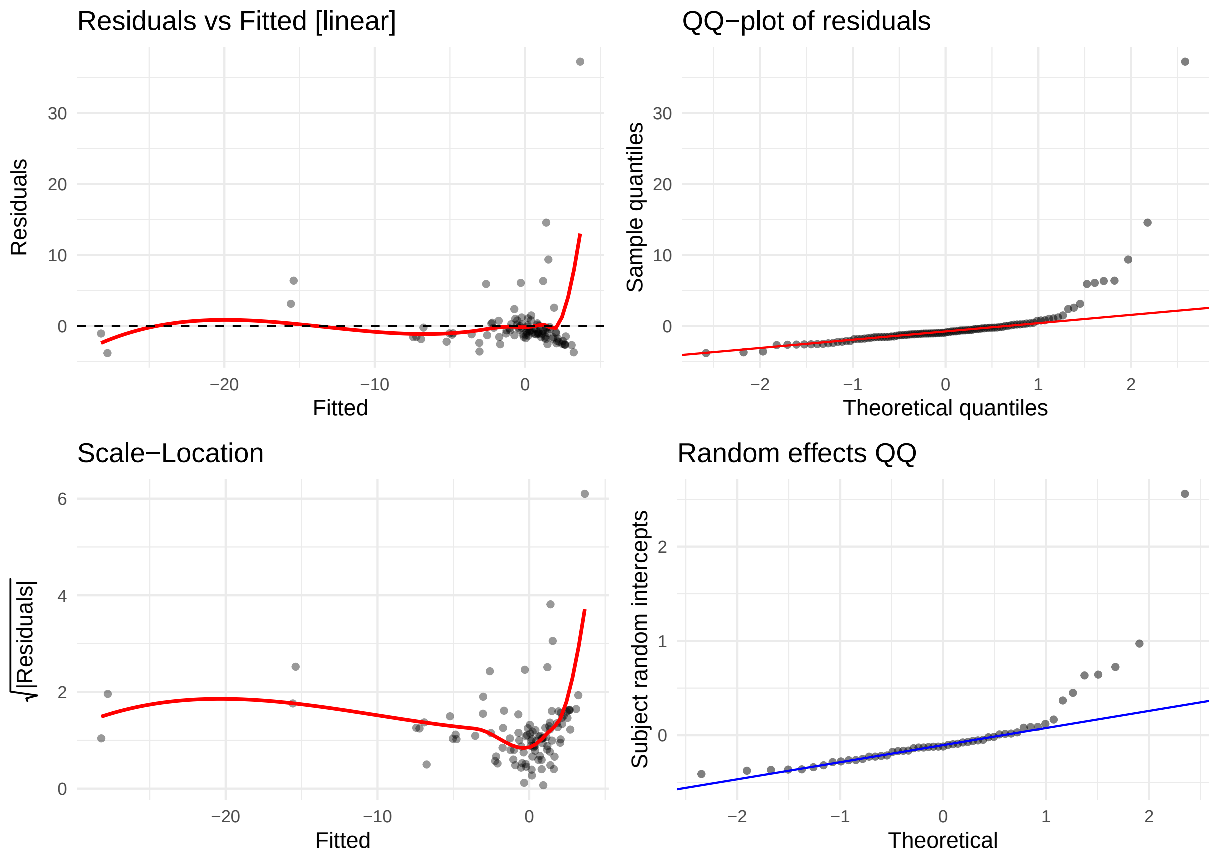


Table 13: Shapiro-Wilk — Granzyme B

Scale	W	p-value	Interpretation
linear	0.6998	0.0000	Strong departure from normality
log-ratio	0.9662	0.0102	Some departure from normality

5.12 IFN-g (IFNG)

AIC linear: 626.2 | **AIC log-ratio:** 272 | **Preferred: log**


Residuals vs Fitted [linear]
Residuals
Fitted
QQ−plot of residuals
Sample quantiles
Theoretical quantiles
Scale−Location
√|Residuals|
Fitted
Random effects QQ
Subject random intercepts
Theoretical

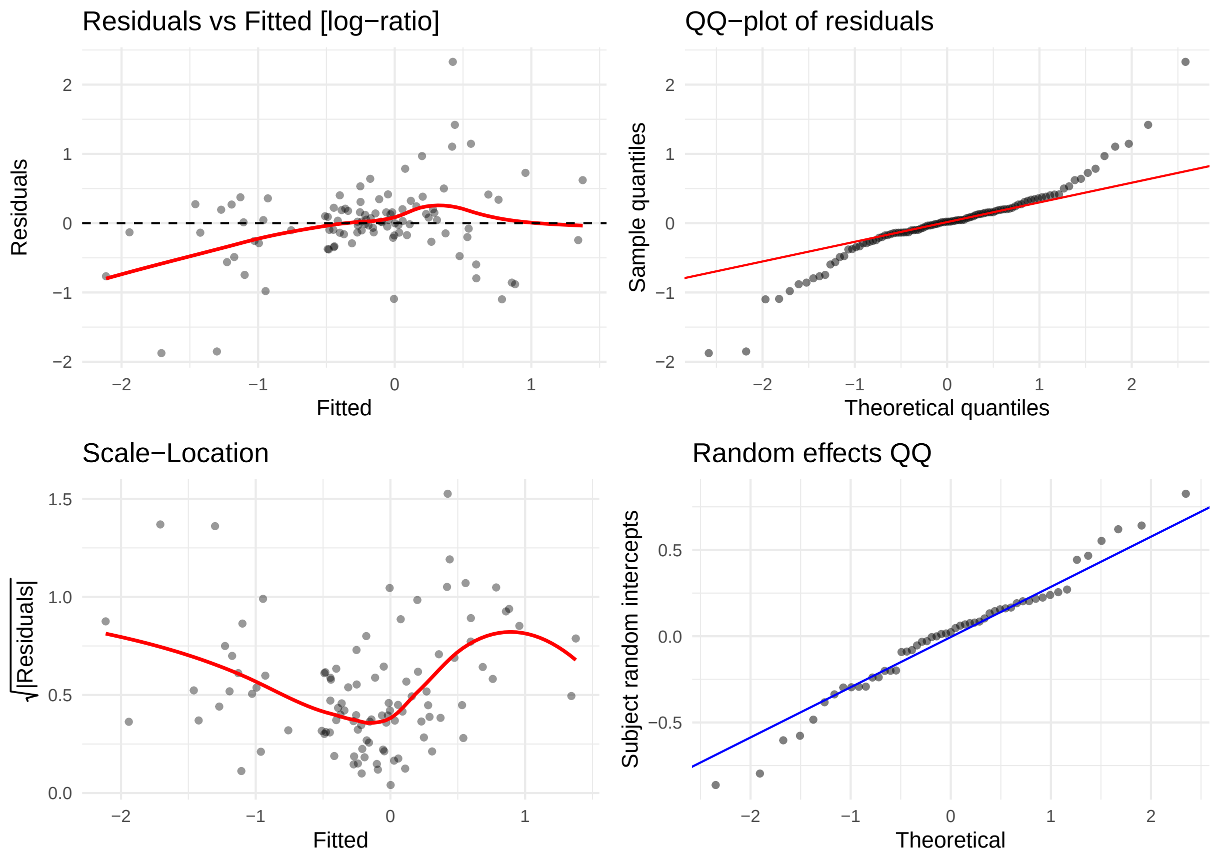
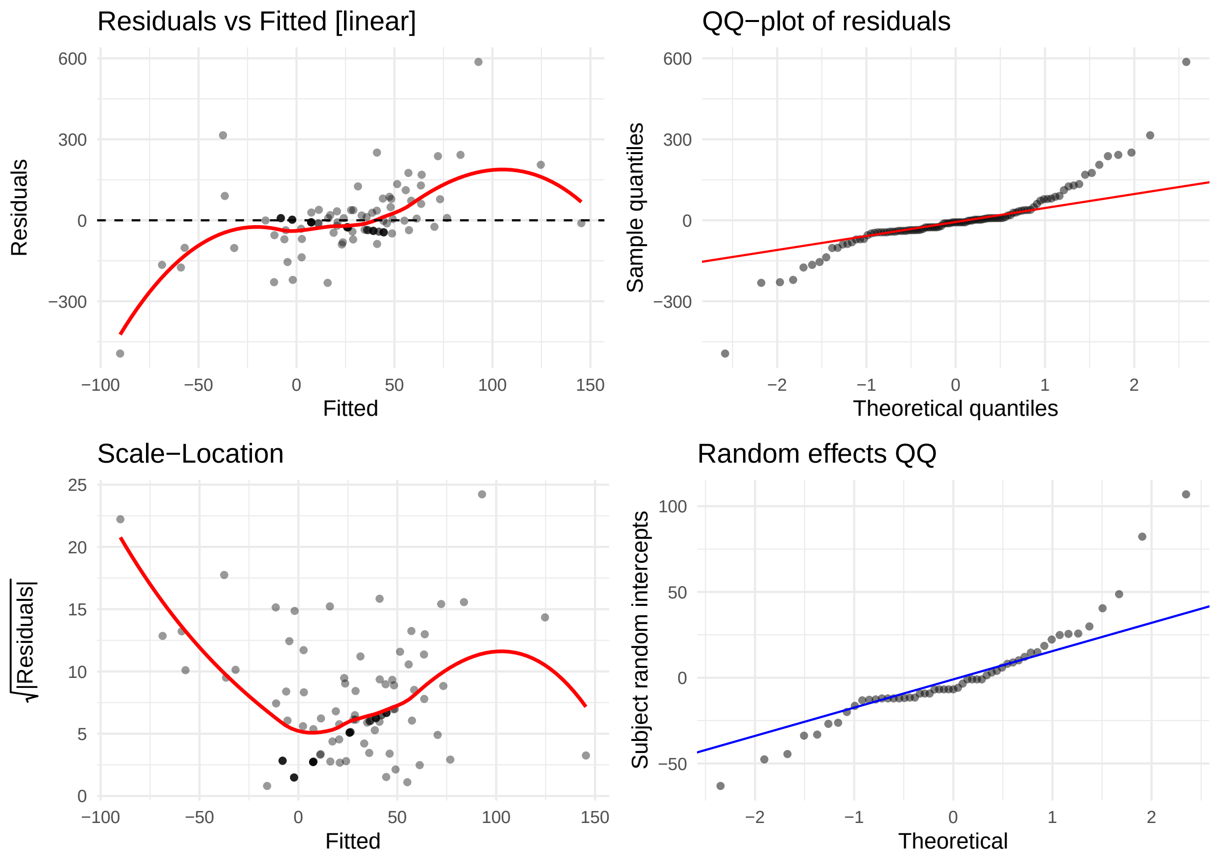


Table 14: Shapiro-Wilk — IFN-g

Scale	W	p-value	Interpretation
linear	0.4426	0	Strong departure from normality
log-ratio	0.9085	0	Strong departure from normality

5.13 IL-12 p70 (IL12P70)

AIC linear: 1279.5 | **AIC log-ratio:** 245.6 | **Preferred: log**



Residuals vs Fitted [linear]
Residuals
Fitted
QQ−plot of residuals
Sample quantiles
Theoretical quantiles
Scale−Location
√|Residuals|
Fitted
Random effects QQ
Subject random intercepts
Theoretical

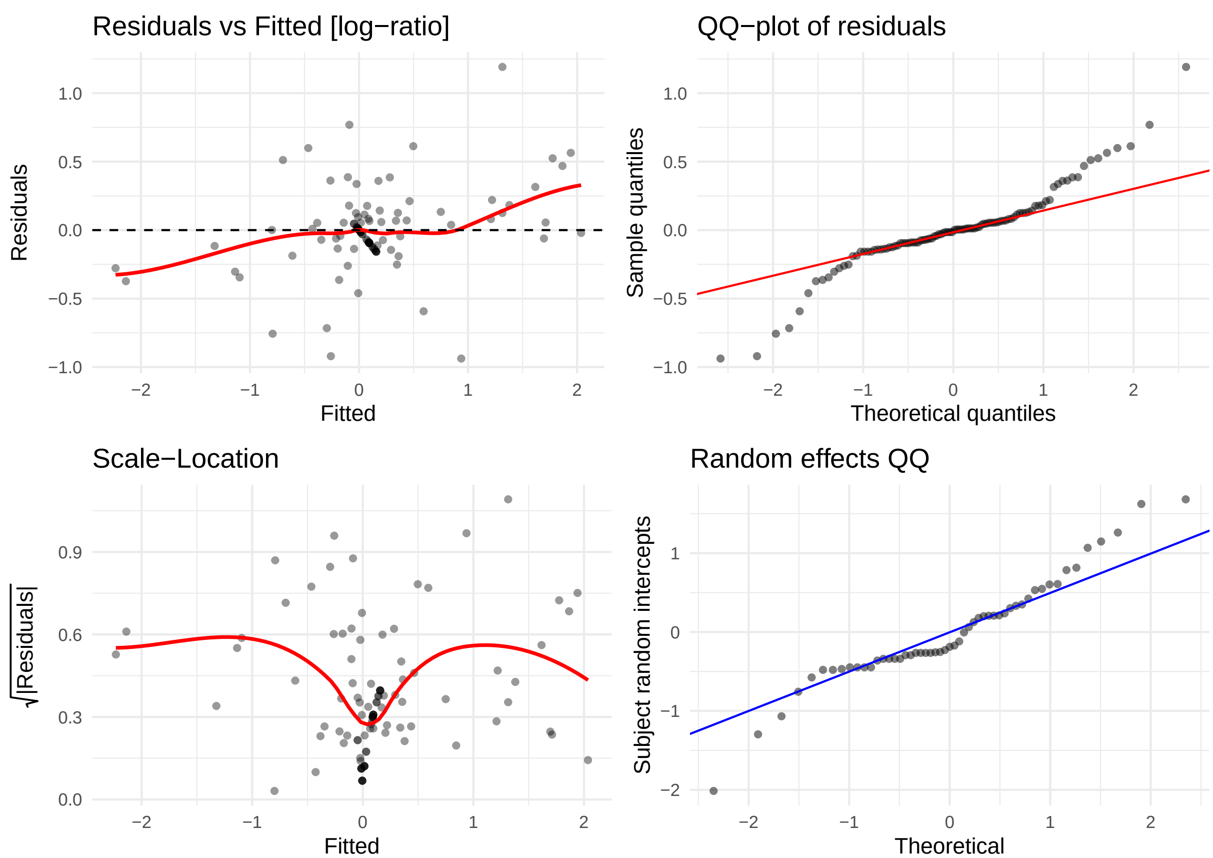
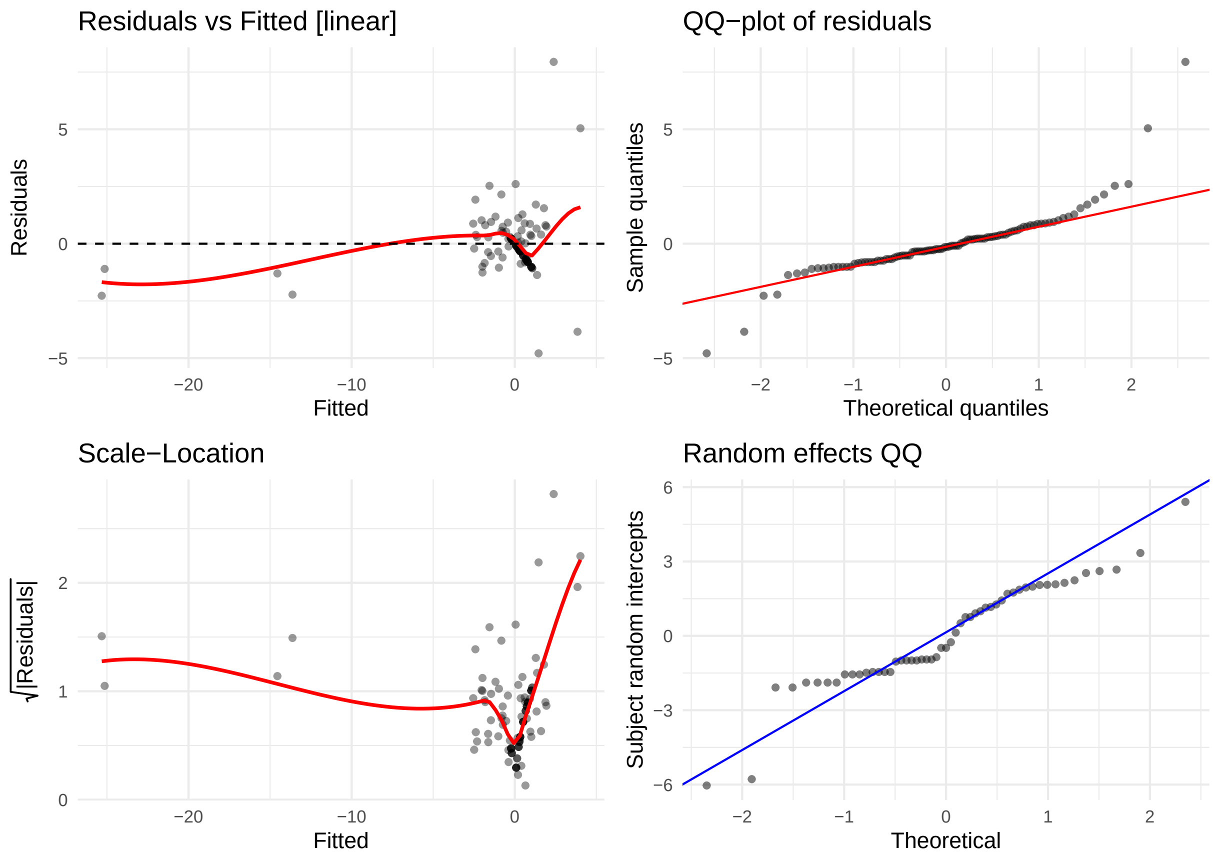


Table 15: Shapiro-Wilk — IL-12 p70

Scale	W	p-value	Interpretation
linear	0.8407	0	Strong departure from normality
log-ratio	0.9124	0	Strong departure from normality

5.14 IL-1b (IL1B)

AIC linear: 511.2 | **AIC log-ratio:** 288 | **Preferred: log**



Residuals vs Fitted [linear]
Residuals
Fitted
QQ−plot of residuals
Sample quantiles
Theoretical quantiles
Scale−Location
√|Residuals|
Fitted
Random effects QQ
Subject random intercepts
Theoretical

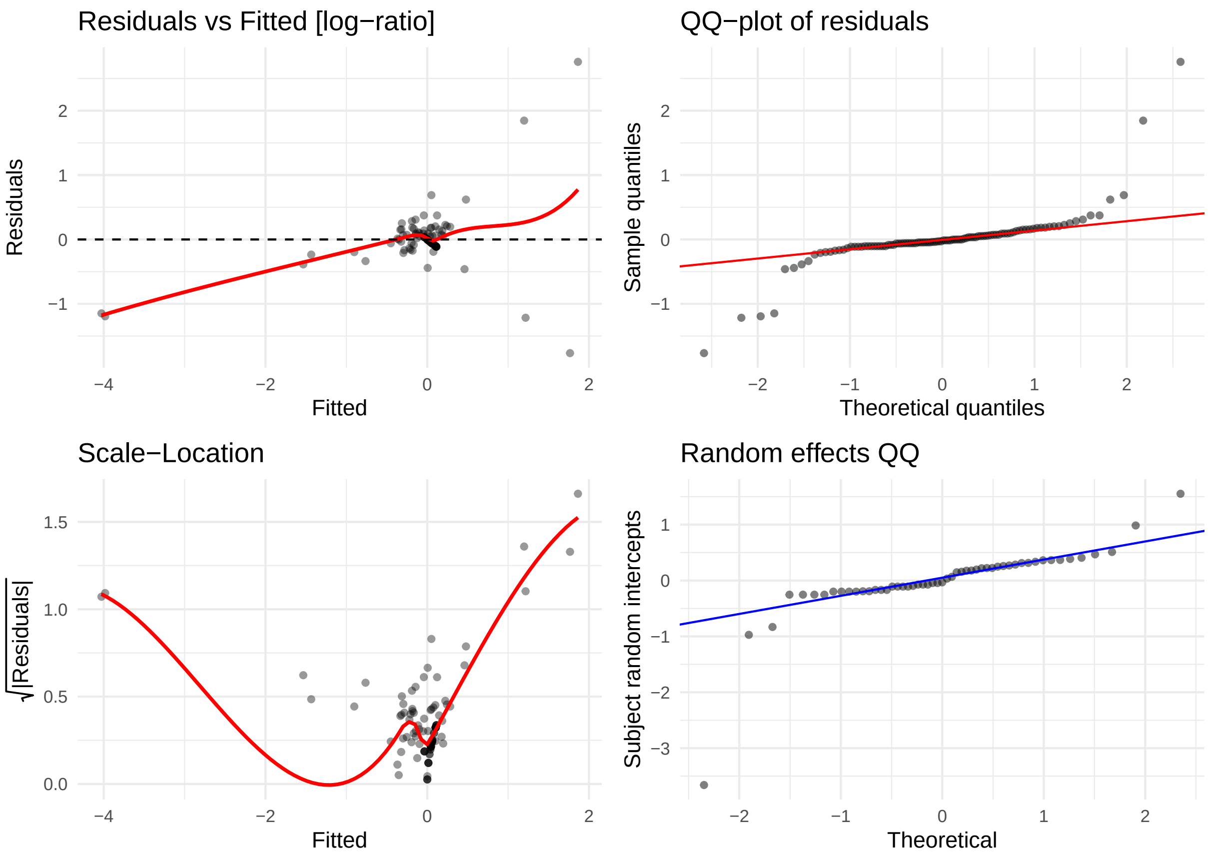
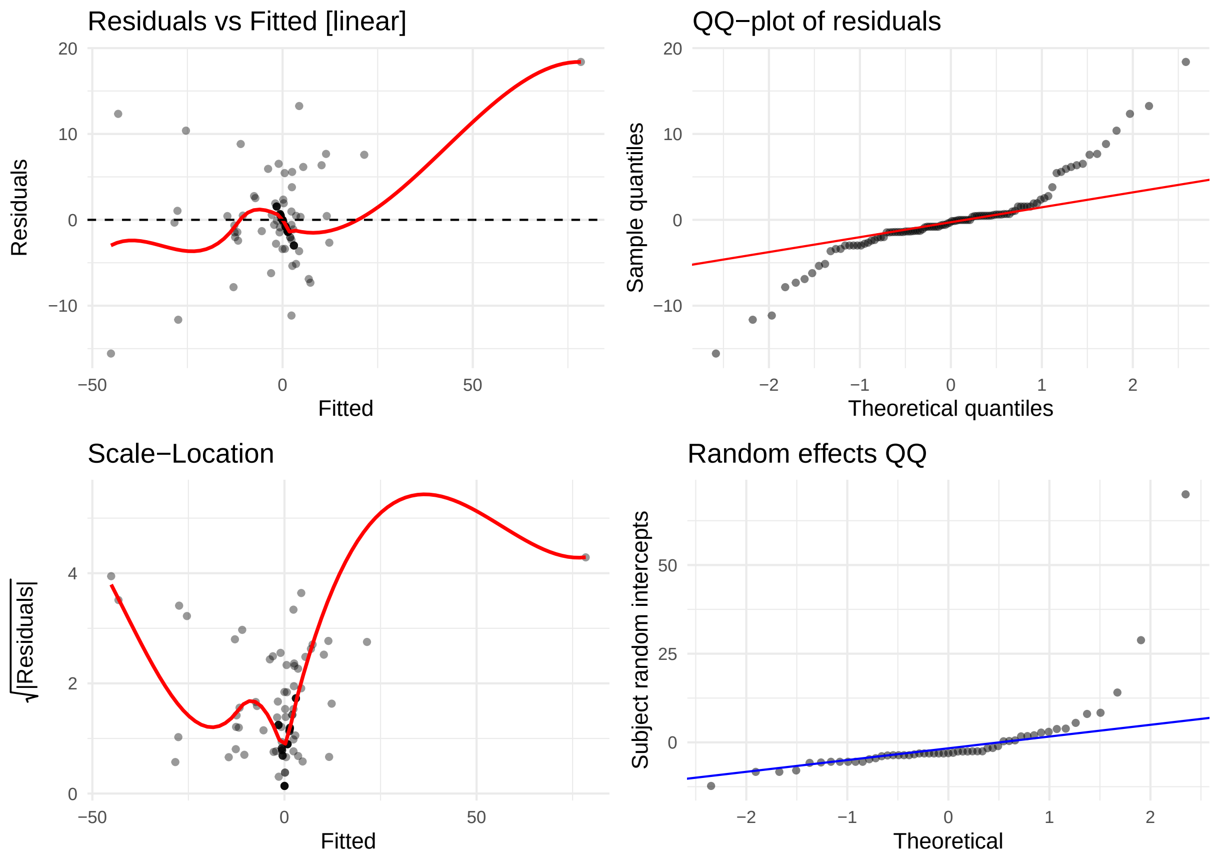


Table 16: Shapiro-Wilk — IL-1b

Scale	W	p-value	Interpretation
linear	0.8121	0	Strong departure from normality
log-ratio	0.6373	0	Strong departure from normality

5.15 IL-2 (`IL2`)

AIC linear: 783 | **AIC log-ratio:** 356.8 | **Preferred: log**



Residuals vs Fitted [linear]
Residuals
Fitted
QQ−plot of residuals
Sample quantiles
Theoretical quantiles
Scale−Location
√|Residuals|
Fitted
Random effects QQ
Subject random intercepts
Theoretical

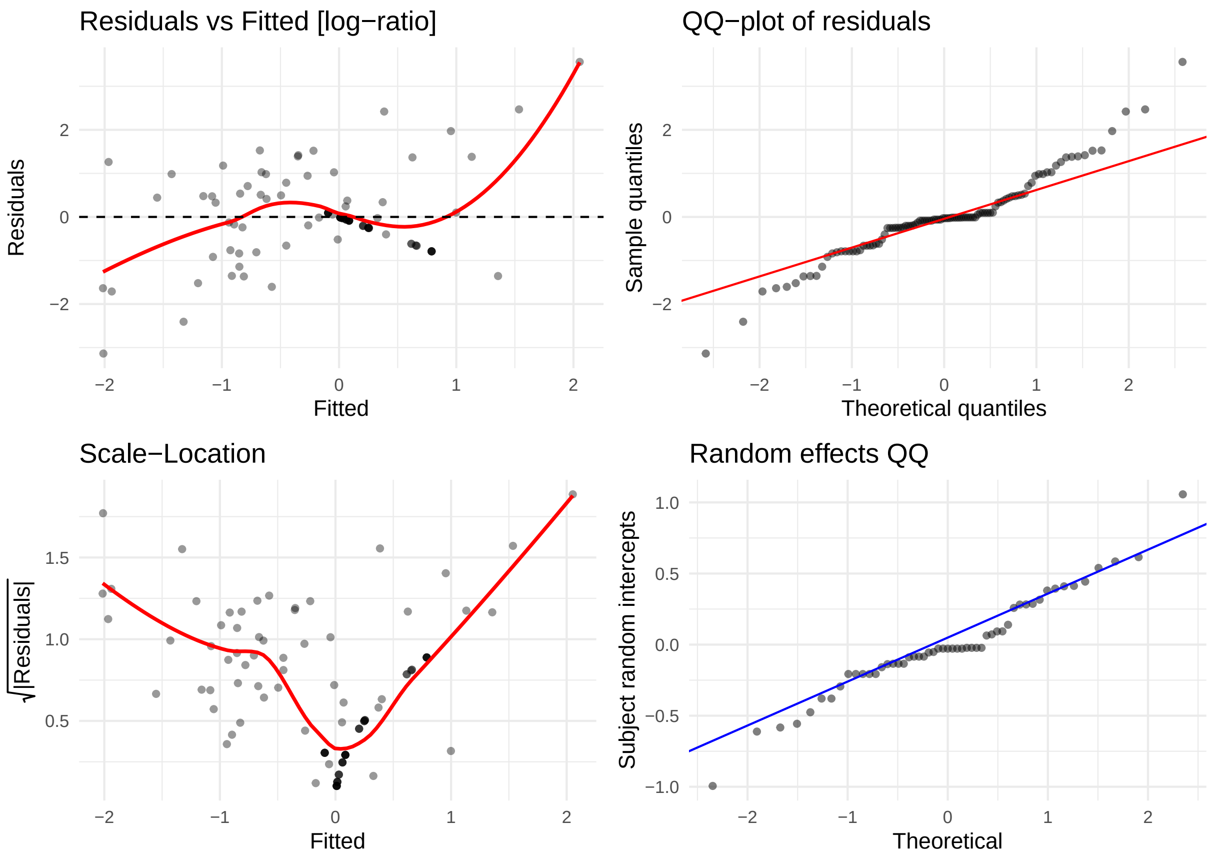
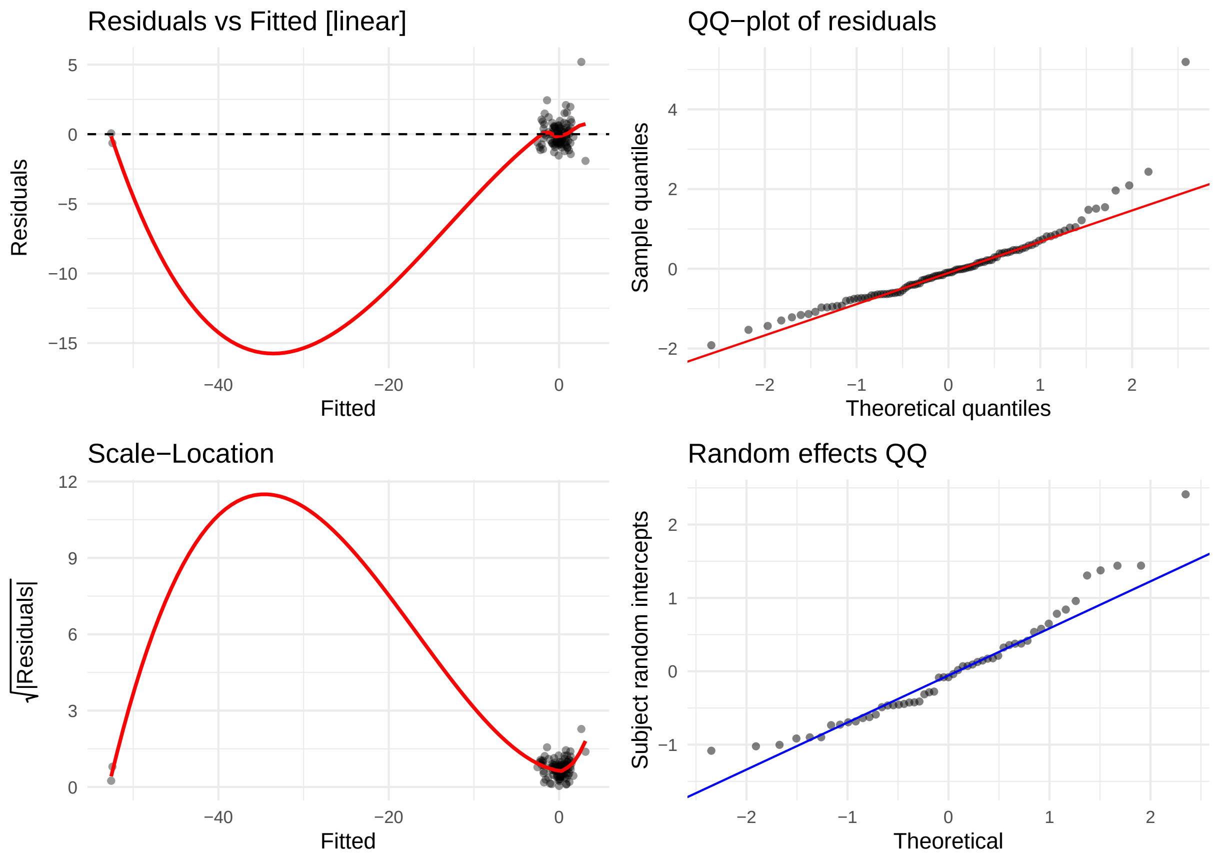


Table 17: Shapiro-Wilk — IL-2

Scale	W	p-value	Interpretation
linear	0.8862	0e+00	Strong departure from normality
log-ratio	0.9438	3e-04	Strong departure from normality

5.16 IL-6 (`IL6`)

AIC linear: 390.6 | **AIC log-ratio:** 364 | **Preferred: log**



Residuals vs Fitted [linear]
Residuals
Fitted
QQ−plot of residuals
Sample quantiles
Theoretical quantiles
Scale−Location
√|Residuals|
Fitted
Random effects QQ
Subject random intercepts
Theoretical

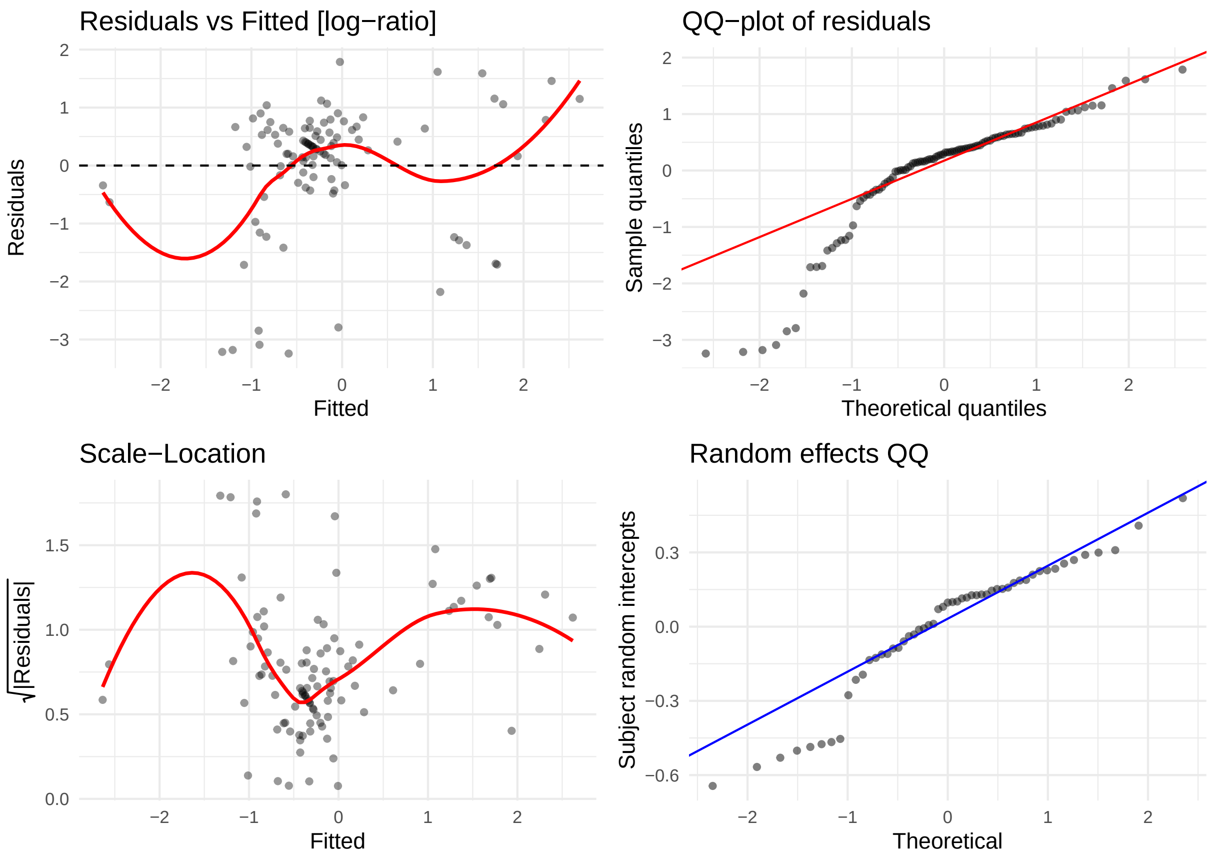
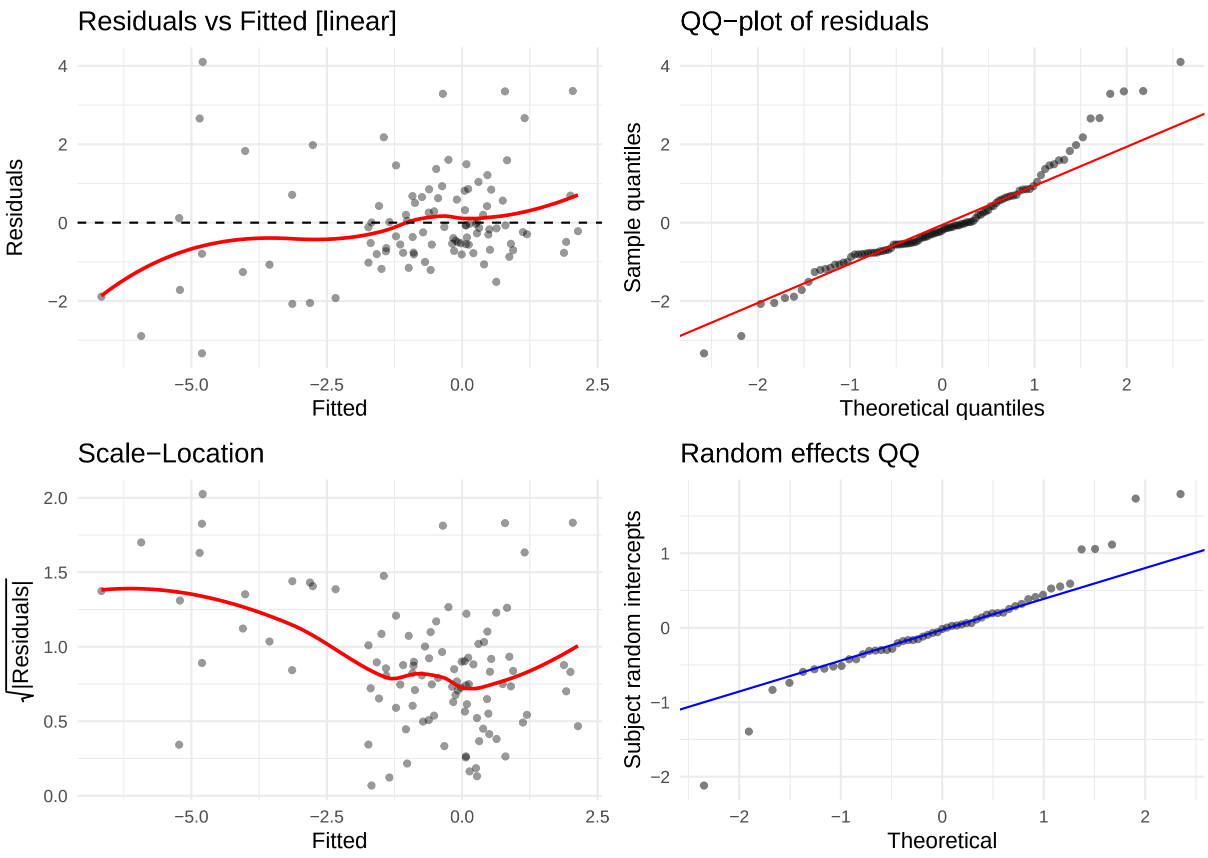


Table 18: Shapiro-Wilk — IL-6

Scale	W	p-value	Interpretation
linear	0.8823	0	Strong departure from normality
log-ratio	0.8496	0	Strong departure from normality

5.17 IL-8 (IL8)

AIC linear: 423.1 | **AIC log-ratio:** 60.3 | **Preferred: log**



Residuals vs Fitted [linear]
Residuals
Fitted
QQ−plot of residuals
Sample quantiles
Theoretical quantiles
Scale−Location
√|Residuals|
Fitted
Random effects QQ
Subject random intercepts
Theoretical

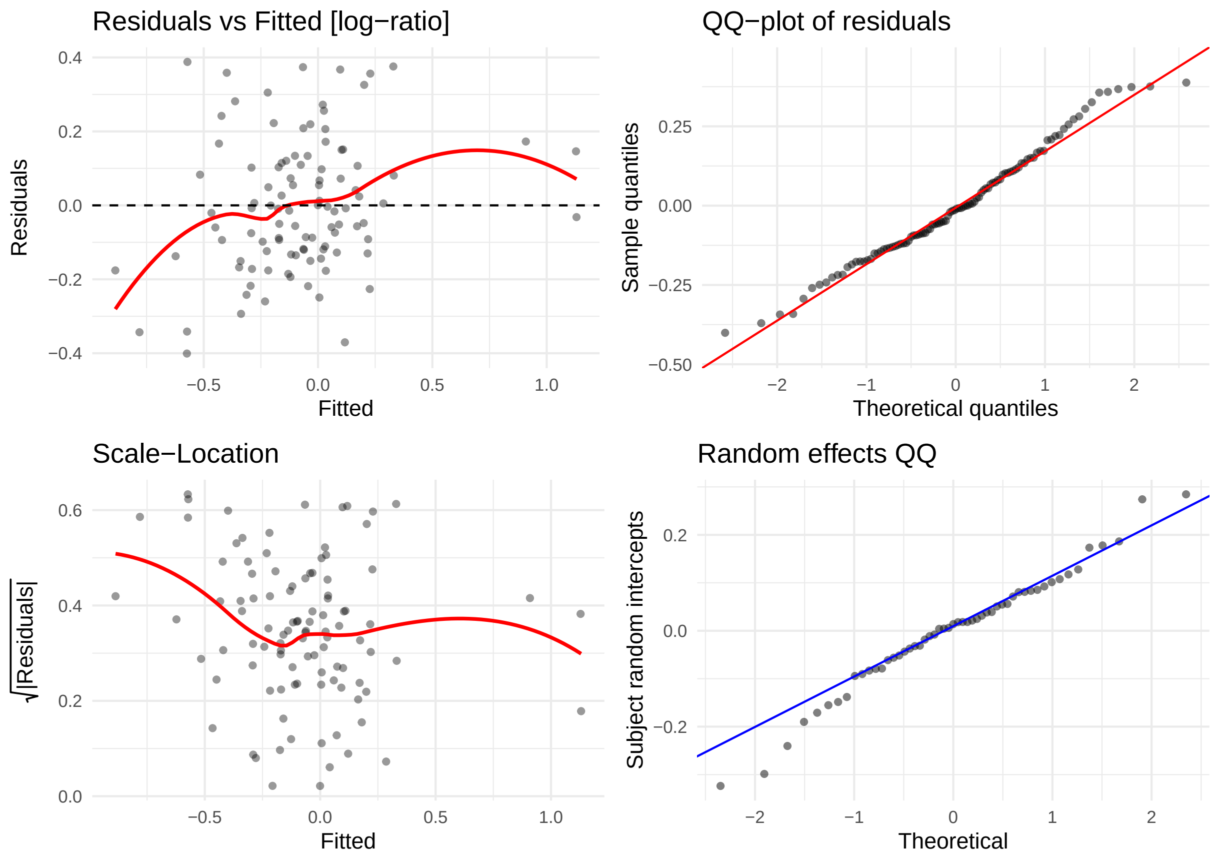
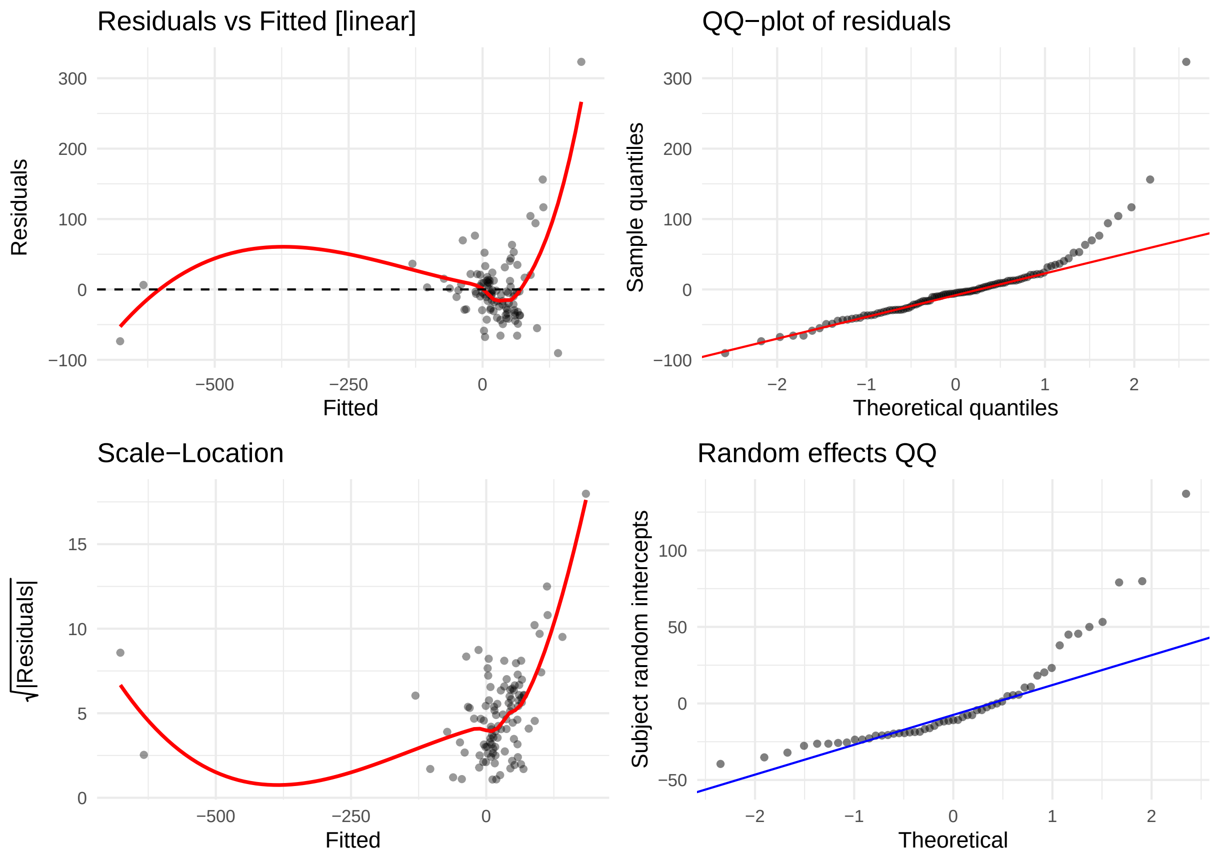


Table 19: Shapiro-Wilk — IL-8

Scale	W	p-value	Interpretation
linear	0.9453	0.0004	Strong departure from normality
log-ratio	0.9824	0.1948	No evidence of non-normality

5.18 IP-10 (CXCL10) (`IP10`)

AIC linear: 1152.5 | **AIC log-ratio:** 173.8 | **Preferred: log**



Residuals vs Fitted [linear]
Residuals
Fitted
QQ−plot of residuals
Sample quantiles
Theoretical quantiles
Scale−Location
√|Residuals|
Fitted
Random effects QQ
Subject random intercepts
Theoretical

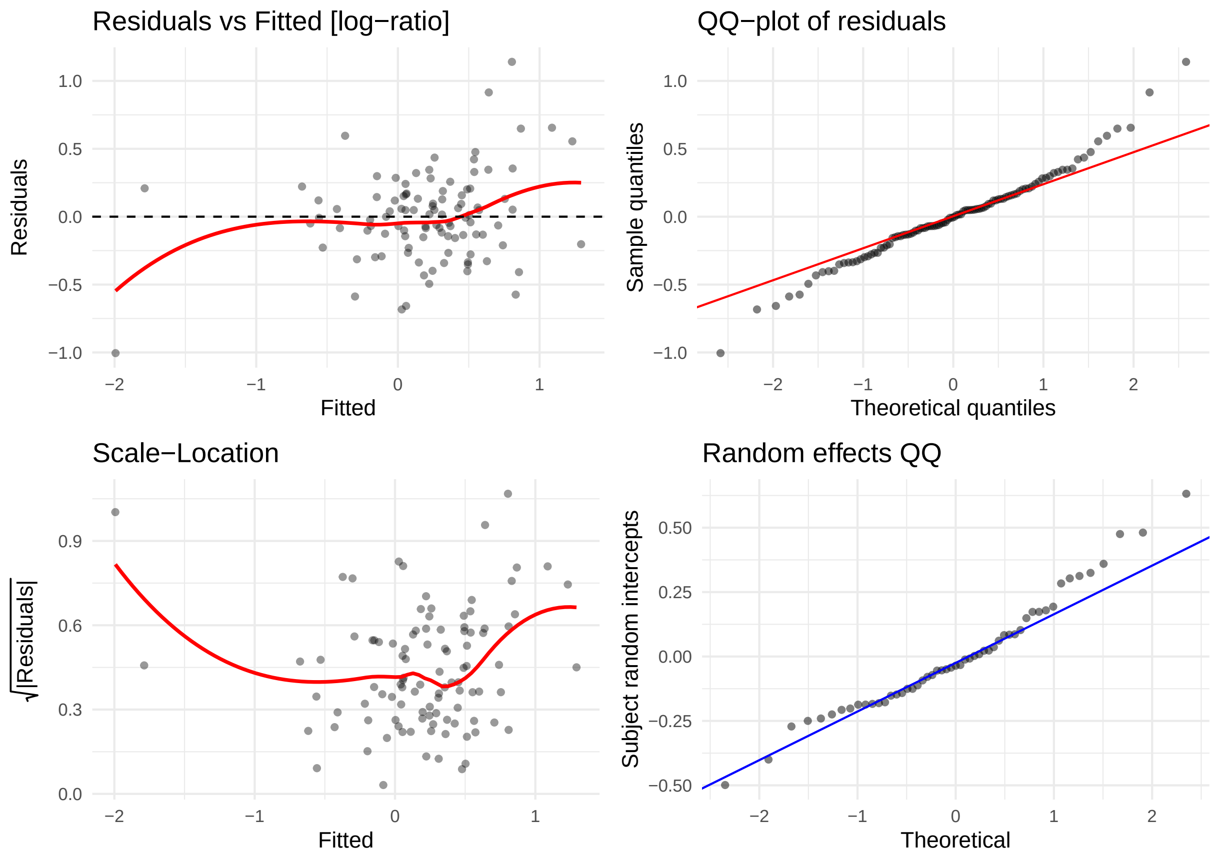
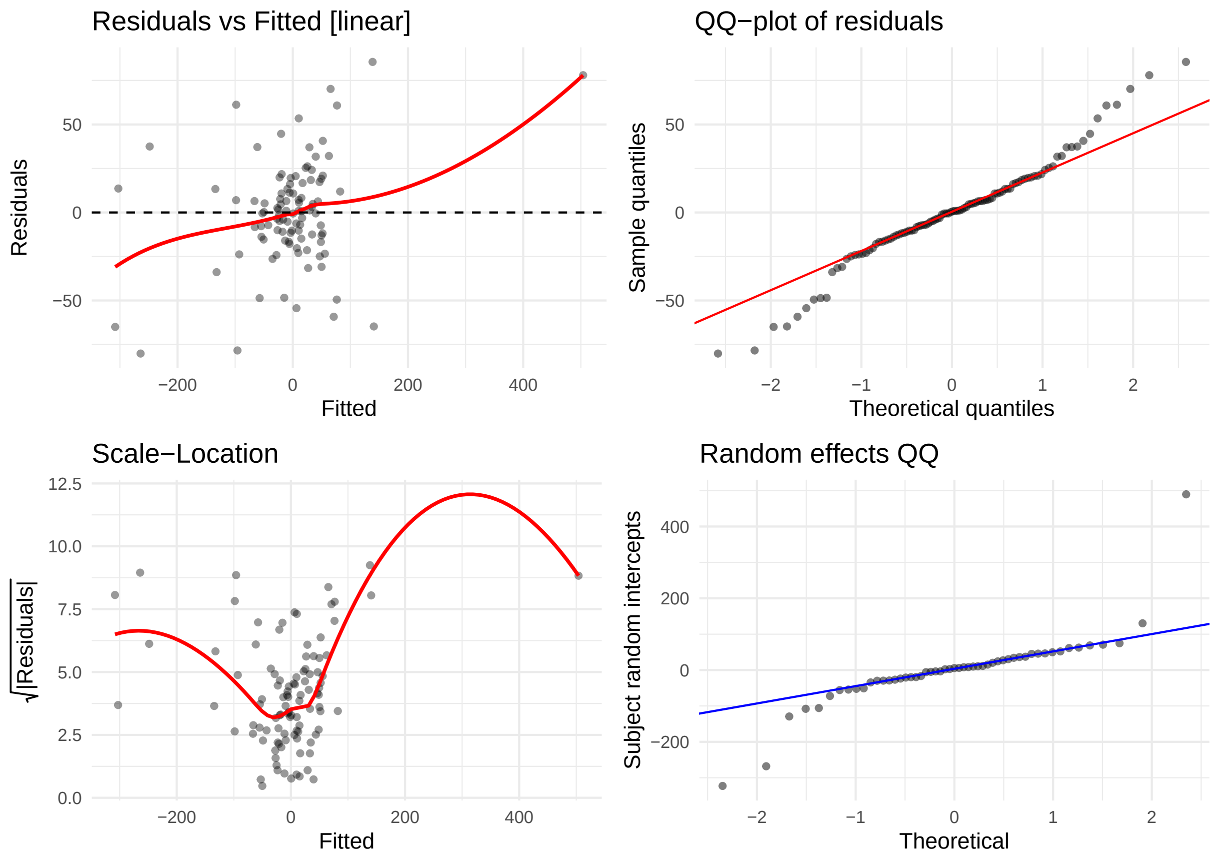


Table 20: Shapiro-Wilk — IP-10 (CXCL10)

Scale	W	p-value	Interpretation
linear	0.7811	0.0000	Strong departure from normality
log-ratio	0.9765	0.0658	Borderline

5.19 I-TAC (CXCL11) (ITAC)

AIC linear: 1174.1 | **AIC log-ratio:** 32.5 | **Preferred: log**



Residuals vs Fitted [linear]
Residuals
Fitted
QQ−plot of residuals
Sample quantiles
Theoretical quantiles
Scale−Location
√|Residuals|
Fitted
Random effects QQ
Subject random intercepts
Theoretical

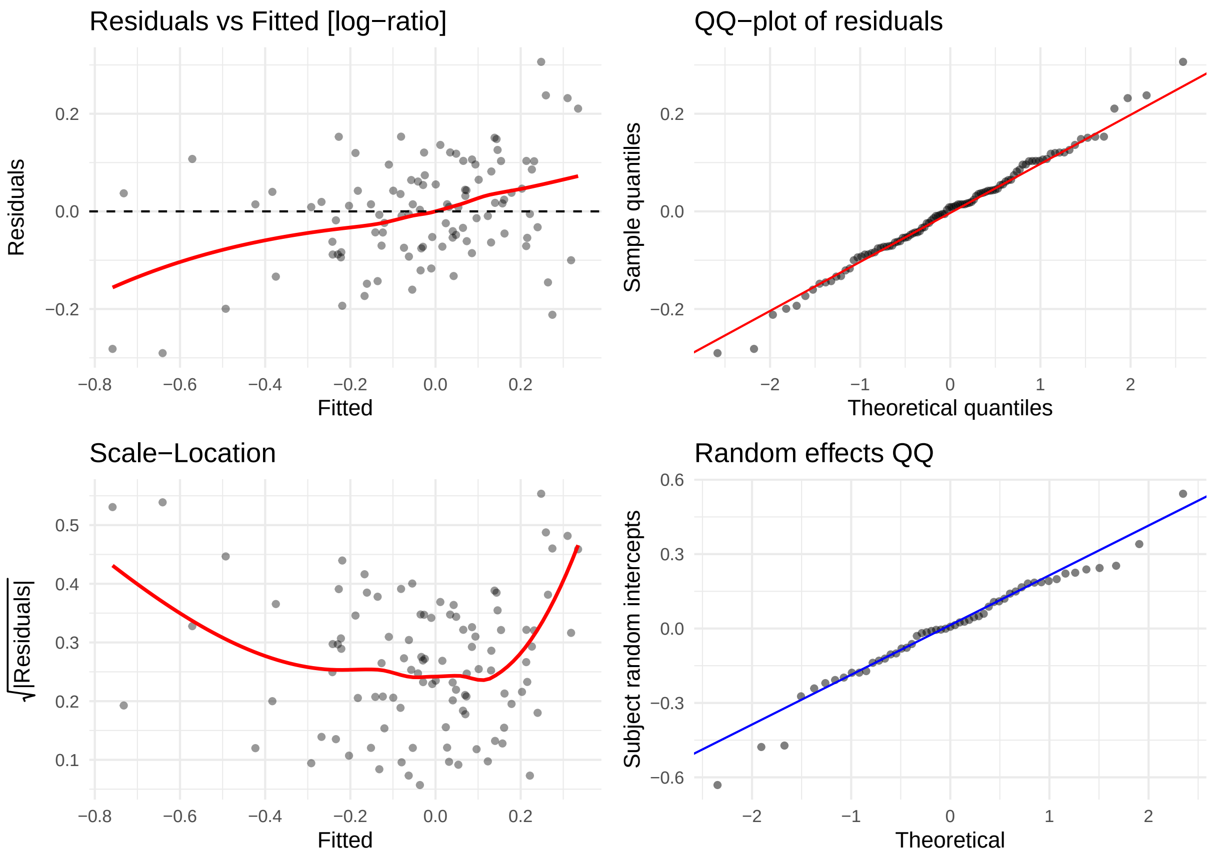
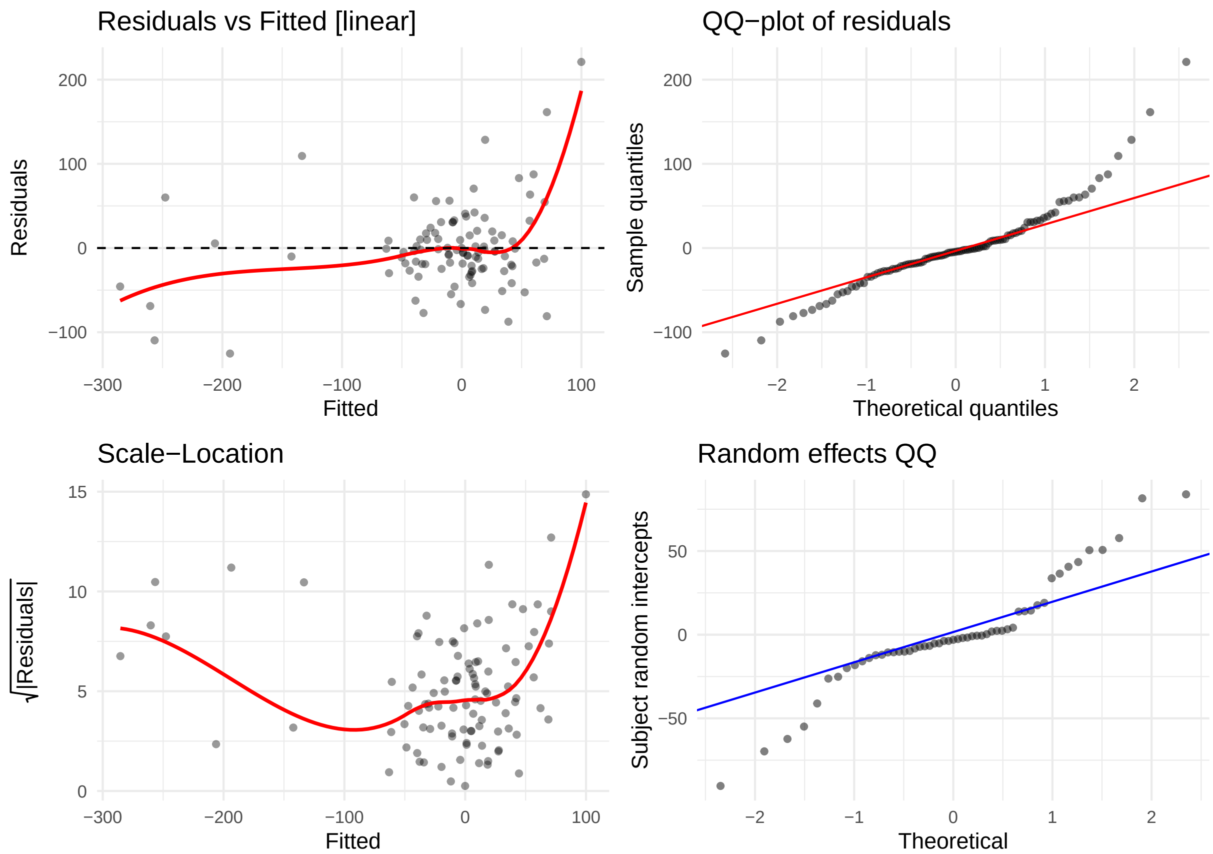


Table 21: Shapiro-Wilk — I-TAC (CXCL11)

Scale	W	p-value	Interpretation
linear	0.9697	0.0191	Some departure from normality
log-ratio	0.9929	0.8770	No evidence of non-normality

5.20 MCP-1 (`MCP1`)

AIC linear: 1148.1 | **AIC log-ratio:** 199.4 | **Preferred: log**


Residuals vs Fitted [linear]
QQ−plot of residuals
Scale−Location
Random effects QQ
Residuals
Fitted
Sample quantiles
Theoretical quantiles
√|Residuals|
Subject random intercepts
Theoretical

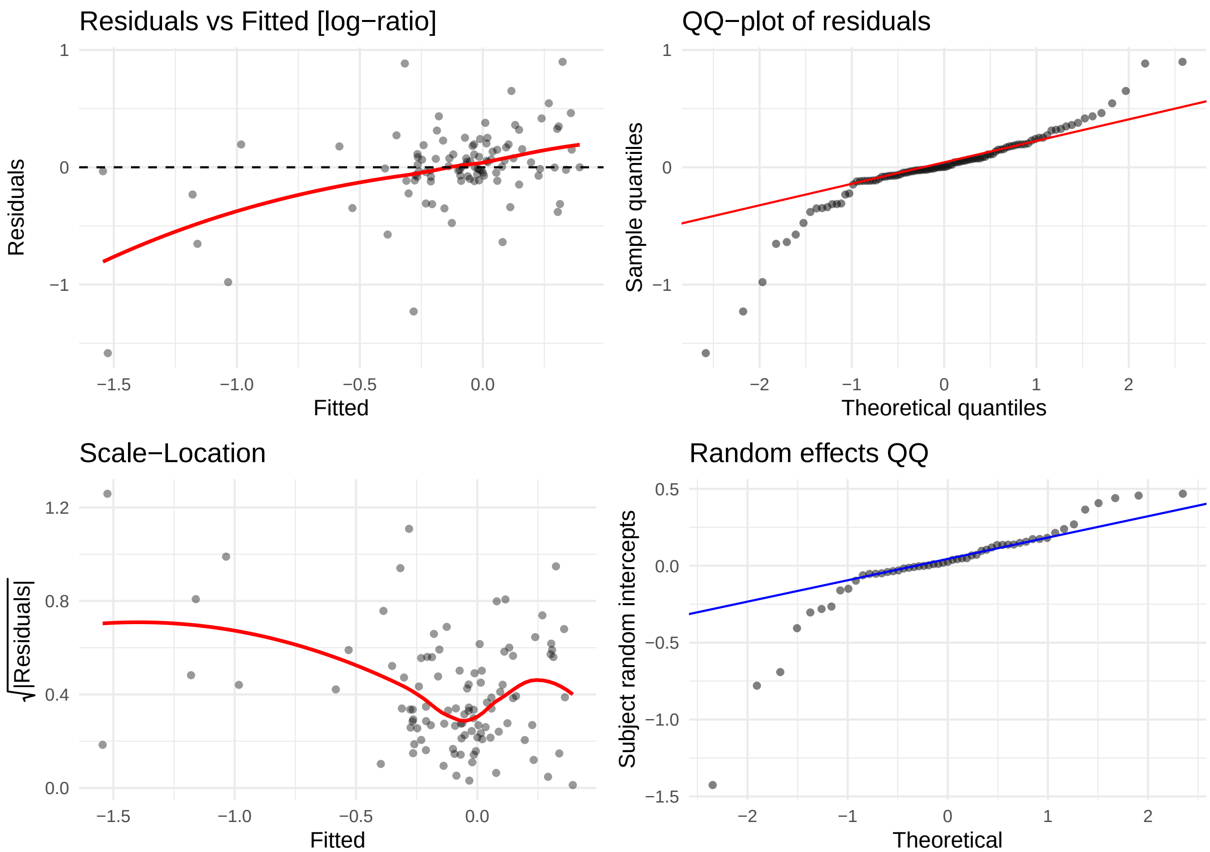
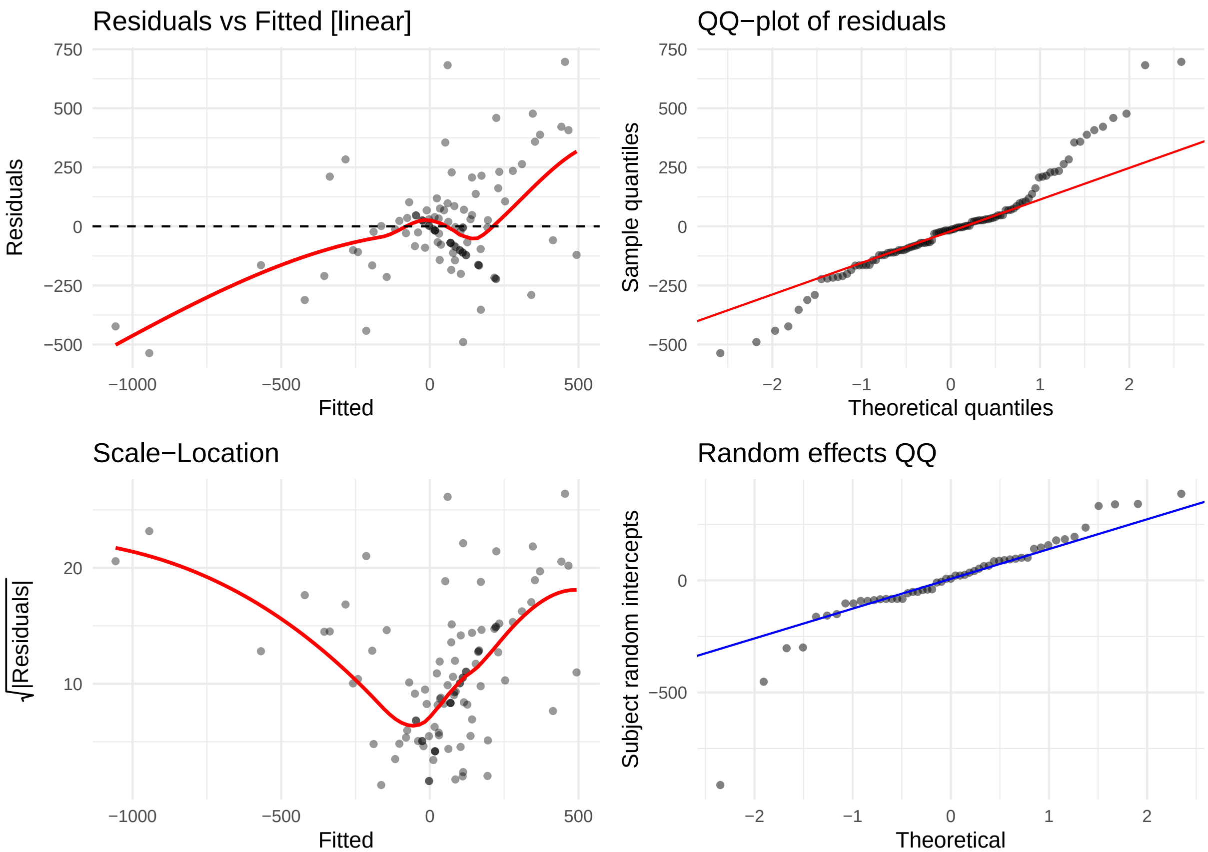


Table 22: Shapiro-Wilk — MCP-1

Scale	W	p-value	Interpretation
linear	0.9190	0	Strong departure from normality
log-ratio	0.8698	0	Strong departure from normality

5.21 MIG (CXCL9) (`MIG`)

AIC linear: 1452.7 | **AIC log-ratio:** 338.2 | **Preferred: log**


Residuals vs Fitted [linear]
Residuals
Fitted
QQ−plot of residuals
Sample quantiles
Theoretical quantiles
Scale−Location
√|Residuals|
Fitted
Random effects QQ
Subject random intercepts
Theoretical

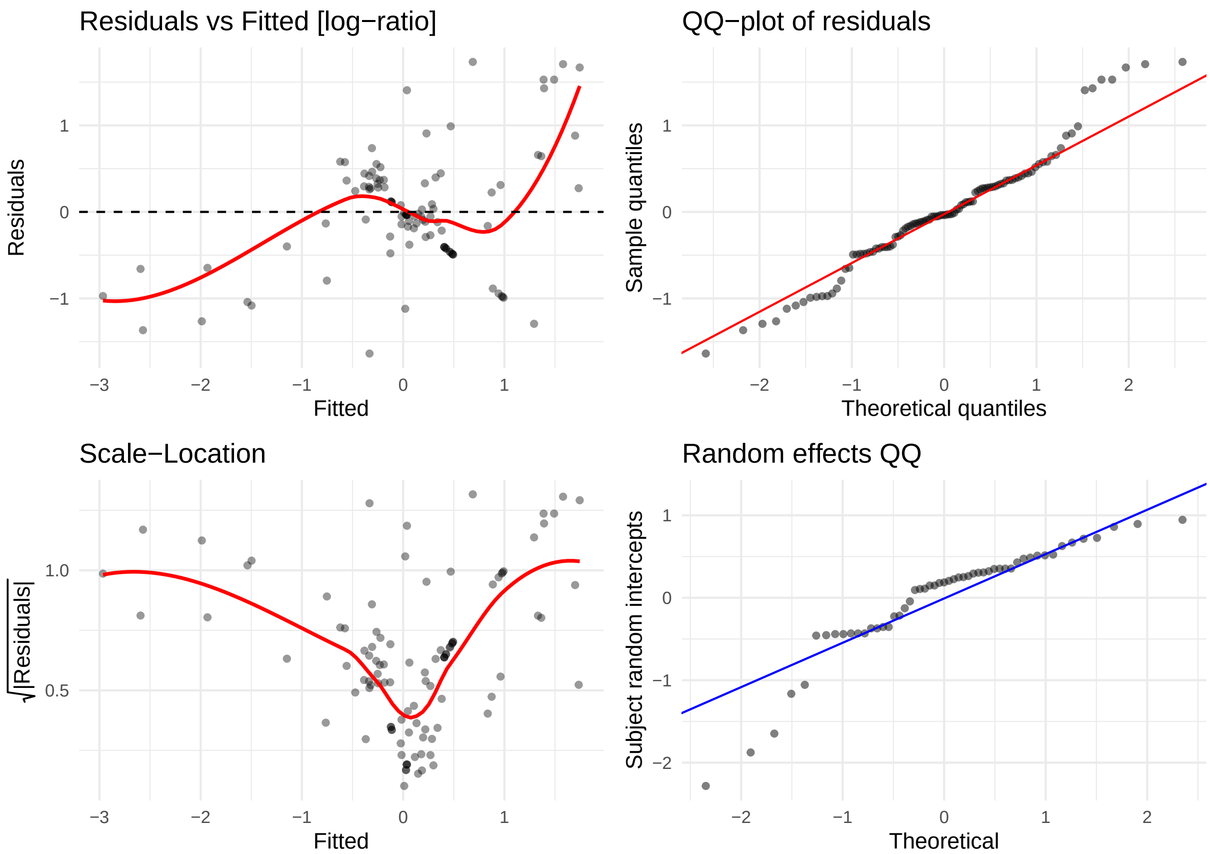
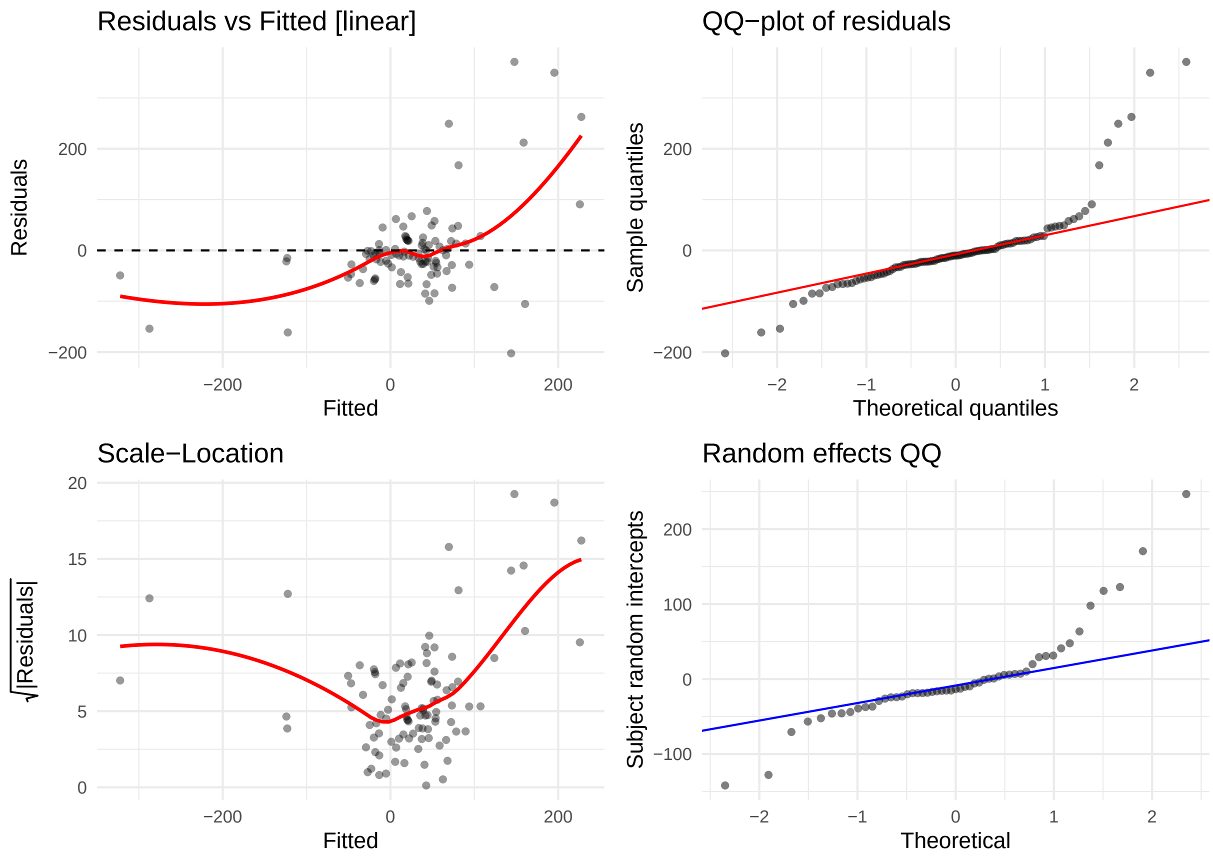


Table 23: Shapiro-Wilk — MIG (CXCL9)

Scale	W	p-value	Interpretation
linear	0.9469	0.0005	Strong departure from normality
log-ratio	0.9698	0.0195	Some departure from normality

5.22 MMP-10 (`MMP10`)

AIC linear: 1254.9 | **AIC log-ratio:** 25.1 | **Preferred: log**



Residuals vs Fitted [linear]
Residuals
Fitted
QQ−plot of residuals
Sample quantiles
Theoretical quantiles
Scale−Location
√|Residuals|
Fitted
Random effects QQ
Subject random intercepts
Theoretical

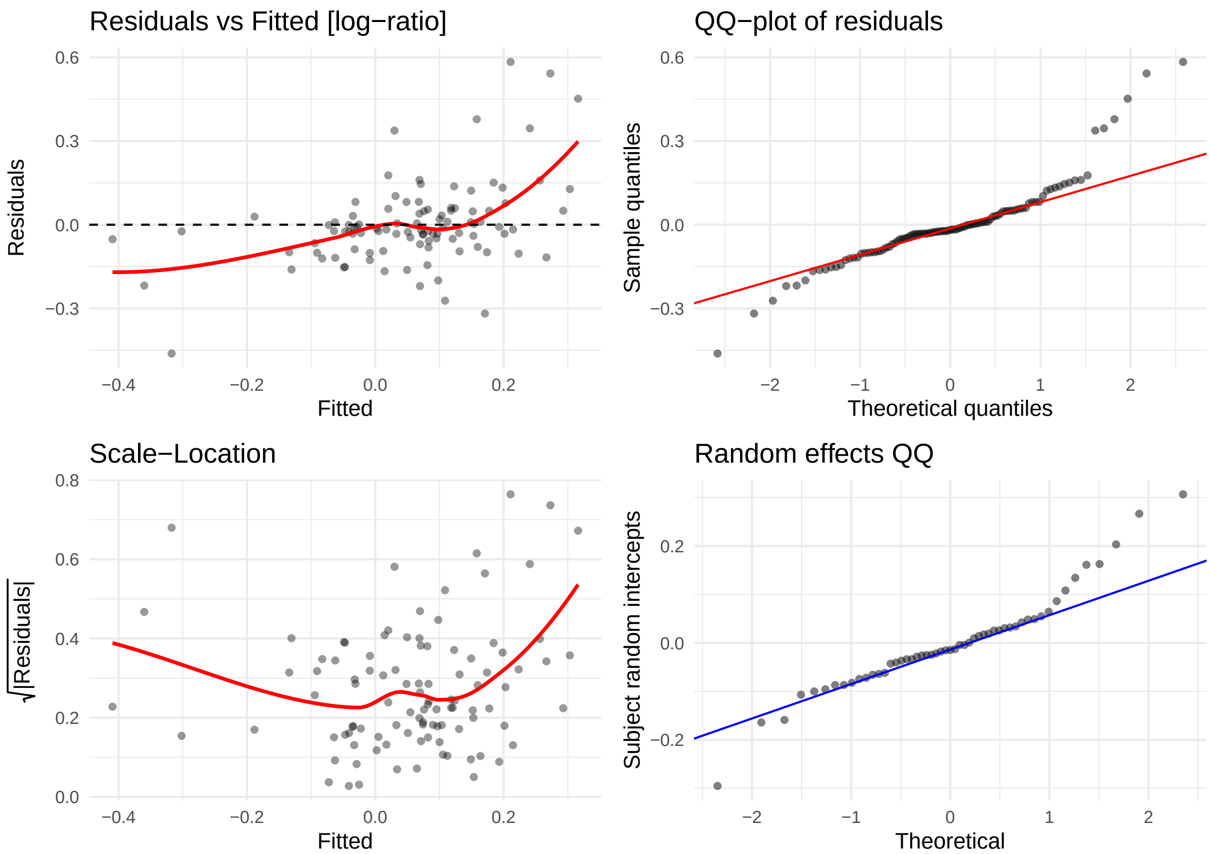
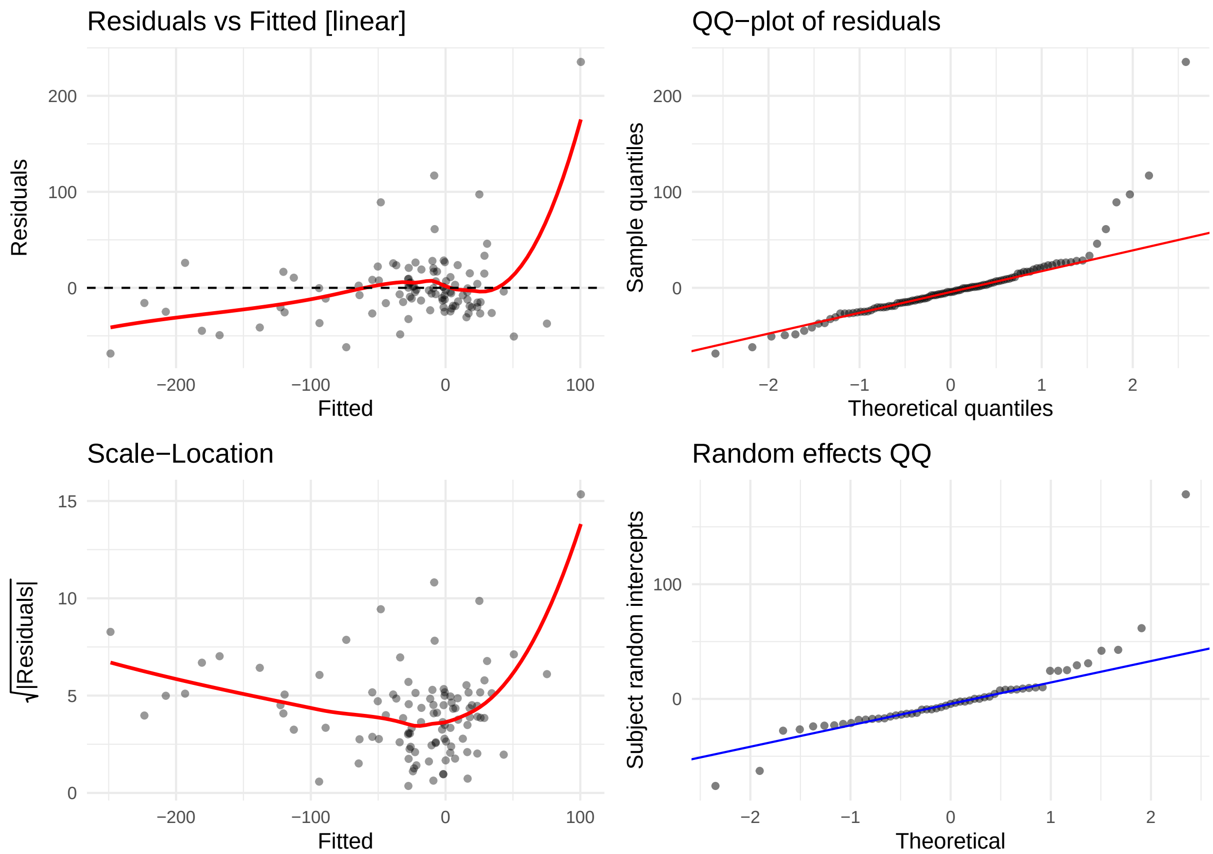


Table 24: Shapiro-Wilk — MMP-10

Scale	W	p-value	Interpretation
linear	0.7799	0	Strong departure from normality
log-ratio	0.8861	0	Strong departure from normality

5.23 PDGF-DD (`PDGFDD`)

AIC linear: 1109.7 | **AIC log-ratio:** 198.7 | **Preferred: log**


Residuals vs Fitted [linear]
Residuals
200
100
0
−200
−100
0
100
Fitted
QQ−plot of residuals
Sample quantiles
200
100
0
−2
−1
0
1
2
Theoretical quantiles
Scale−Location
√|Residuals|
15
10
5
0
−200
−100
0
100
Fitted
Random effects QQ
Subject random intercepts
100
0
−2
−1
0
1
2
Theoretical

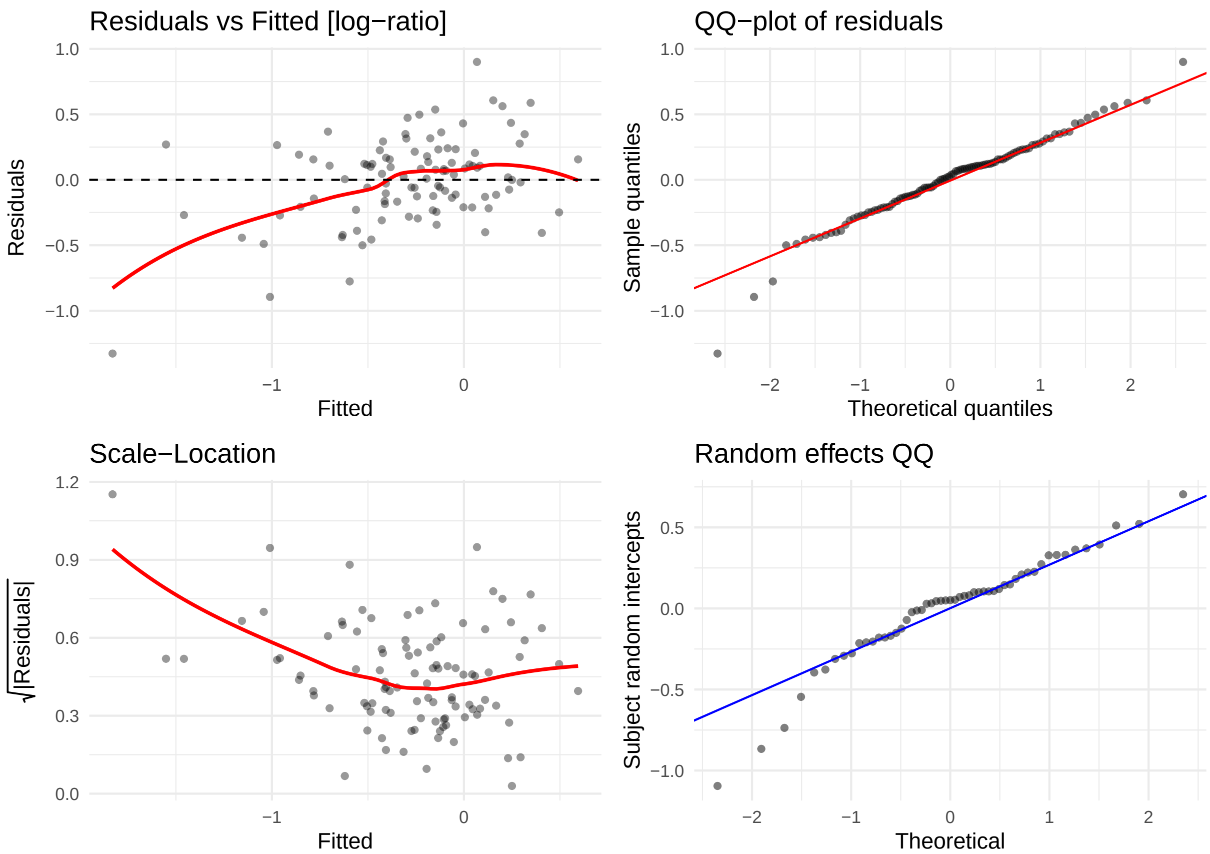
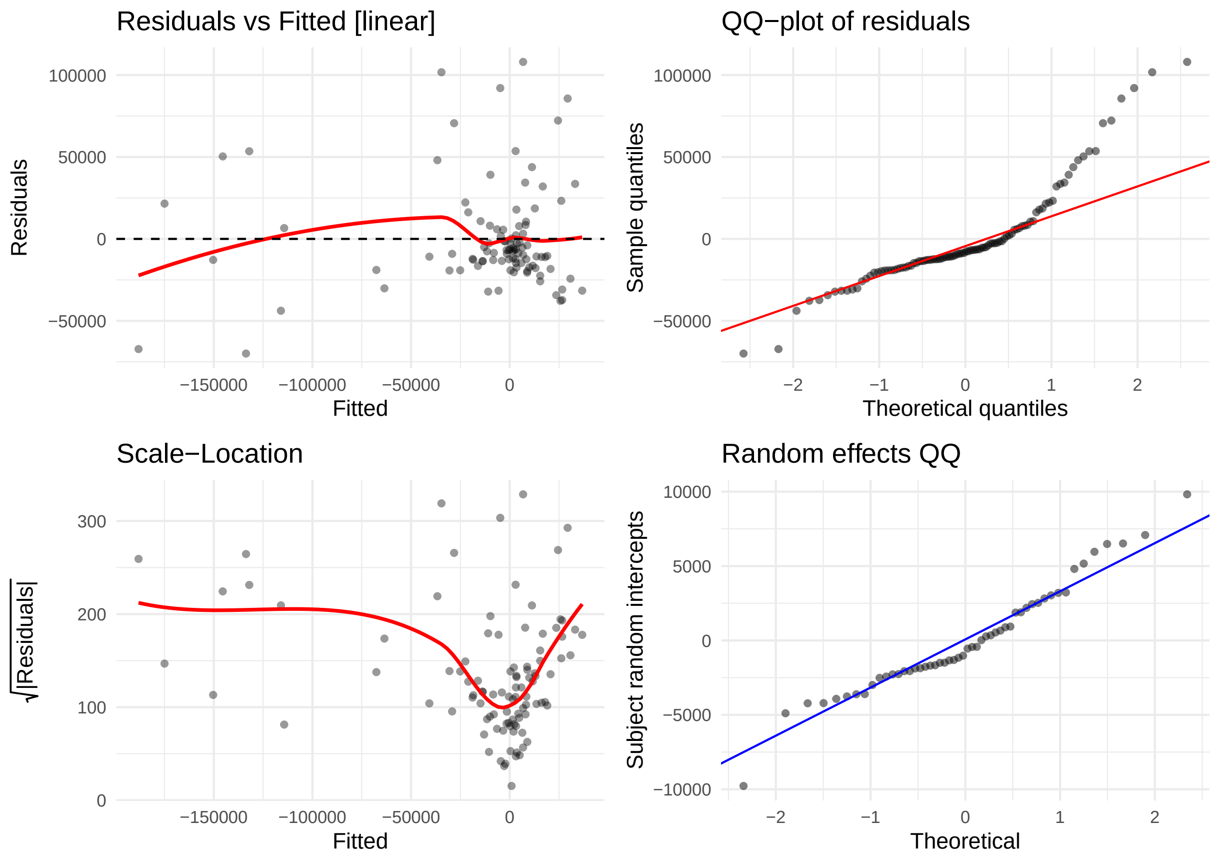


Table 25: Shapiro-Wilk — PDGF-DD

Scale	W	p-value	Interpretation
linear	0.7641	0.0000	Strong departure from normality
log-ratio	0.9663	0.0104	Some departure from normality

5.24 PECAM-1 (PECAM1)

AIC linear: 2298.5 | **AIC log-ratio:** 246.1 | **Preferred: log**



Residuals vs Fitted [linear]
Residuals
100000
50000
0
−50000
−150000
−100000
−50000
0
Fitted
QQ−plot of residuals
Sample quantiles
100000
50000
0
−50000
−2
−1
0
1
2
Theoretical quantiles
Scale−Location
√|Residuals|
300
200
100
0
−150000
−100000
−50000
0
Fitted
Random effects QQ
Subject random intercepts
10000
5000
0
−5000
−10000
−2
−1
0
1
2
Theoretical

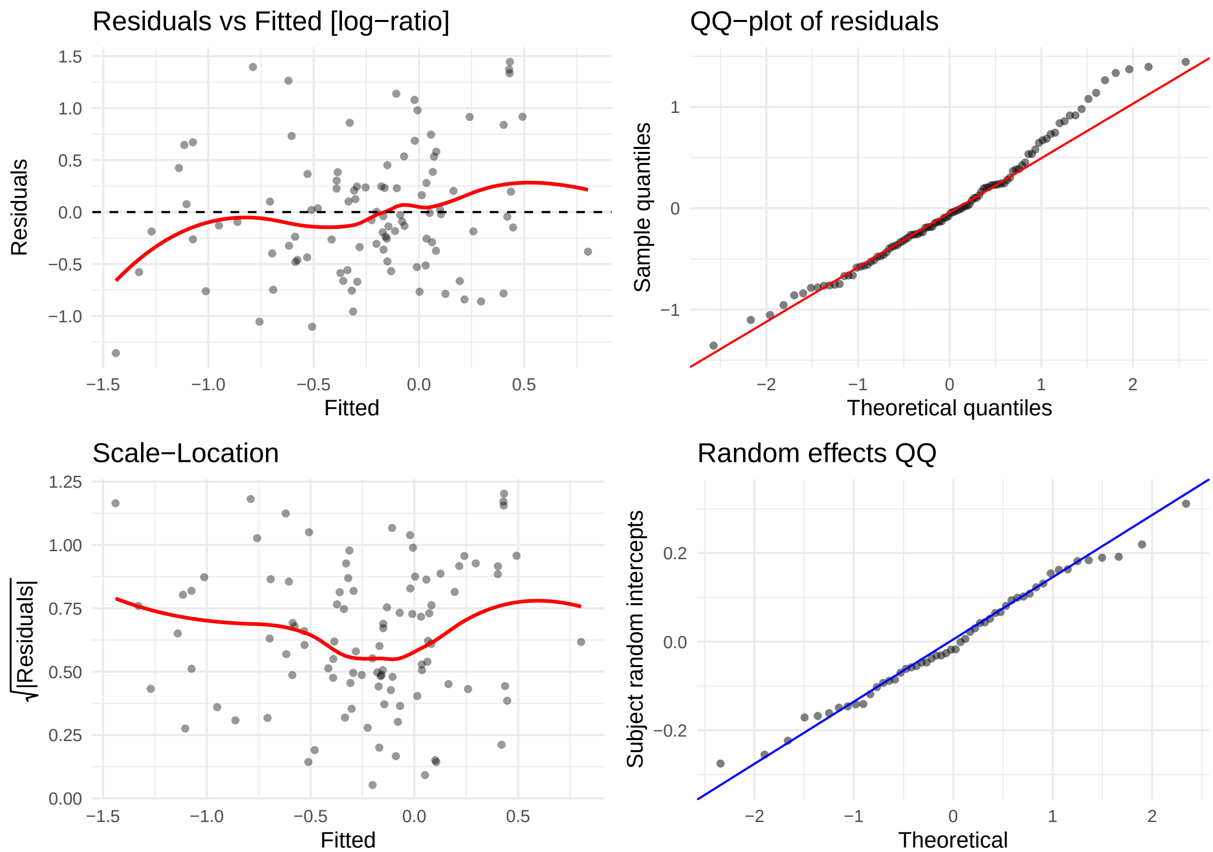
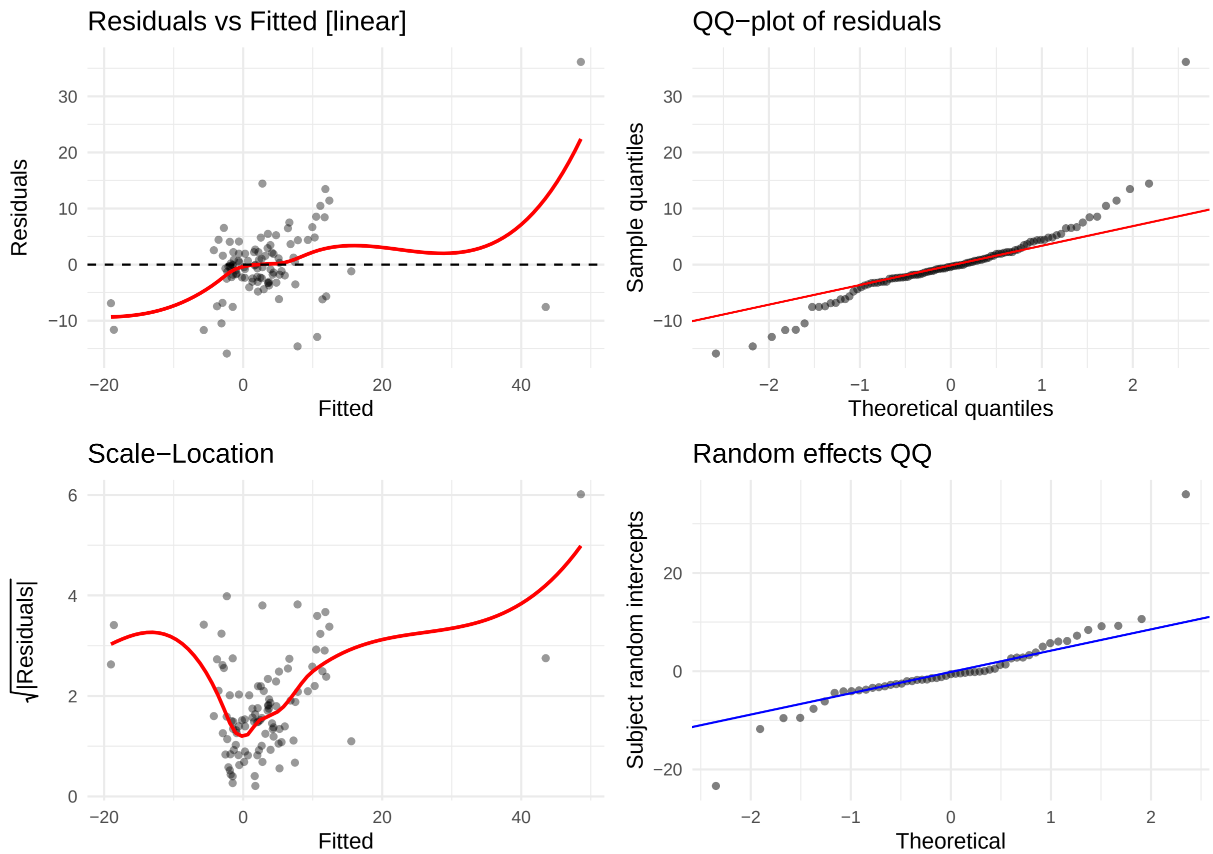


Table 26: Shapiro-Wilk — PECAM-1

Scale	W	p-value	Interpretation
linear	0.8698	0.0000	Strong departure from normality
log-ratio	0.9787	0.1055	No evidence of non-normality

5.25 PlGF (PIGF)

AIC linear: 788.4 | **AIC log-ratio:** -26.8 | **Preferred:** log


Residuals vs Fitted [linear]
Residuals
Fitted
30
20
10
0
−10
−20
0
20
40
QQ−plot of residuals
Sample quantiles
Theoretical quantiles
30
20
10
0
−10
−2
−1
0
1
2
Scale−Location
√|Residuals|
Fitted
6
4
2
0
−20
0
20
40
Random effects QQ
Subject random intercepts
Theoretical
20
0
−20
−2
−1
0
1
2

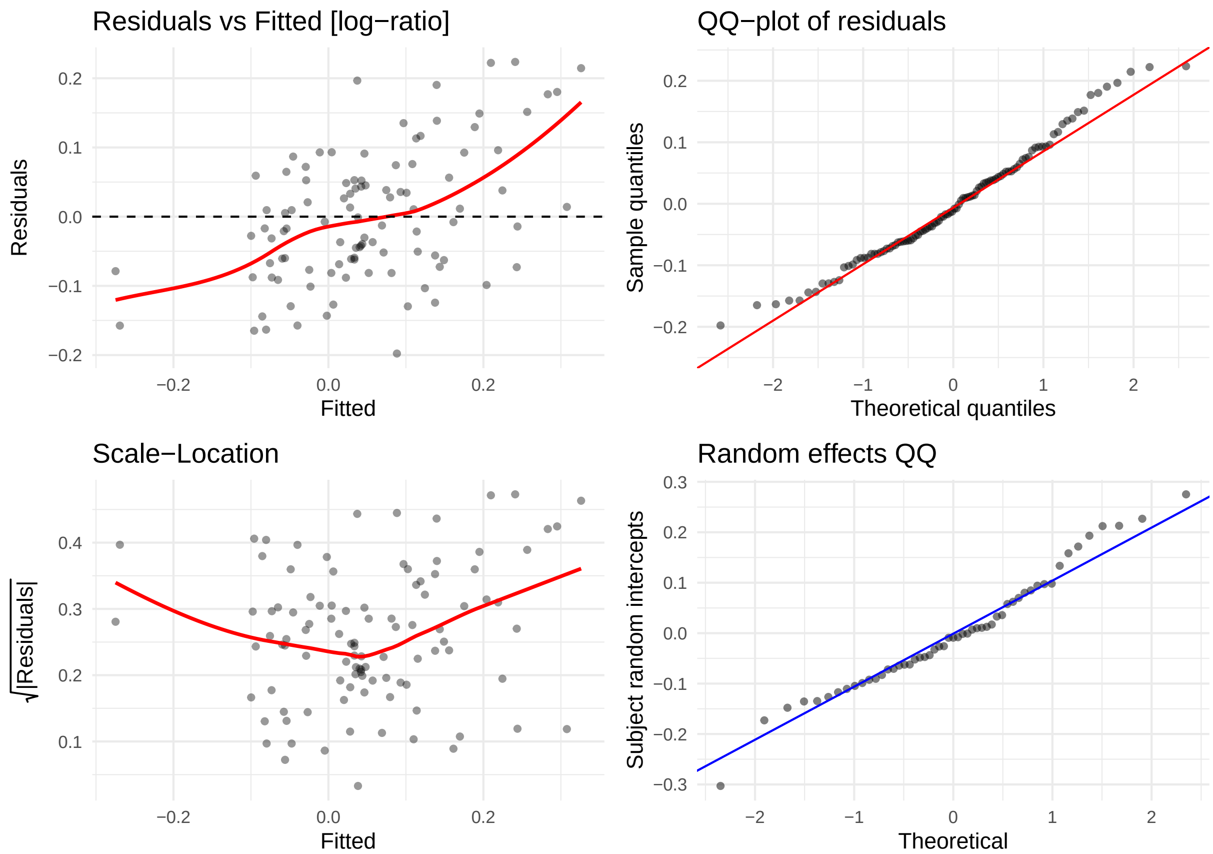
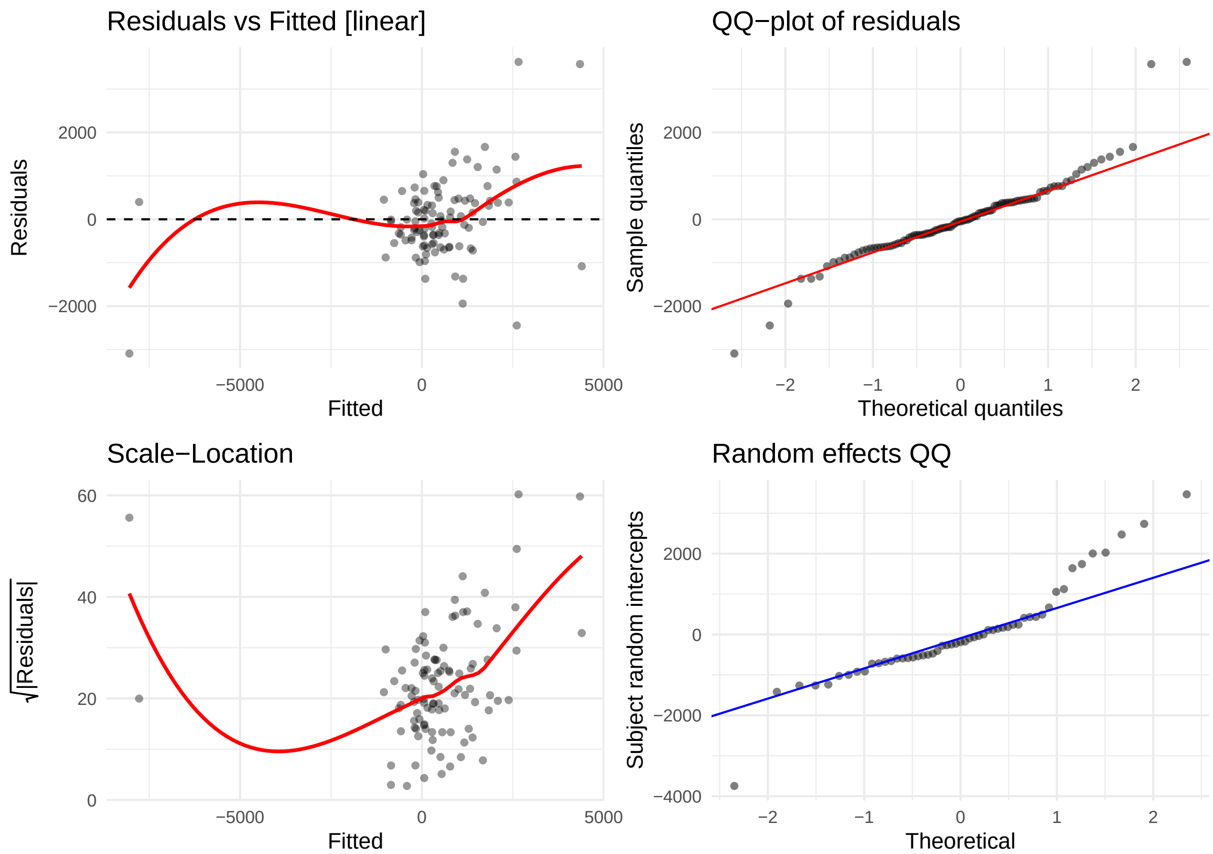


Table 27: Shapiro-Wilk — PIGF

Scale	W	p-value	Interpretation
linear	0.8725	0.0000	Strong departure from normality
log-ratio	0.9804	0.1347	No evidence of non-normality

5.26 Tenascin C (TENASC)

AIC linear: 1756.8 | **AIC log-ratio:** -111.4 | **Preferred: log**


Residuals vs Fitted [linear]
Residuals
Fitted
QQ−plot of residuals
Sample quantiles
Theoretical quantiles
Scale−Location
√|Residuals|
Fitted
Random effects QQ
Subject random intercepts
Theoretical

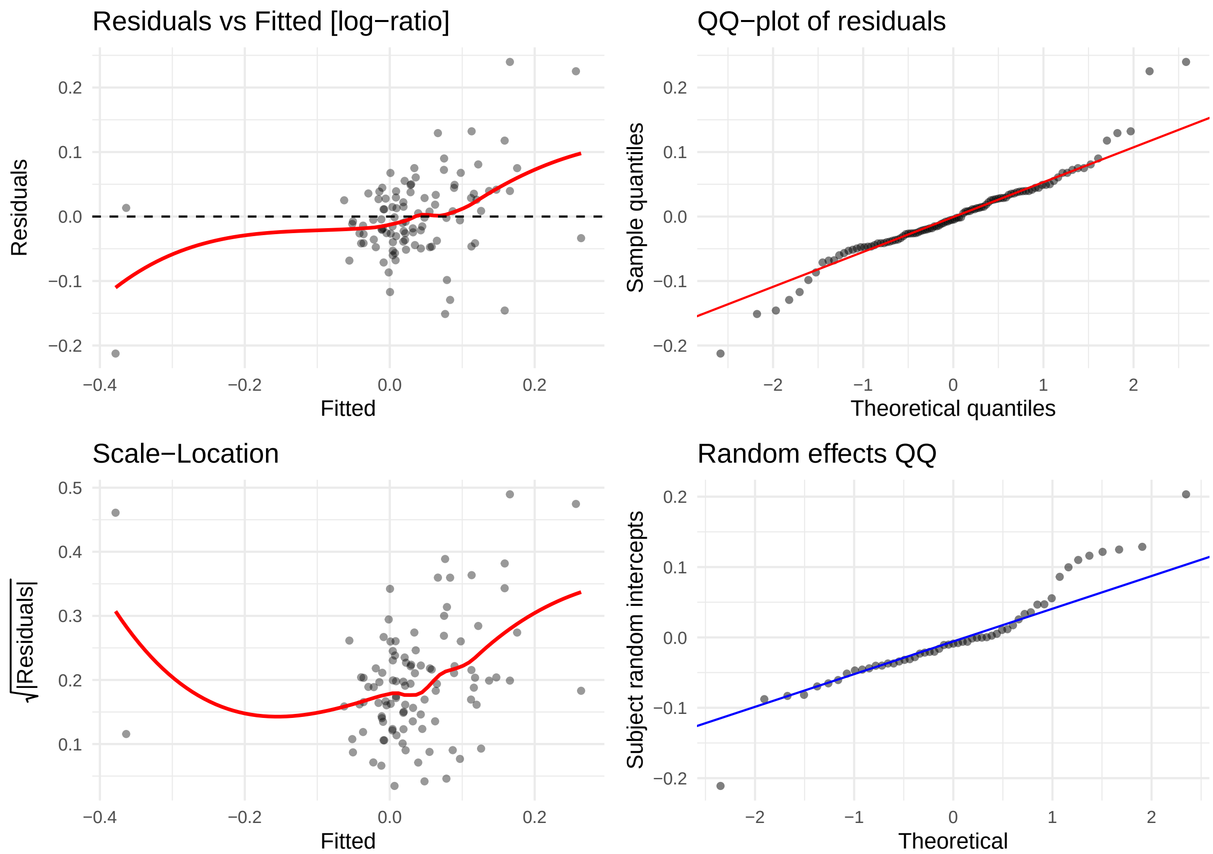
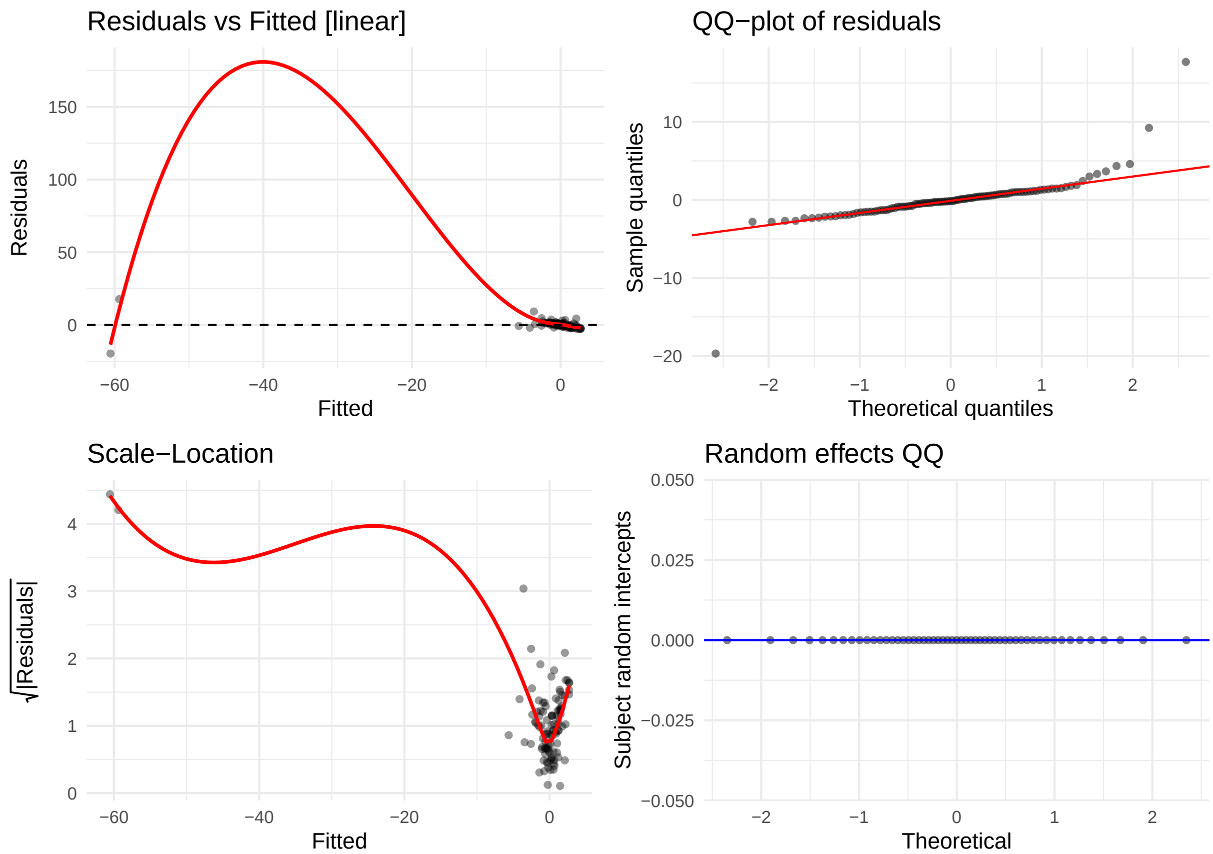


Table 28: Shapiro-Wilk — Tenascin C

Scale	W	p-value	Interpretation
linear	0.9209	0e+00	Strong departure from normality
log-ratio	0.9459	4e-04	Strong departure from normality

5.27 TGF-alpha (TGFA)

AIC linear: 546.5 | **AIC log-ratio:** 262.3 | **Preferred: log**


Residuals vs Fitted [linear]
Residuals
150
100
50
0
−60
−40
−20
0
Fitted
QQ−plot of residuals
Sample quantiles
10
0
−10
−20
−2
−1
0
1
2
Theoretical quantiles
Scale−Location
√|Residuals|
4
3
2
1
0
−60
−40
−20
0
Fitted
Random effects QQ
Subject random intercepts
0.050
0.025
0.000
−0.025
−0.050
−2
−1
0
1
2
Theoretical

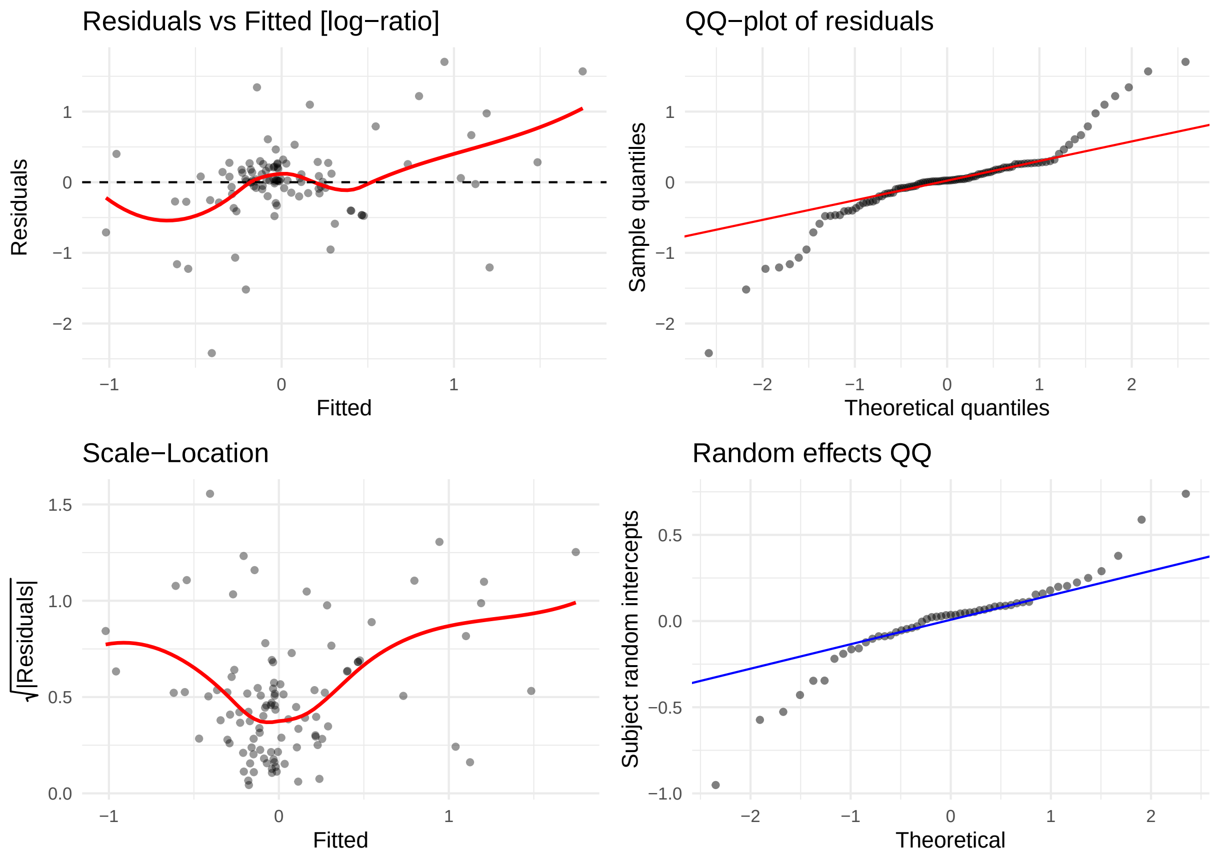
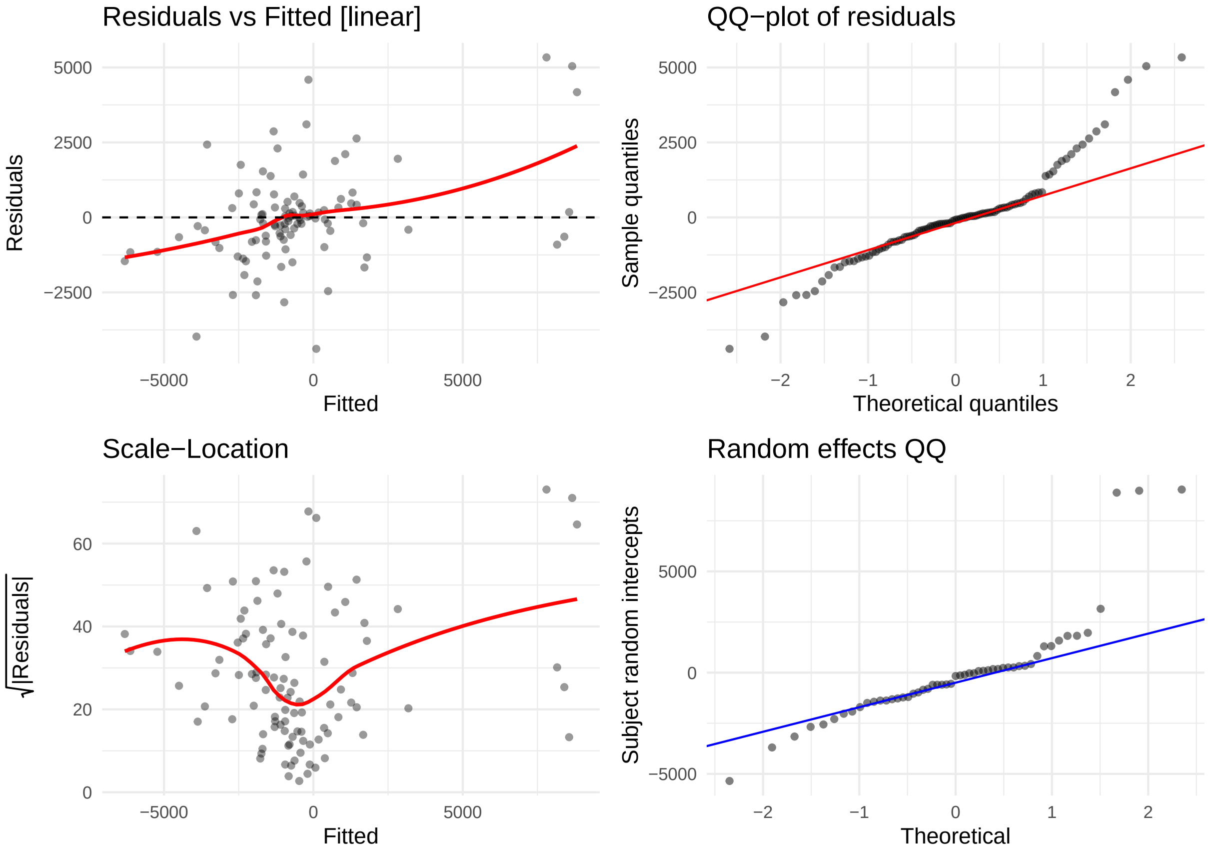


Table 29: Shapiro-Wilk — TGF-alpha

Scale	W	p-value	Interpretation
linear	0.6323	0	Strong departure from normality
log-ratio	0.8833	0	Strong departure from normality

5.28 Thrombospondin-2 (THBS2)

AIC linear: 1882.7 | **AIC log-ratio:** 84.5 | **Preferred: log**


Residuals vs Fitted [linear]
Residuals
Fitted
5000
2500
0
−2500
−5000
0
5000
QQ−plot of residuals
Sample quantiles
Theoretical quantiles
5000
2500
0
−2500
−2
−1
0
1
2
Scale−Location
√|Residuals|
Fitted
60
40
20
0
−5000
0
5000
Random effects QQ
Subject random intercepts
Theoretical
5000
0
−5000
−2
−1
0
1
2

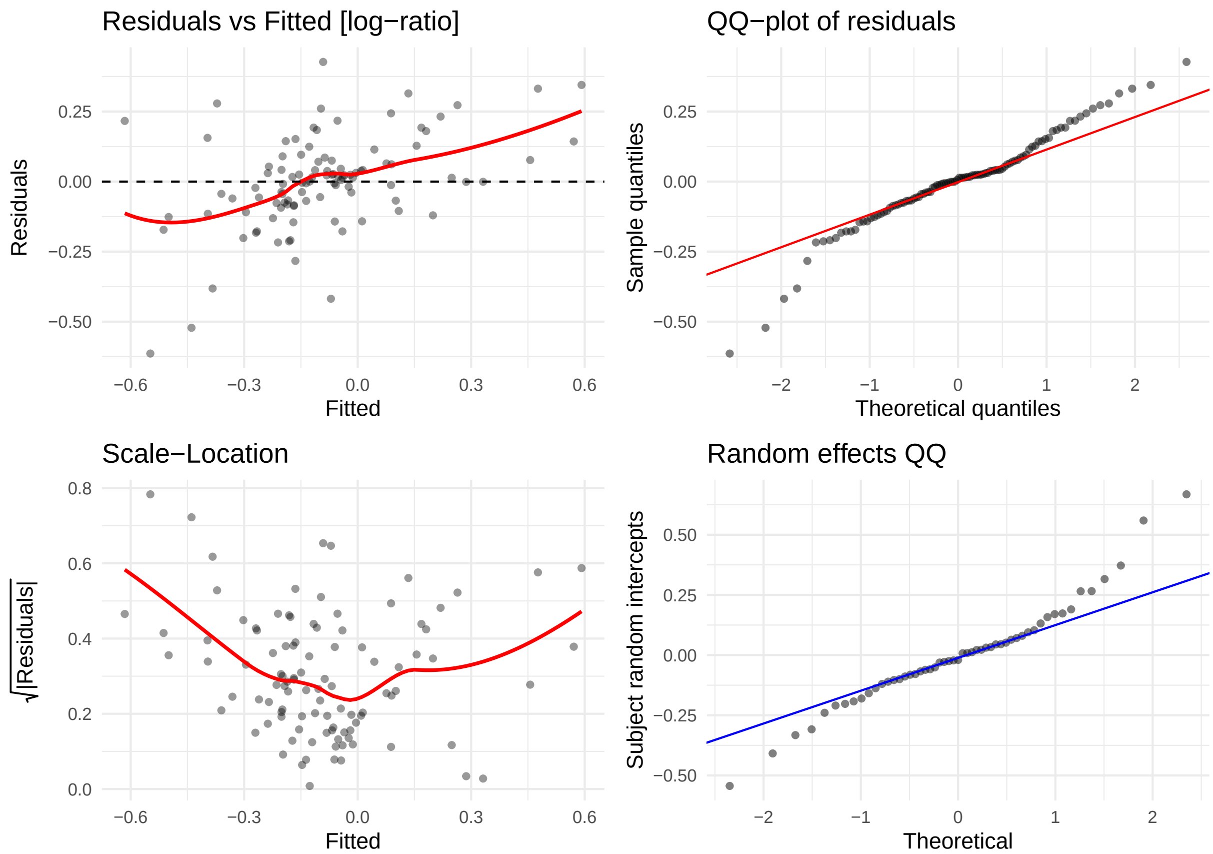
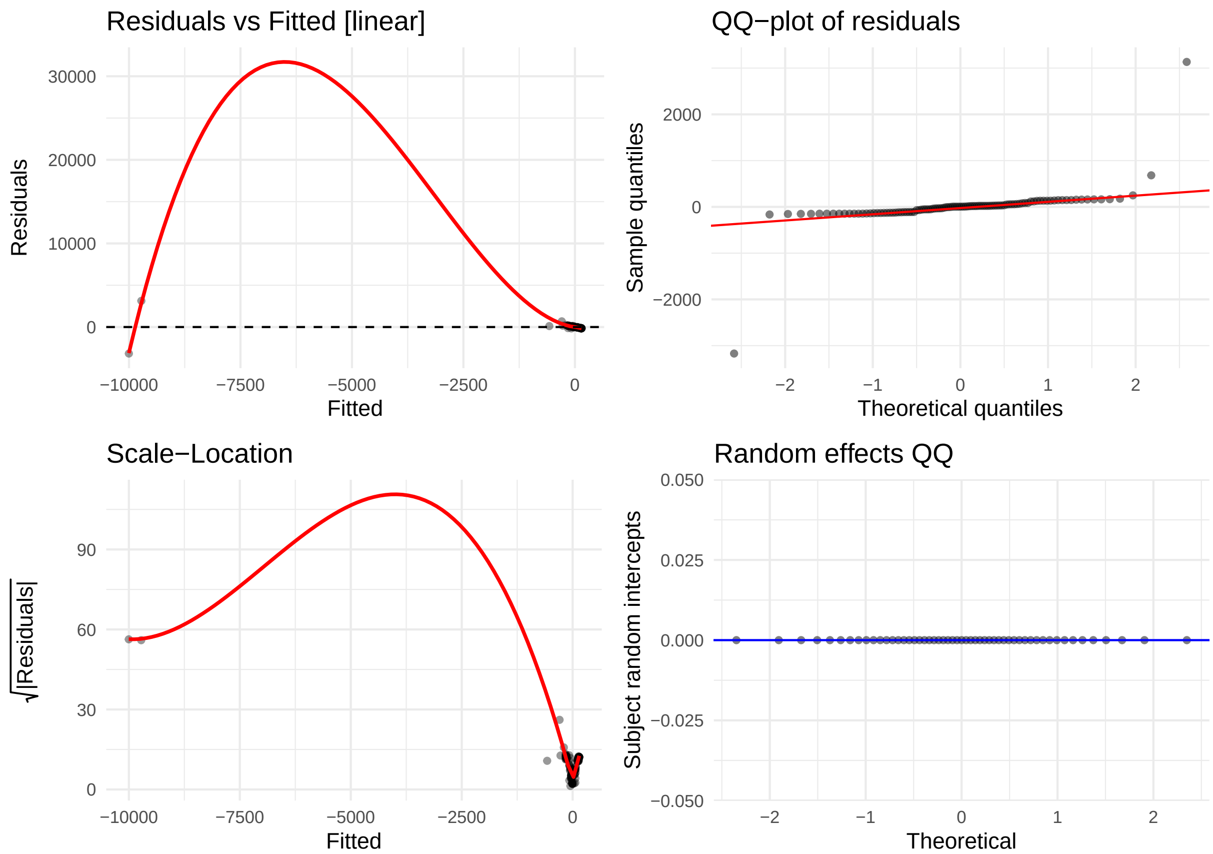


Table 30: Shapiro-Wilk — Thrombospondin-2

Scale	W	p-value	Interpretation
linear	0.9236	0.0000	Strong departure from normality
log-ratio	0.9579	0.0025	Strong departure from normality

5.29 TNFRSF18 (GITR) (`TNFRSF18`)

AIC linear: 1513.7 | **AIC log-ratio:** 223 | **Preferred:** log


Residuals vs Fitted [linear]
Residuals
30000
20000
10000
0
−10000
−7500
−5000
−2500
0
Fitted
QQ−plot of residuals
Sample quantiles
2000
0
−2000
−2
−1
0
1
2
Theoretical quantiles
Scale−Location
√|Residuals|
90
60
30
0
−10000
−7500
−5000
−2500
0
Fitted
Random effects QQ
Subject random intercepts
0.050
0.025
0.000
−0.025
−0.050
−2
−1
0
1
2
Theoretical

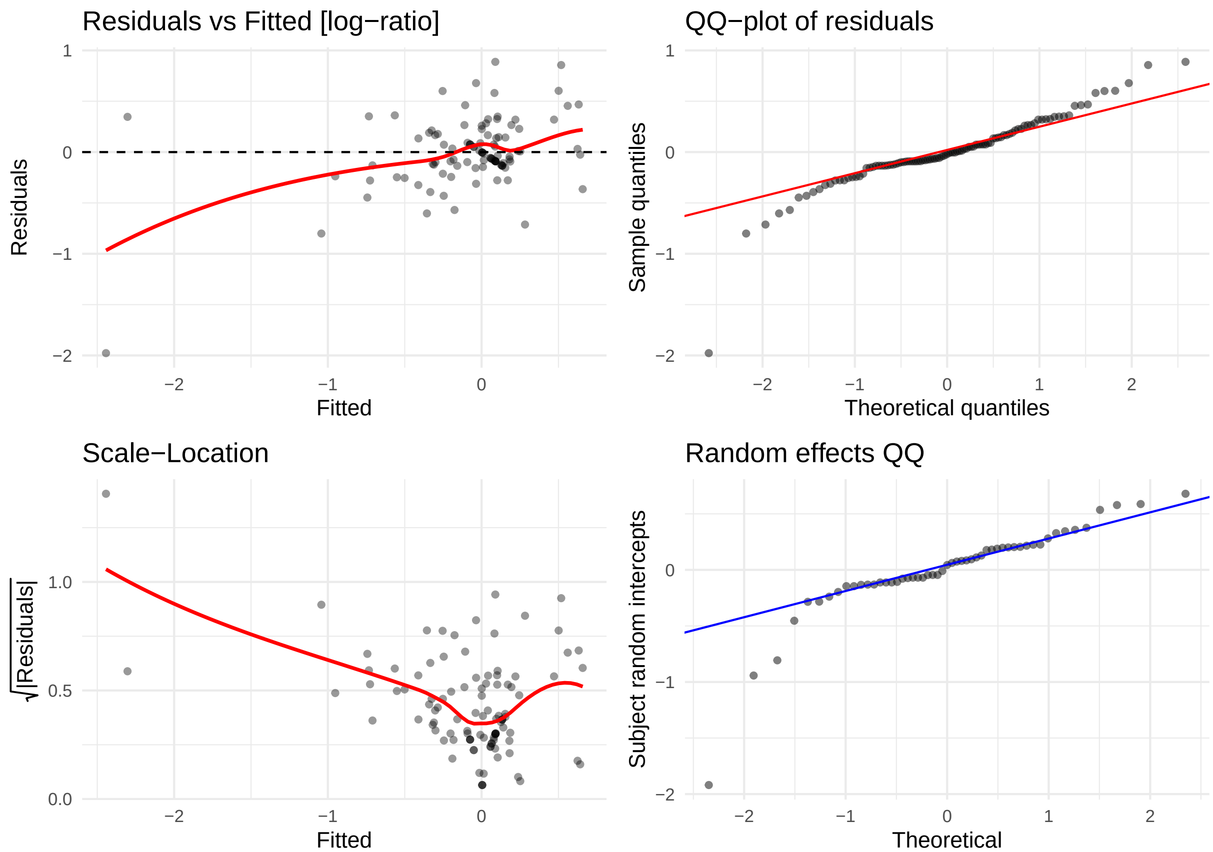
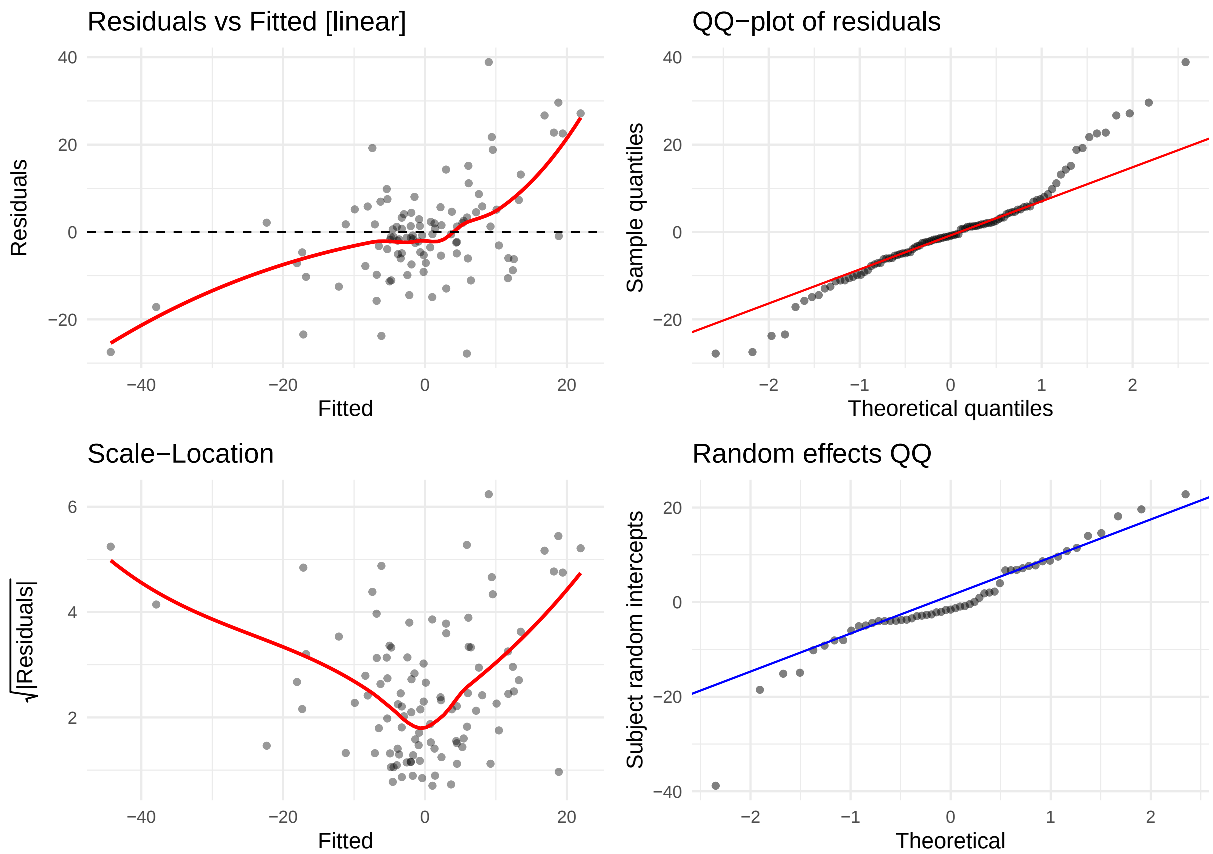


Table 31: Shapiro-Wilk — TNFRSF18 (GITR)

Scale	W	p-value	Interpretation
linear	0.3590	0	Strong departure from normality
log-ratio	0.8813	0	Strong departure from normality

5.30 TNFSF6 (FasL) (`TNFSF6`)

AIC linear: 878.2 | **AIC log-ratio:** 66 | **Preferred: log**



Residuals vs Fitted [linear]
Residuals
Fitted
QQ−plot of residuals
Sample quantiles
Theoretical quantiles
Scale−Location
√|Residuals|
Fitted
Random effects QQ
Subject random intercepts
Theoretical

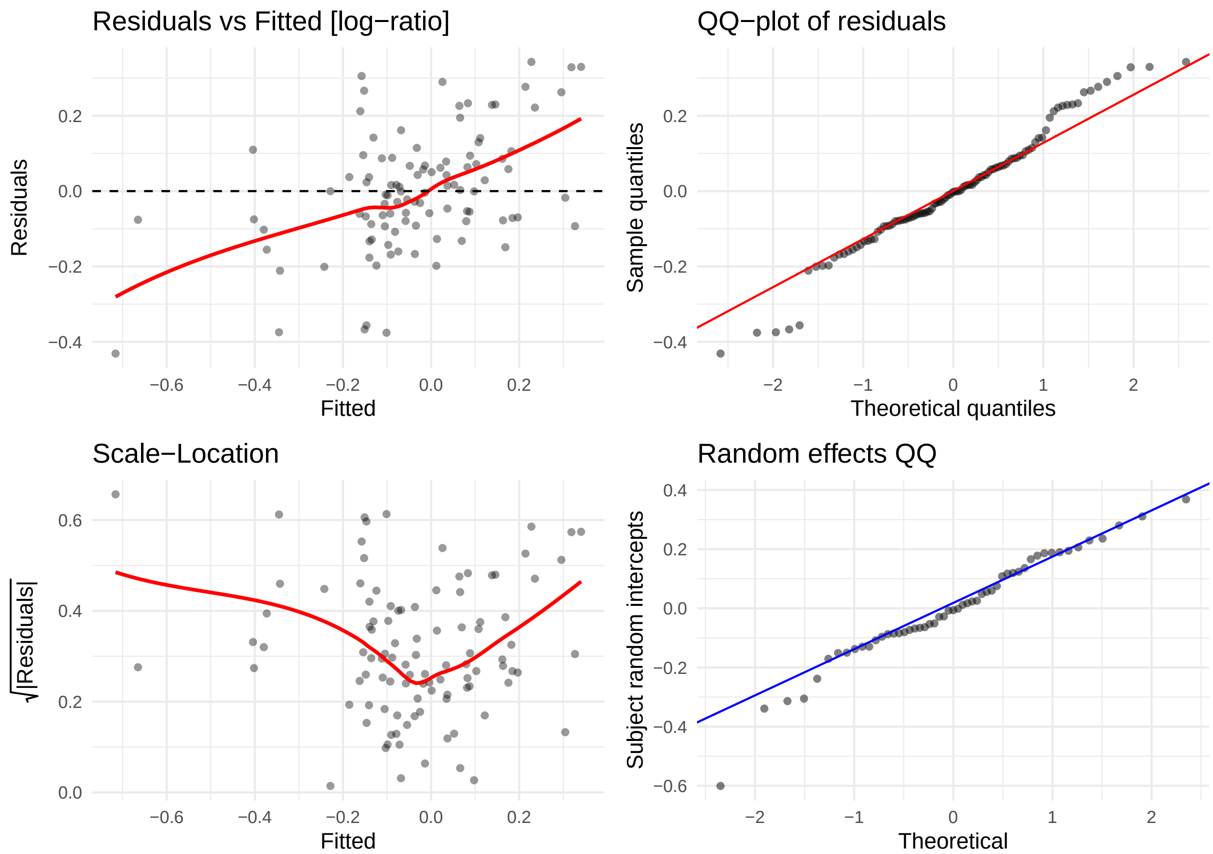
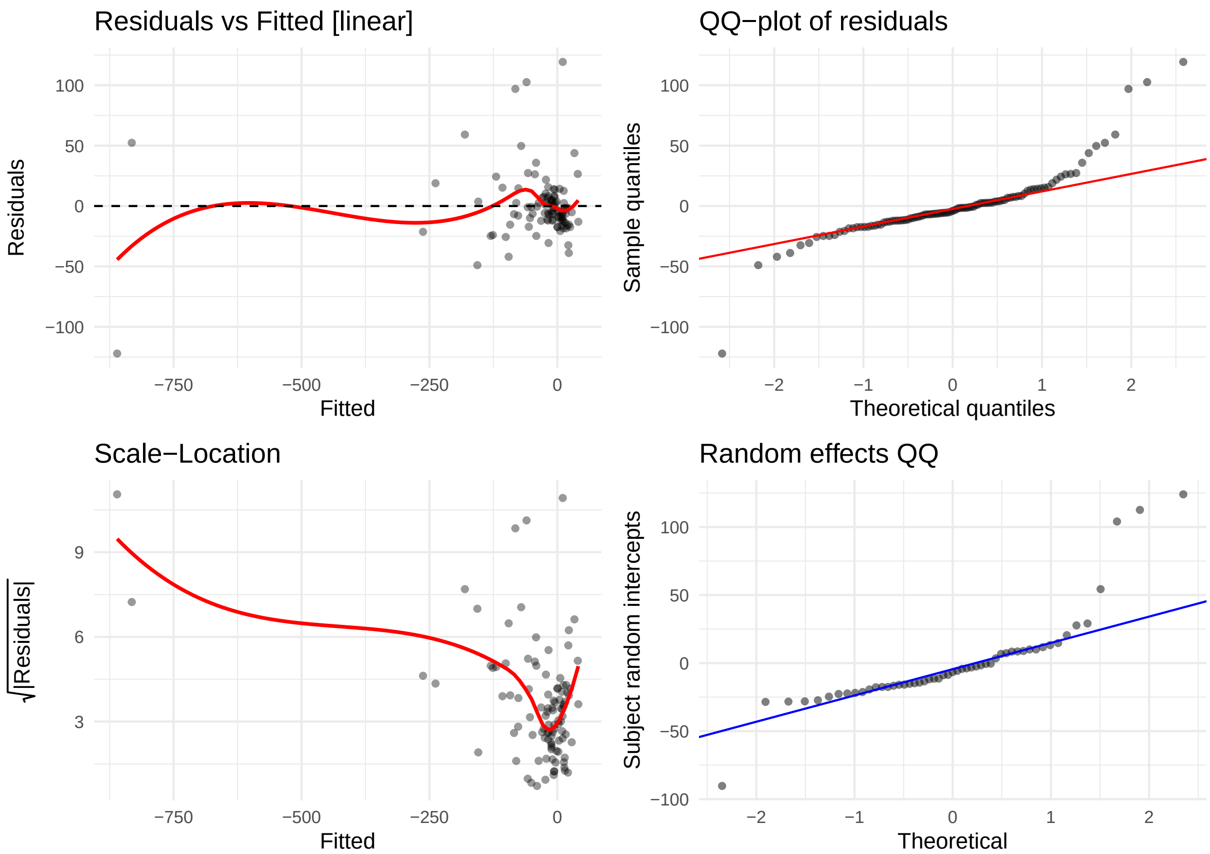


Table 32: Shapiro-Wilk — TNFSF6 (FasL)

Scale	W	p-value	Interpretation
linear	0.9517	0.0009	Strong departure from normality
log-ratio	0.9761	0.0611	Borderline

5.31 VEGF-A (VEGFA)

AIC linear: 1079.9 | **AIC log-ratio:** 193.1 | **Preferred: log**



Residuals vs Fitted [linear]
Residuals
100
50
0
−50
−100
−750
−500
−250
0
Fitted
QQ−plot of residuals
Sample quantiles
100
50
0
−50
−100
−2
−1
0
1
2
Theoretical quantiles
Scale−Location
√|Residuals|
9
6
3
−750
−500
−250
0
Fitted
Random effects QQ
Subject random intercepts
100
50
0
−50
−100
−2
−1
0
1
2
Theoretical

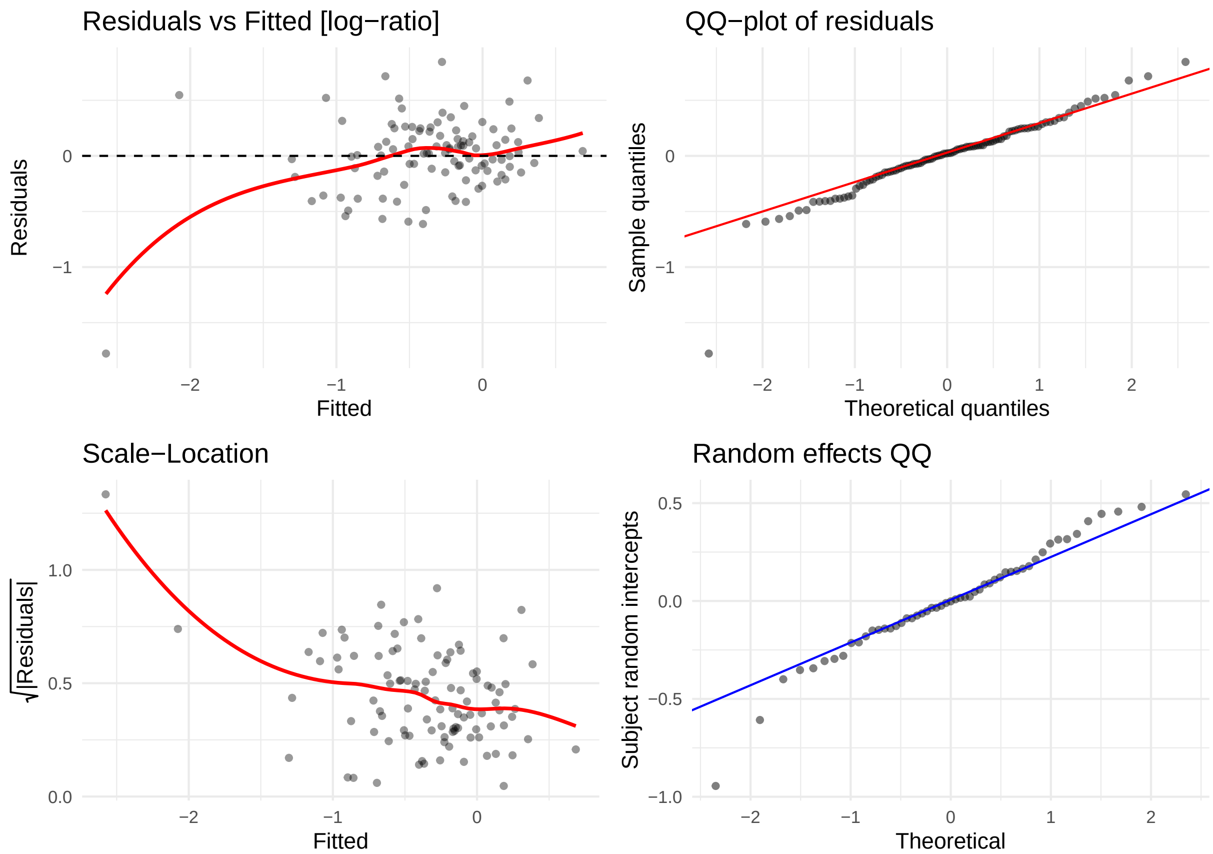
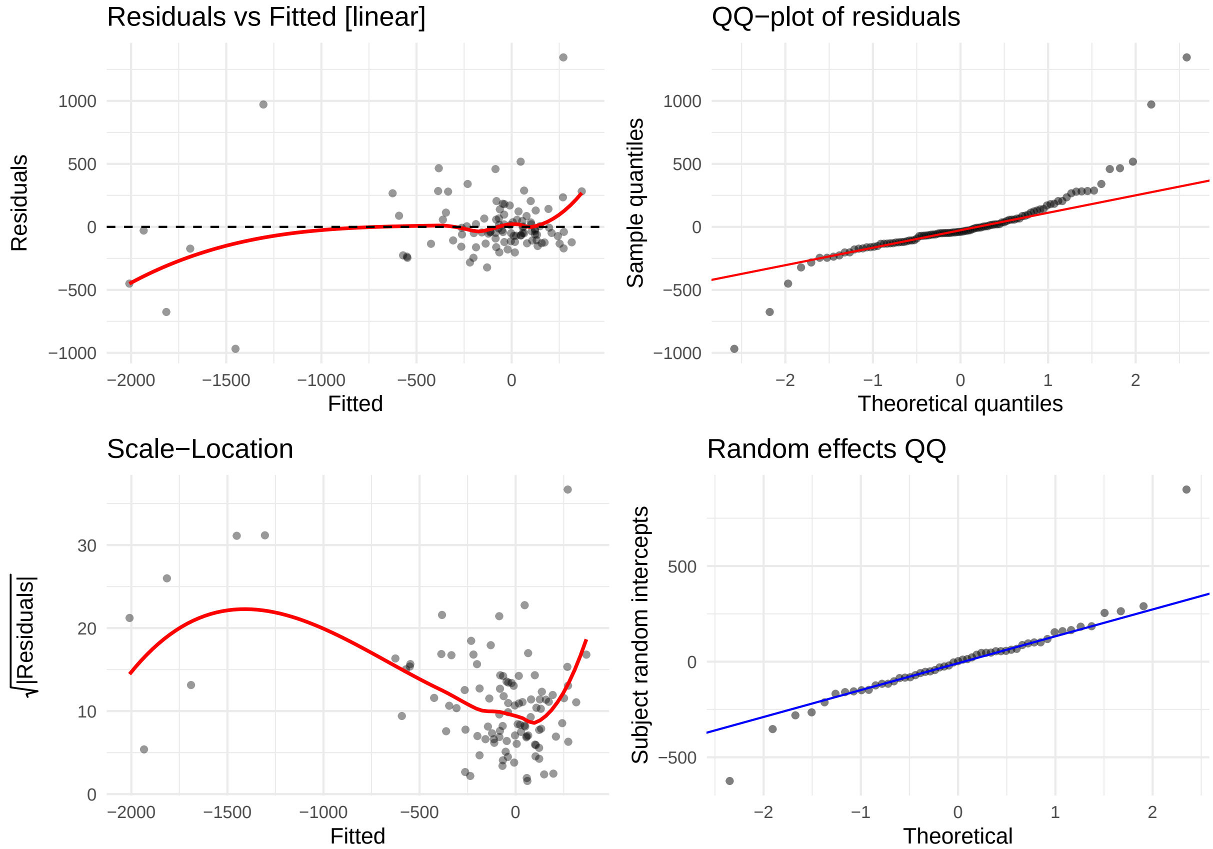


Table 33: Shapiro-Wilk — VEGF-A

Scale	W	p-value	Interpretation
linear	0.8217	0	Strong departure from normality
log-ratio	0.9210	0	Strong departure from normality

5.32 VEGF-C (`VEGFC`)

AIC linear: 1479.2 | **AIC log-ratio:** 140.2 | **Preferred: log**


Residuals vs Fitted [linear]
Residuals
Fitted
QQ−plot of residuals
Sample quantiles
Theoretical quantiles
Scale−Location
√|Residuals|
Fitted
Random effects QQ
Subject random intercepts
Theoretical

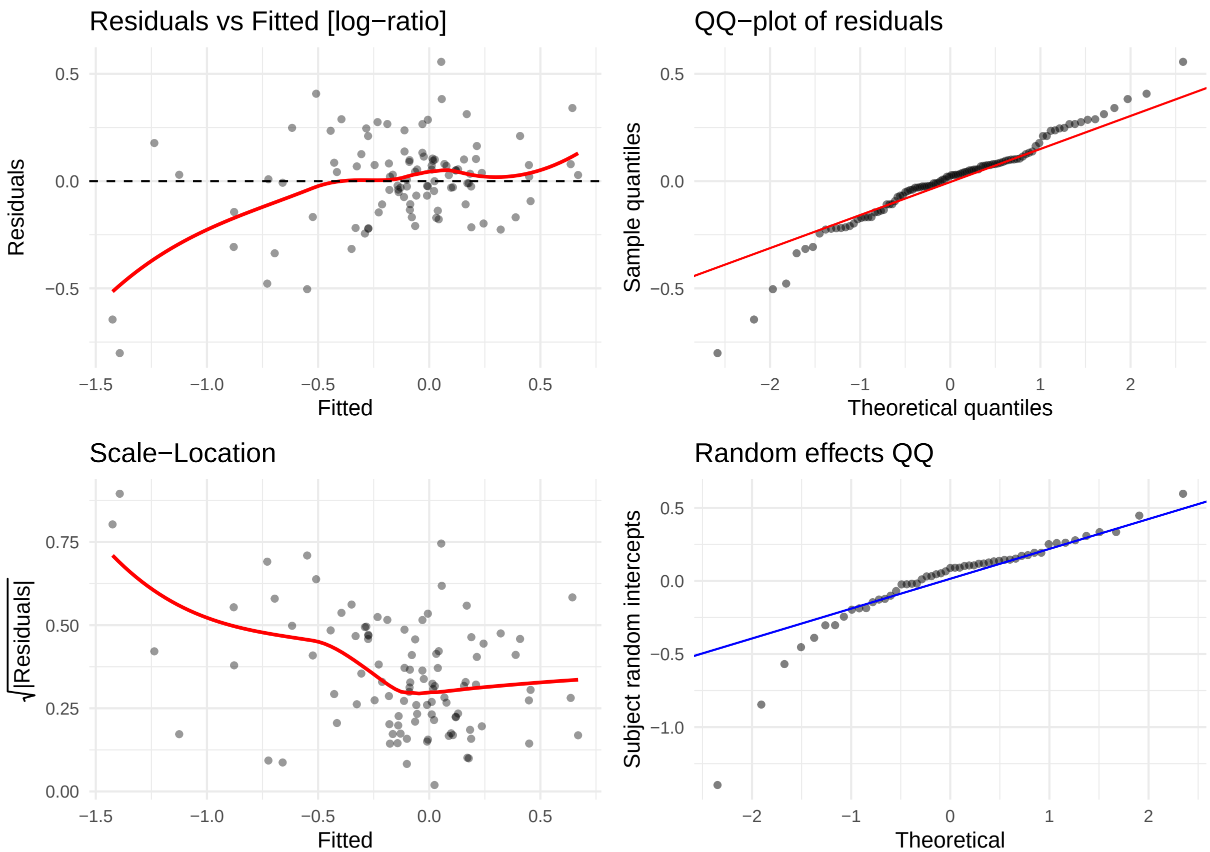
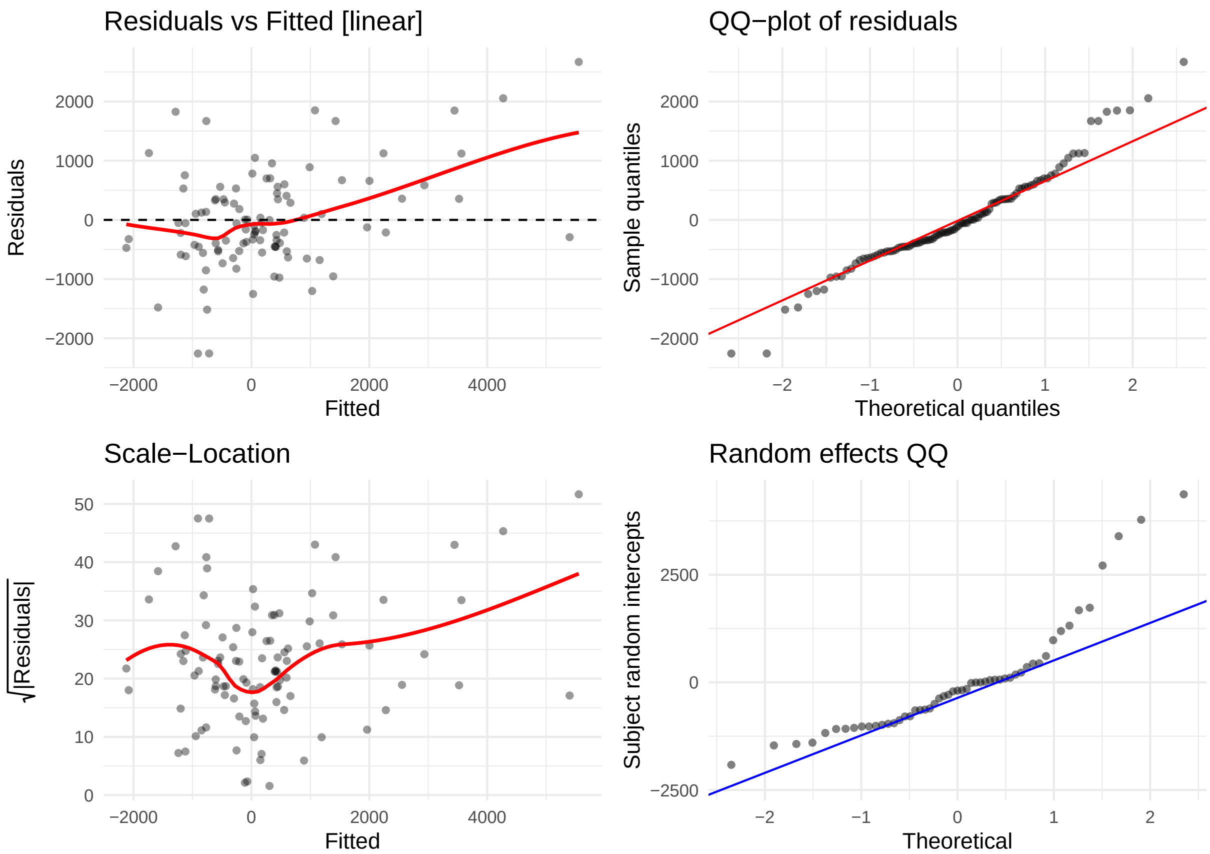


Table 34: Shapiro-Wilk — VEGF-C

Scale	W	p-value	Interpretation
linear	0.8200	0.0000	Strong departure from normality
log-ratio	0.9529	0.0011	Strong departure from normality

5.33 VEGFR2 (`VEGFR2`)

AIC linear: 1752.4 | **AIC log-ratio:** -80.2 | **Preferred: log**



Residuals vs Fitted [linear]
Residuals
Fitted
QQ−plot of residuals
Sample quantiles
Theoretical quantiles
Scale−Location
√|Residuals|
Fitted
Random effects QQ
Subject random intercepts
Theoretical

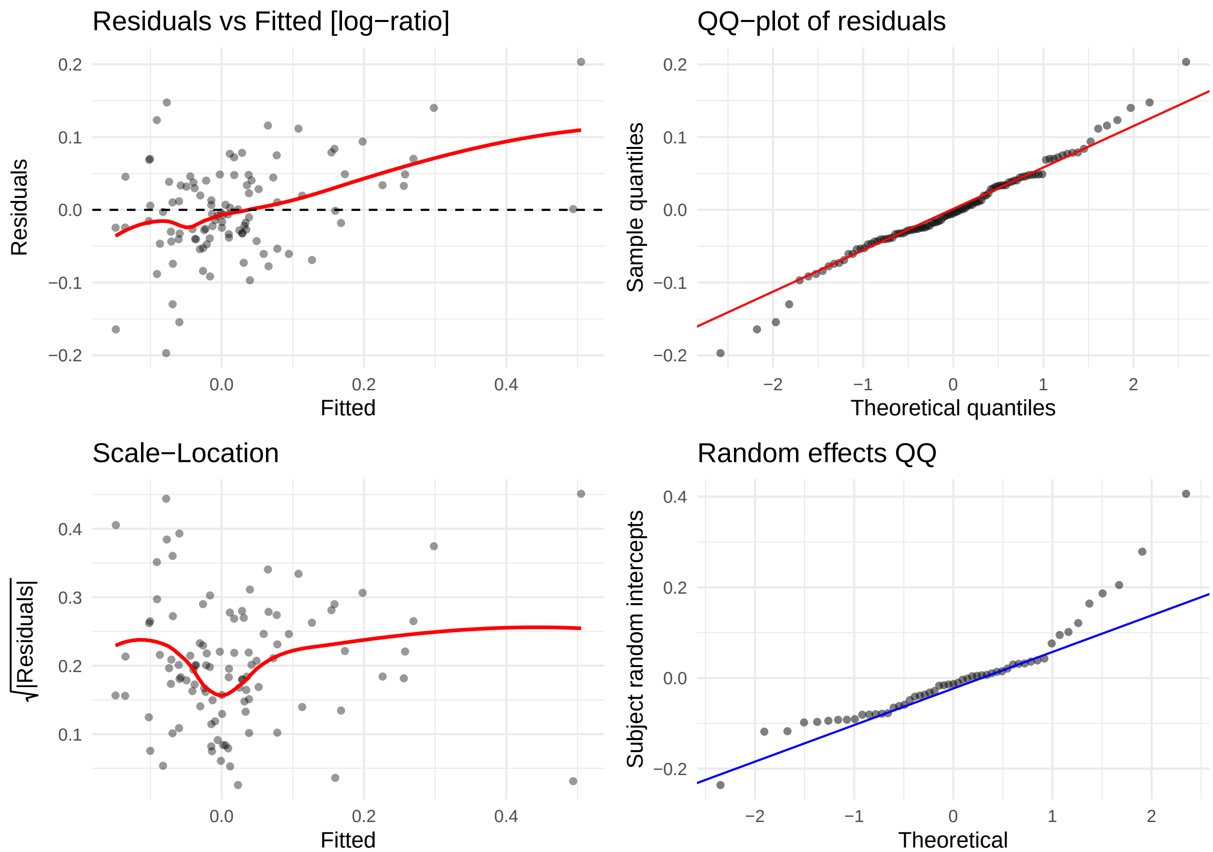


Table 35: Shapiro-Wilk — VEGFR2

Scale	W	p-value	Interpretation
linear	0.967	0.0119	Some departure from normality
log-ratio	0.982	0.1793	No evidence of non-normality

6 AIC comparison summary

```
##
##
## **Summary:** 33 analytes favour log-ratio; 0 favour linear; 0 log models failed to converge.
```

7 Recommendations

7.1 Log-ratio model preferred (ΔAIC > 2)

Back-transform emmeans with `exp()` to obtain **fold-changes** (e.g. exp(0.30) 1.35 = 35% increase over baseline). Report geometric means rather than arithmetic means in tables. Document the per-analyte offset used in `lb_log_offset()`.

7.2 Linear model preferred or comparable fit

Retain the linear model. If this is driven by many OOR values collapsing variability on the log scale, flag the analyte and consider a Tobit LMM.

7.3 High >ULOQ proportion ($> 20\%$)

For analytes with >20% of observations censored above ULOQ: the linear and log-ratio models will underestimate concentrations. Consider:

- Substituting `ULOQ` for `>ULOQ` values (imputation at upper bound) as a sensitivity analysis.
- A **Tobit mixed model** (`censReg` or `crch` packages) for the primary analysis.

7.4 Non-normality not resolved by log-ratio (Shapiro-Wilk $p < 0.01$)

If both models show strong residual non-normality:

1. Check for outliers: flag observations with $|\text{standardised residual}| > 4$ for data-quality review.
2. Report a sensitivity analysis excluding confirmed outliers.
3. Consider a **robust LMM** (`robustlmm::rlmer`) for the primary analysis.

7.5 Heavy tails (excess kurtosis > 7 in `chg`)

Common for VEGF-A, IL-6, IL-8 due to extreme responders. These analytes are strong candidates for the log-ratio model and outlier review.

Generated from `reports/model fit/impress_lb_model_fit.Rmd` — sham-randomised data.